

DETAILED PROJECT REPORT

Augmentation of Piped Water Supply to Sivasagar Town and Adjoining Panchayats

raising capacity from 24 MLD to 46 MLD

Submitted by: Assam Public Health Engineering Department

Implementing agency / SPV: Assam Jal Board

DPR prepared by: Luit Water Infrastructure Consultants Pvt. Ltd.

Prepared in accordance with the KIIFB General DPR Template (chapters 1-18).

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1. SALIENT FEATURES

Sl.	Particular	Detail
1	Title of the project	Augmentation of piped water supply to Sivasagar town and adjoining panchayats
2	District / Taluk / Local body	Sivasagar / Sivasagar / Sivasagar Municipal Board and 9 Gram Panchayats
3	Implementing agency / SPV	Assam Jal Board
4	DPR prepared by	Luit Water Infrastructure Consultants Pvt. Ltd.
5	Project outlay	Rs. 264.80 crore
6	Budget provision	Rs. 88.00 crore in the current financial year
7	Administrative sanction	AS accorded vide G.O. No. PHE/412/2026 dated 5 February 2026
8	Nature of the project	Augmentation of an existing scheme
9	Present status	Existing 24 MLD scheme commissioned in 2003; supply restricted to 90 minutes per day
10	Need for the project	Design population of 3.42 lakh by 2041 against a present service of 1.96 lakh at 78 lpcd
11	Design capacity	46 MLD
12	Service level proposed	135 lpcd, 24x7 supply
13	Source	River Dikhow, intake at Nazira
14	Land required / status	6.40 ha; 4.10 ha in departmental possession, 2.30 ha under acquisition
15	Basis of estimate	Assam PHED Schedule of Rates 2025-26
16	Details of revenue streams	Water tariff; estimated Rs. 11.60 crore per annum at full coverage
17	Cost Benefit Analysis	BCR 1.52
18	Details of project risks	Chapter 12 — 13 risks identified
19	Implementation schedule	24 months from zero date
20	Statutory clearances	Chapter 16
21	O&M arrangements	Chapter 18
22	Attached drawings and other attachments	Annexures I to VII

1.1 Salient Features (continued) — Sheet 1

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Reference: drawing no. PL-716, revision A.

1.1 Salient Features (continued) — Sheet 2

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-113	75.08	1.87	86.12	11.04
N-198	41.14	12.99	49.83	8.69
N-062	41.03	15.37	62.97	21.94
N-111	55.46	11.33	66.42	10.96
N-160	42.40	16.39	61.79	19.39
N-102	73.21	18.82	96.97	23.77
N-191	43.58	16.12	52.04	8.46
N-137	28.44	2.45	44.26	15.82
N-093	82.10	0.42	96.15	14.05
N-009	66.38	10.48	76.29	9.91
N-104	46.89	16.50	68.40	21.51
N-168	77.52	13.74	91.42	13.89
N-131	39.31	7.61	62.34	23.03
N-175	87.16	20.98	107.29	20.13
N-148	92.85	8.43	101.10	8.25
N-215	92.33	19.65	115.97	23.64
N-155	73.96	19.46	92.54	18.59
N-046	81.87	22.20	91.54	9.67
N-141	41.66	10.83	66.38	24.73
N-039	88.49	21.25	98.14	9.65
N-141	50.95	13.30	59.24	8.28
N-050	95.90	6.83	111.70	15.80
N-040	84.07	6.15	92.79	8.72

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

2. EXECUTIVE SUMMARY

The project provides for raising capacity from 24 mld to 46 mld. Sivasagar town is served by a surface water scheme commissioned in 2003 with a design capacity of 24 MLD drawn from the river Dikhow. The distribution network covers 68 per cent of the municipal area and none of the nine

The total project cost is estimated at Rs. 264.80 crore inclusive of all taxes and duties, at the price level of the current financial year. The implementation period is 24 months from the zero date.

2.1 Executive Summary (continued) — Sheet 1

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

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Reference: drawing no. ST-399, revision C.

2.1 Executive Summary (continued) — Sheet 2

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-09	Booster	192.3	67.0	175.67
P-03	Sludge	242.6	79.3	262.02
P-02	Raw water	149.7	50.8	103.52
P-08	Booster	225.4	58.3	179.06
P-09	Clear water	270.4	66.3	244.22
P-08	Raw water	164.7	43.3	97.19
P-06	Sludge	90.2	19.0	23.33
P-12	Booster	224.7	50.8	155.40
P-03	Backwash	146.5	42.3	84.38
P-11	Backwash	78.2	56.7	60.50
P-08	Backwash	147.0	54.4	108.99
P-08	Booster	266.6	60.6	220.09
P-09	Clear water	151.4	59.6	122.83
P-10	Booster	134.2	68.5	125.17
P-09	Backwash	294.1	59.4	237.82
P-08	Clear water	175.9	42.3	101.35
P-09	Sludge	108.7	71.2	105.46
P-03	Clear water	236.5	43.6	140.40
P-03	Clear water	111.0	26.5	40.15
P-10	Sludge	172.3	22.4	52.65
P-04	Sludge	177.2	12.8	30.83
P-03	Raw water	99.0	20.8	28.06
P-12	Sludge	142.4	36.2	70.15
P-04	Backwash	141.4	30.5	58.73
P-01	Sludge	269.2	63.7	233.48

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

3 INTRODUCTION

Sivasagar town is served by a surface water scheme commissioned in 2003 with a design capacity of 24 MLD drawn from the river Dikhow. The distribution network covers 68 per cent of the municipal area and none of the nine adjoining panchayats now proposed for inclusion.

Supply is presently restricted to 90 minutes in the morning at an assessed 78 litres per capita per day against the norm of 135. Non-revenue water is assessed at 38 per cent, largely through leakage in cast iron mains laid before 1990.

3.x Introduction — Working Papers — Sheet 1

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. ST-526, revision C.

3.x Introduction — Working Papers — Sheet 2

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
6,344	500	HDPE	K-9	1220.0
13,639	500	MS	PN 10	428.6
23,468	400	PVC-O	K-7	591.1
22,328	250	PVC-O	PN 10	152.7
13,929	150	MS	K-7	636.3
7,054	400	HDPE	PN 16	1695.6
573	100	PVC-O	PN 16	1707.8
22,707	300	PVC-O	PN 10	1555.2
17,785	150	HDPE	K-7	336.6
21,477	600	DI	PN 6	1259.0
14,531	200	HDPE	PN 16	1384.9
12,740	600	DI	PN 16	1096.9
22,624	250	HDPE	K-9	981.5
22,679	300	MS	K-7	836.3
11,893	150	DI	PN 10	679.0
23,545	250	PVC-O	K-7	1767.3
13,079	500	HDPE	K-7	570.4
3,866	500	HDPE	K-9	1519.2
15,686	600	MS	K-7	56.0
7,577	500	DI	K-7	123.1
11,803	300	MS	K-9	1619.2
18,917	300	MS	K-9	763.3
8,800	200	PVC-O	PN 16	1185.1
21,321	400	PVC-O	K-7	735.2
17,433	250	MS	K-7	1202.1
373	250	PVC-O	K-7	1270.7

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

3.x Introduction — Working Papers — Sheet 3

Design Computation

Velocity = $50.039 \times 7.360 = 368.305$ m/s (permissible 651.442) — safe

Discharge = $80.462 \times 10.027 = 806.787$ lps (permissible 968.612) — safe

Detention time = $55.618 \times 9.100 = 506.118$ hr (permissible 780.965) — revise section

Discharge = $42.556 \times 5.395 = 229.582$ lps (permissible 297.547) — revise section

Head loss = $2.598 \times 3.551 = 9.226$ m (permissible 11.225) — safe

Velocity = $26.743 \times 9.122 = 243.944$ m/s (permissible 311.938) — revise section

Detention time = $12.454 \times 6.208 = 77.319$ hr (permissible 82.189) — safe

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

3.x Introduction — Working Papers — Sheet 4

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. ST-161, revision C.

3.x Introduction — Working Papers — Sheet 5

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Total hardness	mg/l	262	117	200
Total coliform	MPN/100ml	885	Absent	Absent
Residual chlorine	mg/l	Nil	0.62	0.2 min
pH	-	6.9	6.9	6.5-8.5
pH	-	6.5	7.7	6.5-8.5
Total hardness	mg/l	286	185	200
Iron	mg/l	2.81	0.07	1.0
Total hardness	mg/l	477	110	200
Residual chlorine	mg/l	Nil	0.34	0.2 min
Total coliform	MPN/100ml	301	Absent	Absent
pH	-	7.9	7.9	6.5-8.5
Total coliform	MPN/100ml	44	Absent	Absent
Iron	mg/l	4.03	0.20	1.0
pH	-	7.3	7.2	6.5-8.5
Total hardness	mg/l	478	179	200
Residual chlorine	mg/l	Nil	0.67	0.2 min
pH	-	7.8	6.9	6.5-8.5
Iron	mg/l	2.76	0.13	1.0
Iron	mg/l	3.32	0.26	1.0
Turbidity	NTU	137.3	0.40	1
pH	-	6.6	7.7	6.5-8.5
pH	-	8.1	7.3	6.5-8.5

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

3.x Introduction — Working Papers — Sheet 6

Design Computation

Residual pressure = $69.739 \times 9.928 = 692.351$ m (permissible 1178.289) — safe

Velocity = $28.516 \times 0.599 = 17.088$ m/s (permissible 31.008) — safe

Velocity = $77.078 \times 10.153 = 782.547$ m/s (permissible 914.616) — revise section

Discharge = $57.712 \times 0.912 = 52.638$ lps (permissible 95.714) — safe

Residual pressure = $59.693 \times 10.397 = 620.646$ m (permissible 983.618) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

3.x Introduction — Working Papers — Sheet 7

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Reference: drawing no. ST-422, revision B.

3.x Introduction — Working Papers — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
17.9	Medium sand	39	SPT
5.4	Clayey silt	35	DS
26.5	Clayey silt	48	DS
23.1	Coarse sand	31	DS
19.1	Medium sand	16	UDS
13.7	Coarse sand	22	SPT
22.0	Gravelly sand	24	DS
19.0	Gravelly sand	53	UDS
27.7	Clayey silt	49	SPT
27.5	Weathered rock	21	DS
16.7	Medium sand	23	UDS
5.5	Medium sand	59	UDS
23.2	Clayey silt	49	UDS
11.2	Coarse sand	23	UDS
16.6	Silty clay	58	SPT
0.9	Coarse sand	5	DS
18.7	Weathered rock	25	DS
26.2	Clayey silt	52	UDS
18.5	Clayey silt	56	UDS
11.2	Weathered rock	55	SPT
2.7	Gravelly sand	60	UDS
7.0	Silty clay	48	SPT
9.6	Weathered rock	31	DS
24.3	Silty clay	57	DS
11.4	Weathered rock	58	SPT

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

3.x Introduction — Working Papers — Sheet 9

Design Computation

Discharge = $70.208 \times 6.126 = 430.085$ lps (permissible 463.502) — safe

Discharge = $3.267 \times 9.001 = 29.405$ lps (permissible 42.912) — safe

Residual pressure = $88.842 \times 10.577 = 939.679$ m (permissible 1290.674) — revise section

Residual pressure = $37.961 \times 4.738 = 179.852$ m (permissible 282.388) — safe

Discharge = $1.314 \times 0.553 = 0.727$ lps (permissible 1.112) — safe

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

3.x Introduction — Working Papers — Sheet 10

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. WS-460, revision C.

4 STATUS OF FEASIBILITY STUDIES

A feasibility study was carried out in 2023 under the state water sector programme, which examined three source options — augmentation from the Dikhow, a well field in the Disang valley, and conjunctive use. Surface augmentation was recommended on yield reliability and capital cost.

4.x Status Of Feasibility Studies — Working Papers — Sheet 1

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Reference: drawing no. PL-471, revision A.

4.x Status Of Feasibility Studies — Working Papers — Sheet 2

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Iron	mg/l	3.19	0.20	1.0
Turbidity	NTU	128.7	0.49	1
Total hardness	mg/l	92	167	200
pH	-	7.7	8.0	6.5-8.5
Total coliform	MPN/100ml	494	Absent	Absent
Total coliform	MPN/100ml	482	Absent	Absent
Total coliform	MPN/100ml	142	Absent	Absent
Total hardness	mg/l	149	116	200
Residual chlorine	mg/l	Nil	0.72	0.2 min
pH	-	8.0	7.3	6.5-8.5
Total hardness	mg/l	419	97	200
Residual chlorine	mg/l	Nil	0.35	0.2 min
pH	-	7.7	7.4	6.5-8.5
Turbidity	NTU	238.7	0.52	1
pH	-	7.3	7.7	6.5-8.5
Iron	mg/l	3.07	0.18	1.0
Residual chlorine	mg/l	Nil	0.57	0.2 min
Iron	mg/l	3.69	0.16	1.0
Residual chlorine	mg/l	Nil	0.67	0.2 min
Total coliform	MPN/100ml	581	Absent	Absent
Total hardness	mg/l	143	138	200
Iron	mg/l	3.55	0.27	1.0
Total hardness	mg/l	267	150	200

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

4.x Status Of Feasibility Studies — Working Papers — Sheet 3

Design Computation

Detention time = $36.730 \times 6.977 = 256.280$ hr (permissible 436.917) — safe

Discharge = $76.284 \times 2.156 = 164.505$ lps (permissible 266.497) — safe

Detention time = $3.918 \times 0.651 = 2.549$ hr (permissible 3.769) — safe

Discharge = $68.533 \times 10.623 = 728.047$ lps (permissible 1191.963) — safe

Residual pressure = $45.626 \times 5.737 = 261.771$ m (permissible 491.401) — safe

Discharge = $7.187 \times 9.294 = 66.801$ lps (permissible 88.842) — safe

Residual pressure = $43.079 \times 8.737 = 376.391$ m (permissible 523.370) — safe

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

4.x Status Of Feasibility Studies — Working Papers — Sheet 4

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

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Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. PL-250, revision C.

4.x Status Of Feasibility Studies — Working Papers — Sheet 5

Bore Log Summary

Depth (m)	Strata	N value	Sample
6.1	Coarse sand	11	UDS
26.0	Medium sand	53	SPT
10.1	Coarse sand	37	UDS
16.8	Gravelly sand	25	SPT
18.1	Medium sand	60	SPT
24.5	Clayey silt	33	DS
10.2	Clayey silt	22	UDS
12.2	Medium sand	59	UDS
27.3	Silty clay	35	DS
10.6	Silty clay	35	UDS
14.3	Silty clay	24	DS
3.2	Clayey silt	4	SPT
10.6	Silty clay	30	UDS
23.9	Medium sand	44	SPT
1.7	Medium sand	40	SPT
9.3	Weathered rock	40	UDS

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

4.x Status Of Feasibility Studies — Working Papers — Sheet 6

Design Computation

Detention time = $63.459 \times 4.411 = 279.940$ hr (permissible 525.124) — safe

Residual pressure = $89.180 \times 2.498 = 222.791$ m (permissible 342.070) — safe

Velocity = $77.293 \times 5.048 = 390.161$ m/s (permissible 617.926) — safe

Discharge = $79.203 \times 10.270 = 813.424$ lps (permissible 1164.927) — safe

Velocity = $24.987 \times 8.666 = 216.537$ m/s (permissible 279.580) — safe

Discharge = $88.978 \times 1.832 = 163.023$ lps (permissible 224.184) — safe

Detention time = $89.667 \times 8.918 = 799.612$ hr (permissible 1283.796) — safe

Velocity = $74.481 \times 3.065 = 228.269$ m/s (permissible 372.457) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

4.x Status Of Feasibility Studies — Working Papers — Sheet 7

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

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Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. ST-505, revision B.

4.x Status Of Feasibility Studies — Working Papers — Sheet 8

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	56.4	0.38	1
Total hardness	mg/l	267	169	200
Total coliform	MPN/100ml	33	Absent	Absent
pH	-	7.6	7.0	6.5-8.5
pH	-	7.0	7.2	6.5-8.5
Turbidity	NTU	178.7	0.42	1
Residual chlorine	mg/l	Nil	0.59	0.2 min
Total coliform	MPN/100ml	564	Absent	Absent
Residual chlorine	mg/l	Nil	0.37	0.2 min
pH	-	6.4	7.1	6.5-8.5
Turbidity	NTU	54.5	0.85	1
Turbidity	NTU	69.2	0.32	1
Total hardness	mg/l	124	92	200
Turbidity	NTU	191.7	0.20	1
Turbidity	NTU	193.2	0.42	1
Iron	mg/l	3.22	0.18	1.0
Total coliform	MPN/100ml	229	Absent	Absent
Total hardness	mg/l	340	152	200
Total hardness	mg/l	321	174	200
pH	-	7.5	8.0	6.5-8.5
pH	-	7.5	7.2	6.5-8.5
Total hardness	mg/l	416	87	200
Total hardness	mg/l	407	108	200

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

5 REQUIREMENT / DEMAND ANALYSIS

Population has been projected by the geometric increase method from the 2011 and 2021 census figures, cross-checked against the arithmetic and incremental increase methods. The design population for 2041 is 3.42 lakh including the nine panchayats.

Demand at 135 lpcd with 15 per cent allowance for losses works out to 53.1 MLD at the design year. Phase I capacity of 46 MLD meets demand to 2036, with the intake and clear water main sized for the full 53 MLD to avoid a second disruption.

5.x Requirement / Demand Analysis — Working Papers — Sheet 1

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. HY-344, revision B.

5.x Requirement / Demand Analysis — Working Papers — Sheet 2

Population Projection

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2046	Ward 10	21,568	100	2.157
2036	Ward 14	34,235	135	4.622
2031	Ward 15	24,002	150	3.600
2051	Ward 2	26,980	135	3.642
2031	Ward 20	35,904	135	4.847
2051	Ward 16	16,203	135	2.187
2036	Ward 14	32,124	100	3.212
2051	Ward 7	45,613	135	6.158
2026	Ward 15	14,469	70	1.013
2051	Ward 19	13,526	70	0.947
2031	Ward 1	21,654	100	2.165
2056	Ward 3	12,793	150	1.919
2046	Ward 22	35,645	70	2.495
2031	Ward 11	41,225	70	2.886
2031	Ward 2	37,368	135	5.045
2051	Ward 22	32,404	70	2.268
2046	Ward 8	40,681	100	4.068

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

5.x Requirement / Demand Analysis — Working Papers — Sheet 3

Design Computation

Velocity = $10.487 \times 9.814 = 102.919$ m/s (permissible 177.611) — safe

Velocity = $32.985 \times 9.539 = 314.642$ m/s (permissible 565.238) — safe

Velocity = $44.679 \times 7.905 = 353.169$ m/s (permissible 464.107) — safe

Residual pressure = $23.231 \times 0.579 = 13.448$ m (permissible 16.465) — safe

Discharge = $51.856 \times 4.476 = 232.133$ lps (permissible 250.731) — safe

Discharge = $41.245 \times 7.334 = 302.480$ lps (permissible 463.757) — safe

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

5.x Requirement / Demand Analysis — Working Papers — Sheet 4

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. PL-593, revision A.

5.x Requirement / Demand Analysis — Working Papers — Sheet 5

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-08	Booster	102.1	85.8	119.38
P-05	Sludge	186.9	85.4	217.40
P-10	Booster	131.8	16.0	28.67
P-08	Clear water	257.3	58.9	206.52
P-05	Booster	305.9	80.5	335.42
P-02	Backwash	31.5	79.1	33.98
P-03	Clear water	319.7	81.8	356.26
P-11	Sludge	25.7	31.5	11.04
P-01	Backwash	172.4	66.2	155.55
P-06	Backwash	169.9	83.3	192.69
P-10	Backwash	62.4	64.2	54.55
P-10	Raw water	105.7	70.4	101.33
P-07	Backwash	82.9	26.8	30.22
P-02	Clear water	239.9	39.0	127.55
P-07	Sludge	66.9	22.7	20.68
P-03	Clear water	97.7	33.2	44.14
P-02	Backwash	30.9	14.1	5.93
P-10	Raw water	145.2	60.7	120.16

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

5.x Requirement / Demand Analysis — Working Papers — Sheet 6

Design Computation

Detention time = $46.390 \times 2.456 = 113.920$ hr (permissible 136.677) — safe

Residual pressure = $46.118 \times 3.220 = 148.519$ m (permissible 225.606) — safe

Head loss = $33.988 \times 7.471 = 253.912$ m (permissible 388.939) — safe

Detention time = $28.405 \times 0.919 = 26.096$ hr (permissible 48.063) — safe

Head loss = $60.798 \times 2.727 = 165.768$ m (permissible 283.356) — safe

Discharge = $39.759 \times 6.479 = 257.598$ lps (permissible 397.244) — safe

Discharge = $73.406 \times 5.251 = 385.442$ lps (permissible 534.183) — revise section

Detention time = $64.708 \times 3.138 = 203.027$ hr (permissible 283.063) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

5.x Requirement / Demand Analysis — Working Papers — Sheet 7

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. ST-729, revision C.

5.x Requirement / Demand Analysis — Working Papers — Sheet 8

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-024	35.82	16.88	49.99	14.17
N-168	62.40	3.94	78.87	16.48
N-034	52.53	9.49	60.04	7.51
N-157	65.32	5.53	72.33	7.01
N-200	77.81	13.17	90.01	12.21
N-217	48.33	7.36	64.27	15.95
N-058	52.56	21.13	61.44	8.88
N-057	95.13	3.72	118.73	23.60
N-114	80.74	5.61	97.05	16.30
N-011	95.54	3.27	112.36	16.82
N-209	29.04	19.99	45.78	16.74
N-021	59.53	22.38	77.52	17.99
N-169	95.75	18.27	116.50	20.75
N-132	34.88	16.18	45.04	10.16
N-094	78.00	21.17	91.93	13.93
N-124	30.24	9.77	38.82	8.58
N-008	77.00	5.91	100.81	23.81
N-029	84.15	4.98	105.08	20.93
N-147	62.19	21.10	83.25	21.06
N-057	94.45	16.08	103.52	9.06
N-190	46.60	0.86	68.08	21.48
N-112	83.60	19.35	96.00	12.40
N-164	95.38	7.66	116.05	20.67
N-186	83.33	15.77	105.10	21.77
N-124	45.34	2.84	61.50	16.16

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

5.x Requirement / Demand Analysis — Working Papers — Sheet 9

Design Computation

Detention time = $33.292 \times 3.474 = 115.664$ hr (permissible 215.112) — safe

Residual pressure = $69.609 \times 8.563 = 596.068$ m (permissible 645.385) — safe

Discharge = $6.481 \times 3.888 = 25.200$ lps (permissible 32.329) — safe

Discharge = $10.150 \times 3.445 = 34.966$ lps (permissible 46.001) — safe

Discharge = $23.151 \times 2.025 = 46.892$ lps (permissible 77.089) — safe

Velocity = $20.372 \times 1.107 = 22.542$ m/s (permissible 40.090) — safe

Detention time = $17.206 \times 2.683 = 46.167$ hr (permissible 76.413) — safe

Residual pressure = $73.518 \times 9.959 = 732.185$ m (permissible 801.067) — revise section

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

5.x Requirement / Demand Analysis — Working Papers — Sheet 10

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. WS-259, revision C.

5.x Requirement / Demand Analysis — Working Papers — Sheet 11

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
22,442	600	PVC-O	K-9	323.9
5,776	300	PVC-O	K-9	894.8
15,769	400	DI	PN 6	1352.0
7,396	400	DI	PN 6	1246.3
11,473	250	PVC-O	PN 10	665.6
13,254	600	MS	PN 10	1607.5
8,451	600	MS	PN 10	1644.9
10,436	500	HDPE	K-7	105.8
20,085	300	MS	K-7	329.9
11,783	500	DI	PN 10	600.0
10,280	250	MS	K-7	329.3
16,430	600	MS	PN 6	1514.3
13,105	250	HDPE	PN 6	1130.2
5,002	500	PVC-O	PN 10	1321.3
7,406	600	PVC-O	PN 6	1728.3
15,453	200	MS	K-9	1407.2
22,395	600	PVC-O	K-7	1763.0
8,161	150	DI	PN 16	1050.3
22,983	600	DI	K-9	774.5
15,713	250	MS	K-7	1390.4
14,548	100	DI	PN 10	478.6
18,998	300	DI	PN 6	1470.1

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

5.x Requirement / Demand Analysis — Working Papers — Sheet 12

Design Computation

Detention time = $70.266 \times 6.155 = 432.515$ hr (permissible 746.750) — safe

Detention time = $45.809 \times 9.535 = 436.775$ hr (permissible 731.540) — safe

Velocity = $24.045 \times 9.924 = 238.614$ m/s (permissible 341.406) — safe

Head loss = $4.017 \times 11.687 = 46.941$ m (permissible 50.257) — safe

Discharge = $45.752 \times 6.984 = 319.520$ lps (permissible 341.018) — safe

Residual pressure = $18.666 \times 10.331 = 192.832$ m (permissible 297.848) — safe

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

5.x Requirement / Demand Analysis — Working Papers — Sheet 13

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. WS-191, revision B.

5.x Requirement / Demand Analysis — Working Papers — Sheet 14

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-07	Clear water	148.3	84.4	170.58
P-06	Sludge	43.8	51.0	30.43
P-09	Booster	212.3	49.9	144.30
P-08	Backwash	47.8	18.7	12.19
P-08	Backwash	257.4	54.9	192.54
P-02	Sludge	45.3	36.3	22.38
P-01	Clear water	209.9	32.6	93.17
P-07	Backwash	176.7	84.5	203.39
P-04	Backwash	140.8	20.8	39.86
P-05	Raw water	241.3	23.1	75.87
P-08	Clear water	209.5	33.2	94.66
P-11	Sludge	36.0	48.3	23.70
P-03	Clear water	113.3	21.4	33.04
P-09	Sludge	154.4	43.6	91.73
P-01	Backwash	219.2	79.6	237.60
P-05	Clear water	268.0	13.6	49.55
P-07	Backwash	161.7	56.3	124.01
P-04	Raw water	150.3	47.4	97.08
P-09	Raw water	214.2	14.3	41.63
P-07	Raw water	223.8	52.8	160.97
P-01	Booster	315.5	59.6	256.40
P-03	Backwash	268.3	70.5	257.82
P-06	Raw water	123.4	25.0	42.04
P-02	Booster	316.2	57.3	246.68

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

6 FUNCTIONAL DESIGN

The scheme comprises an intake well and jack well at Nazira, a raw water rising main of 1,100 mm diameter over 8.4 km, a conventional water treatment plant of 46 MLD, a clear water transmission main, four service reservoirs totalling 12.4 ML, and 214 km of distribution network.

6.x Functional Design — Working Papers — Sheet 1

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. WS-257, revision C.

6.x Functional Design — Working Papers — Sheet 2

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	128.0	0.61	1
Residual chlorine	mg/l	Nil	0.34	0.2 min
Residual chlorine	mg/l	Nil	0.55	0.2 min
Iron	mg/l	0.93	0.10	1.0
Total coliform	MPN/100ml	113	Absent	Absent
Iron	mg/l	0.45	0.11	1.0
Iron	mg/l	0.84	0.22	1.0
Total coliform	MPN/100ml	425	Absent	Absent
Total hardness	mg/l	464	135	200
Turbidity	NTU	110.1	0.62	1
Total hardness	mg/l	109	60	200
Iron	mg/l	3.23	0.27	1.0
Residual chlorine	mg/l	Nil	0.40	0.2 min
Total hardness	mg/l	369	70	200
Iron	mg/l	2.22	0.26	1.0
Turbidity	NTU	113.6	0.87	1
Total hardness	mg/l	473	148	200
pH	-	6.6	7.2	6.5-8.5
Residual chlorine	mg/l	Nil	0.24	0.2 min
Total coliform	MPN/100ml	307	Absent	Absent
Residual chlorine	mg/l	Nil	0.46	0.2 min
Iron	mg/l	2.95	0.22	1.0
Residual chlorine	mg/l	Nil	0.48	0.2 min
Iron	mg/l	4.08	0.12	1.0
pH	-	8.1	7.3	6.5-8.5
Total coliform	MPN/100ml	41	Absent	Absent

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

6.x Functional Design — Working Papers — Sheet 3

Design Computation

Head loss = $79.581 \times 4.871 = 387.664$ m (permissible 597.287) — safe

Residual pressure = $15.411 \times 8.827 = 136.039$ m (permissible 176.058) — safe

Discharge = $17.078 \times 5.034 = 85.967$ lps (permissible 94.916) — safe

Discharge = $67.525 \times 10.078 = 680.492$ lps (permissible 1239.513) — safe

Detention time = $51.617 \times 7.906 = 408.100$ hr (permissible 533.670) — safe

Discharge = $79.699 \times 11.020 = 878.294$ lps (permissible 1213.169) — safe

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

6.x Functional Design — Working Papers — Sheet 4

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

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Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Reference: drawing no. HY-803, revision C.

6.x Functional Design — Working Papers — Sheet 5

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
pH	-	6.4	7.3	6.5-8.5
pH	-	8.3	8.0	6.5-8.5
pH	-	8.2	7.6	6.5-8.5
Iron	mg/l	3.47	0.09	1.0
Iron	mg/l	2.51	0.03	1.0
Iron	mg/l	3.57	0.02	1.0
Total coliform	MPN/100ml	680	Absent	Absent
Total hardness	mg/l	210	119	200
pH	-	7.5	7.1	6.5-8.5
Total coliform	MPN/100ml	368	Absent	Absent
Iron	mg/l	1.50	0.11	1.0
Total hardness	mg/l	388	176	200
Residual chlorine	mg/l	Nil	0.36	0.2 min
Iron	mg/l	2.31	0.09	1.0
pH	-	7.2	7.4	6.5-8.5
Total hardness	mg/l	160	188	200
Iron	mg/l	1.06	0.04	1.0
Residual chlorine	mg/l	Nil	0.35	0.2 min
Residual chlorine	mg/l	Nil	0.30	0.2 min
Total coliform	MPN/100ml	814	Absent	Absent
Total coliform	MPN/100ml	372	Absent	Absent

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

6.x Functional Design — Working Papers — Sheet 6

Design Computation

Residual pressure = $52.230 \times 5.782 = 302.020$ m (permissible 399.347) — safe

Velocity = $41.604 \times 3.317 = 138.003$ m/s (permissible 210.793) — safe

Velocity = $20.294 \times 8.812 = 178.824$ m/s (permissible 298.473) — safe

Detention time = $75.786 \times 10.138 = 768.342$ hr (permissible 1133.720) — safe

Head loss = $50.128 \times 6.455 = 323.556$ m (permissible 517.590) — safe

Head loss = $14.813 \times 4.728 = 70.041$ m (permissible 124.496) — safe

Residual pressure = $63.734 \times 2.373 = 151.213$ m (permissible 242.736) — safe

Residual pressure = $18.535 \times 4.790 = 88.777$ m (permissible 147.351) — safe

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

6.x Functional Design — Working Papers — Sheet 7

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. WS-823, revision B.

6.x Functional Design — Working Papers — Sheet 8

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Iron	mg/l	3.81	0.23	1.0
Turbidity	NTU	11.4	0.75	1
Total coliform	MPN/100ml	840	Absent	Absent
Turbidity	NTU	185.0	0.52	1
Turbidity	NTU	31.8	0.43	1
Iron	mg/l	0.76	0.21	1.0
pH	-	7.9	8.0	6.5-8.5
Iron	mg/l	1.11	0.21	1.0
Residual chlorine	mg/l	Nil	0.23	0.2 min
Residual chlorine	mg/l	Nil	0.61	0.2 min
Total hardness	mg/l	375	68	200
Residual chlorine	mg/l	Nil	0.50	0.2 min
Total coliform	MPN/100ml	461	Absent	Absent
pH	-	7.8	7.8	6.5-8.5
pH	-	7.5	7.0	6.5-8.5
Iron	mg/l	1.00	0.02	1.0
Total hardness	mg/l	235	182	200
Turbidity	NTU	175.0	0.63	1
Total hardness	mg/l	370	102	200
pH	-	7.6	6.9	6.5-8.5
Turbidity	NTU	68.1	0.34	1
Turbidity	NTU	82.6	0.21	1
Residual chlorine	mg/l	Nil	0.75	0.2 min
pH	-	8.2	7.7	6.5-8.5

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

6.x Functional Design — Working Papers — Sheet 9

Design Computation

Detention time = $11.142 \times 8.437 = 94.001$ hr (permissible 157.531) — safe

Detention time = $89.147 \times 9.544 = 850.824$ hr (permissible 1585.279) — safe

Discharge = $67.416 \times 3.477 = 234.382$ lps (permissible 310.848) — safe

Discharge = $81.878 \times 5.650 = 462.571$ lps (permissible 824.938) — safe

Head loss = $77.726 \times 1.154 = 89.702$ m (permissible 133.322) — safe

Head loss = $45.859 \times 9.158 = 419.954$ m (permissible 581.775) — safe

Velocity = $48.558 \times 10.559 = 512.743$ m/s (permissible 795.077) — safe

Discharge = $14.899 \times 4.559 = 67.926$ lps (permissible 109.548) — safe

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

6.x Functional Design — Working Papers — Sheet 10

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. WS-405, revision A.

6.x Functional Design — Working Papers — Sheet 11

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-057	76.00	9.14	97.58	21.58
N-007	93.90	7.32	108.59	14.69
N-121	89.74	20.88	97.30	7.57
N-081	58.09	14.05	74.17	16.09
N-097	30.09	8.92	40.43	10.34
N-184	88.03	23.32	103.78	15.75
N-112	44.21	6.94	66.98	22.76
N-102	73.38	14.93	93.92	20.54
N-201	29.56	15.82	47.14	17.57
N-185	77.53	5.55	88.92	11.39
N-205	71.73	7.71	83.24	11.51
N-177	82.26	16.43	104.08	21.81
N-169	85.18	19.39	108.32	23.14
N-155	90.86	23.21	114.53	23.67
N-085	42.49	2.22	52.74	10.25
N-217	28.81	9.54	52.33	23.52
N-077	48.89	18.03	60.75	11.87
N-102	94.86	5.59	107.76	12.90
N-159	47.78	19.79	55.07	7.28

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

6.x Functional Design — Working Papers — Sheet 12

Design Computation

Velocity = $68.195 \times 3.472 = 236.741$ m/s (permissible 349.192) — safe

Discharge = $20.807 \times 0.835 = 17.374$ lps (permissible 22.577) — safe

Head loss = $46.683 \times 1.357 = 63.342$ m (permissible 83.850) — safe

Detention time = $45.702 \times 10.813 = 494.164$ hr (permissible 710.189) — safe

Detention time = $58.879 \times 10.909 = 642.327$ hr (permissible 1164.146) — safe

Velocity = $37.136 \times 9.087 = 337.442$ m/s (permissible 439.640) — revise section

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

7 ENGINEERING DESIGN

The treatment plant follows a conventional train of cascade aeration, flash mixing, clariflocculation, rapid gravity filtration and chlorination, designed to CPHEEO Manual on Water Supply and Treatment 1999 as amended. Hydraulic design of the network is by the Hardy Cross method verified in EPANET for the design year demand under fire flow conditions.

7.x Engineering Design — Working Papers — Sheet 1

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Reference: drawing no. HY-872, revision C.

7.x Engineering Design — Working Papers — Sheet 2

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-087	90.36	11.69	102.76	12.40
N-011	65.20	6.97	74.92	9.73
N-125	83.53	17.82	93.81	10.28
N-038	67.93	12.79	78.07	10.14
N-202	87.01	22.80	108.85	21.84
N-004	84.42	17.19	104.59	20.17
N-201	63.70	21.36	82.28	18.58
N-060	76.62	6.23	95.00	18.38
N-141	95.30	20.71	118.61	23.31
N-048	60.46	20.98	80.73	20.26
N-104	68.47	20.34	93.72	25.25
N-114	66.94	22.76	87.61	20.67
N-035	40.80	22.22	57.76	16.96
N-139	50.61	15.34	62.49	11.89
N-108	39.75	14.54	59.25	19.50
N-013	29.75	22.96	39.33	9.58
N-177	89.24	9.14	101.76	12.52
N-107	83.01	4.27	95.62	12.62
N-155	32.23	23.85	42.91	10.68
N-004	50.21	19.40	74.21	24.00
N-131	84.84	18.69	104.82	19.98
N-152	40.54	13.06	55.81	15.28
N-046	76.35	7.51	95.74	19.39
N-131	88.75	22.12	102.10	13.34
N-040	37.54	22.83	56.56	19.02
N-062	34.97	19.20	59.71	24.74

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

7.x Engineering Design — Working Papers — Sheet 3

Design Computation

Residual pressure = $80.561 \times 1.464 = 117.953 \text{ m}$ (permissible 172.717) — safe

Head loss = $82.593 \times 0.718 = 59.321 \text{ m}$ (permissible 107.353) — safe

Detention time = $71.737 \times 1.431 = 102.643 \text{ hr}$ (permissible 160.550) — safe

Velocity = $8.077 \times 10.132 = 81.842 \text{ m/s}$ (permissible 145.408) — safe

Detention time = $66.086 \times 1.812 = 119.771 \text{ hr}$ (permissible 158.608) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

7.x Engineering Design — Working Papers — Sheet 4

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. HY-347, revision B.

7.x Engineering Design — Working Papers — Sheet 5

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
2,773	400	MS	PN 10	683.1
344	100	HDPE	PN 10	1735.5
6,269	600	HDPE	PN 10	708.5
6,327	600	MS	PN 16	1111.1
9,412	500	MS	K-7	544.1
7,244	600	HDPE	PN 10	1689.2
20,201	500	PVC-O	PN 10	1584.2
11,007	250	HDPE	PN 6	557.7
15,946	250	PVC-O	PN 10	1102.2
14,662	100	HDPE	K-7	835.8
23,514	150	MS	K-7	1554.6
7,849	400	DI	PN 16	498.2
9,725	300	DI	K-7	844.0
14,891	400	MS	PN 16	129.2
18,804	300	MS	PN 10	1247.9
2,304	250	MS	PN 16	1229.2
5,439	400	MS	PN 16	1616.5
20,648	600	HDPE	K-7	1404.1
12,384	250	PVC-O	PN 6	1011.2
17,341	200	HDPE	PN 16	568.9
19,344	500	PVC-O	PN 6	1530.7
10,265	100	HDPE	K-9	235.8
18,746	150	DI	K-9	1619.3
19,212	200	PVC-O	PN 6	1716.8
21,256	250	PVC-O	K-7	252.1

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

7.x Engineering Design — Working Papers — Sheet 6

Design Computation

Head loss = $67.470 \times 8.924 = 602.083$ m (permissible 831.670) — safe

Head loss = $62.518 \times 3.738 = 233.722$ m (permissible 379.682) — safe

Discharge = $23.611 \times 9.908 = 233.935$ lps (permissible 359.027) — revise section

Head loss = $15.328 \times 9.741 = 149.308$ m (permissible 159.746) — safe

Velocity = $51.573 \times 2.234 = 115.202$ m/s (permissible 123.287) — safe

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

7.x Engineering Design — Working Papers — Sheet 7

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. WS-204, revision C.

7.x Engineering Design — Working Papers — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
22.4	Weathered rock	9	DS
27.9	Weathered rock	7	DS
24.4	Weathered rock	22	DS
16.5	Silty clay	42	DS
2.1	Weathered rock	50	UDS
29.7	Silty clay	58	SPT
5.7	Coarse sand	53	DS
25.1	Gravelly sand	48	SPT
27.3	Coarse sand	9	SPT
19.8	Clayey silt	58	UDS
3.1	Clayey silt	59	DS
17.5	Coarse sand	30	DS
10.2	Gravelly sand	56	SPT
28.8	Clayey silt	5	SPT
6.1	Medium sand	5	SPT
19.3	Weathered rock	5	UDS
9.3	Weathered rock	11	DS
25.4	Silty clay	36	DS
19.9	Coarse sand	46	SPT
13.7	Clayey silt	41	UDS
15.9	Gravelly sand	49	SPT

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

7.x Engineering Design — Working Papers — Sheet 9

Design Computation

Detention time = $39.900 \times 4.296 = 171.403$ hr (permissible 277.959) — safe

Discharge = $80.586 \times 2.787 = 224.604$ lps (permissible 256.391) — safe

Discharge = $19.460 \times 6.247 = 121.559$ lps (permissible 224.092) — safe

Residual pressure = $2.097 \times 11.974 = 25.113$ m (permissible 38.231) — safe

Detention time = $87.922 \times 5.937 = 522.007$ hr (permissible 621.306) — safe

Residual pressure = $59.196 \times 6.083 = 360.086$ m (permissible 662.652) — revise section

Velocity = $10.903 \times 1.398 = 15.241$ m/s (permissible 17.102) — safe

Detention time = $18.413 \times 8.883 = 163.573$ hr (permissible 287.685) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

7.x Engineering Design — Working Papers — Sheet 10

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Reference: drawing no. PL-295, revision B.

7.x Engineering Design — Working Papers — Sheet 11

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Total coliform	MPN/100ml	128	Absent	Absent
Residual chlorine	mg/l	Nil	0.20	0.2 min
Turbidity	NTU	81.8	0.84	1
Iron	mg/l	0.91	0.09	1.0
Total hardness	mg/l	238	120	200
pH	-	6.5	7.8	6.5-8.5
Residual chlorine	mg/l	Nil	0.39	0.2 min
Turbidity	NTU	48.1	0.60	1
Residual chlorine	mg/l	Nil	0.65	0.2 min
Turbidity	NTU	89.9	0.56	1
pH	-	6.7	7.3	6.5-8.5
Total coliform	MPN/100ml	625	Absent	Absent
Turbidity	NTU	16.9	0.21	1
Turbidity	NTU	101.0	0.50	1
Total hardness	mg/l	383	63	200
Residual chlorine	mg/l	Nil	0.44	0.2 min
Turbidity	NTU	227.2	0.31	1

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

7.x Engineering Design — Working Papers — Sheet 12

Design Computation

Discharge = $30.935 \times 5.904 = 182.649$ lps (permissible 244.054) — safe

Residual pressure = $52.101 \times 9.462 = 492.993$ m (permissible 757.352) — safe

Head loss = $43.532 \times 3.462 = 150.700$ m (permissible 193.804) — safe

Discharge = $14.274 \times 8.440 = 120.478$ lps (permissible 157.201) — safe

Detention time = $66.861 \times 5.259 = 351.588$ hr (permissible 396.221) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

7.x Engineering Design — Working Papers — Sheet 13

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Reference: drawing no. ST-586, revision B.

7.x Engineering Design — Working Papers — Sheet 14

Population Projection

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2036	Ward 3	3,546	150	0.532
2056	Ward 22	28,094	135	3.793
2056	Ward 8	27,433	100	2.743
2026	Ward 4	43,159	150	6.474
2056	Ward 19	31,981	150	4.797
2051	Ward 24	21,677	70	1.517
2036	Ward 7	8,487	100	0.849
2051	Ward 10	43,205	135	5.833
2051	Ward 2	36,130	70	2.529
2041	Ward 5	37,836	135	5.108
2036	Ward 8	8,347	70	0.584
2046	Ward 17	18,275	135	2.467
2031	Ward 1	24,342	100	2.434
2036	Ward 5	31,327	70	2.193
2051	Ward 24	10,347	70	0.724
2026	Ward 7	1,888	70	0.132
2041	Ward 7	8,869	70	0.621
2051	Ward 18	12,548	150	1.882
2036	Ward 11	15,982	70	1.119
2056	Ward 21	21,642	100	2.164
2051	Ward 24	5,482	135	0.740
2051	Ward 3	6,830	70	0.478
2036	Ward 14	39,667	150	5.950

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

7.x Engineering Design — Working Papers — Sheet 15

Design Computation

Residual pressure = $30.470 \times 5.419 = 165.126 \text{ m}$ (permissible 259.047) — safe

Velocity = $22.113 \times 9.803 = 216.774 \text{ m/s}$ (permissible 337.504) — safe

Residual pressure = $77.606 \times 7.149 = 554.813 \text{ m}$ (permissible 875.980) — safe

Detention time = $67.177 \times 10.298 = 691.775 \text{ hr}$ (permissible 797.139) — safe

Head loss = $81.925 \times 2.833 = 232.108 \text{ m}$ (permissible 282.176) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

7.x Engineering Design — Working Papers — Sheet 16

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. WS-485, revision B.

7.x Engineering Design — Working Papers — Sheet 17

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
11,324	600	HDPE	K-7	492.6
10,405	100	PVC-O	K-7	1033.6
21,982	400	PVC-O	K-9	831.9
23,316	200	MS	PN 16	1631.9
10,042	600	DI	PN 6	227.7
2,028	600	HDPE	K-9	58.4
8,030	200	PVC-O	PN 6	1189.4
2,435	100	DI	PN 6	97.1
603	250	DI	K-9	511.8
10,534	100	MS	K-7	480.5
7,520	150	PVC-O	PN 6	1162.4
13,034	100	DI	K-9	1243.5
7,931	600	DI	PN 6	1017.9
22,209	500	PVC-O	PN 10	1702.3
21,053	300	MS	PN 16	1615.0
19,390	150	DI	K-7	1664.9
8,345	600	PVC-O	PN 10	916.2
10,092	150	PVC-O	PN 10	770.8
3,807	150	DI	PN 6	1733.1
2,394	500	DI	K-7	1640.9
11,799	200	HDPE	PN 6	245.6
21,675	250	MS	PN 16	602.4
16,671	250	PVC-O	K-9	276.3
9,592	500	MS	K-9	385.2

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

7.x Engineering Design — Working Papers — Sheet 18

Design Computation

Discharge = $71.959 \times 6.964 = 501.160$ lps (permissible 806.137) — safe

Detention time = $28.063 \times 8.254 = 231.636$ hr (permissible 394.884) — safe

Detention time = $32.650 \times 10.335 = 337.441$ hr (permissible 379.410) — safe

Head loss = $27.114 \times 8.021 = 217.486$ m (permissible 306.977) — safe

Residual pressure = $55.114 \times 11.266 = 620.904$ m (permissible 1135.994) — safe

Detention time = $79.738 \times 6.263 = 499.437$ hr (permissible 590.703) — revise section

Residual pressure = $87.938 \times 9.898 = 870.446$ m (permissible 1634.987) — safe

Detention time = $63.515 \times 2.565 = 162.888$ hr (permissible 198.329) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

7.x Engineering Design — Working Papers — Sheet 19

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. ST-255, revision A.

7.x Engineering Design — Working Papers — Sheet 20

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	226.5	0.31	1
Iron	mg/l	2.09	0.05	1.0
Iron	mg/l	3.47	0.14	1.0
Residual chlorine	mg/l	Nil	0.68	0.2 min
Total hardness	mg/l	96	187	200
pH	-	7.1	7.6	6.5-8.5
pH	-	8.0	7.9	6.5-8.5
Iron	mg/l	3.17	0.16	1.0
Total coliform	MPN/100ml	828	Absent	Absent
Total coliform	MPN/100ml	754	Absent	Absent
Residual chlorine	mg/l	Nil	0.21	0.2 min
Iron	mg/l	1.04	0.16	1.0
Iron	mg/l	2.70	0.04	1.0
Total hardness	mg/l	393	143	200
Residual chlorine	mg/l	Nil	0.69	0.2 min
Total coliform	MPN/100ml	176	Absent	Absent
Total hardness	mg/l	126	158	200
Turbidity	NTU	60.9	0.21	1
Residual chlorine	mg/l	Nil	0.27	0.2 min

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

7.x Engineering Design — Working Papers — Sheet 21

Design Computation

Residual pressure = $50.440 \times 1.574 = 79.372$ m (permissible 128.067) — safe

Detention time = $77.736 \times 2.914 = 226.559$ hr (permissible 422.684) — safe

Head loss = $5.070 \times 8.112 = 41.132$ m (permissible 46.182) — safe

Velocity = $51.342 \times 6.618 = 339.756$ m/s (permissible 512.696) — safe

Head loss = $18.878 \times 3.816 = 72.043$ m (permissible 81.018) — safe

Detention time = $33.166 \times 3.805 = 126.189$ hr (permissible 227.988) — revise section

Discharge = $66.457 \times 1.722 = 114.455$ lps (permissible 171.729) — safe

Discharge = $13.567 \times 9.218 = 125.062$ lps (permissible 231.068) — safe

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

7.x Engineering Design — Working Papers — Sheet 22

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. HY-301, revision A.

7.x Engineering Design — Working Papers — Sheet 23

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	224.7	0.62	1
Residual chlorine	mg/l	Nil	0.29	0.2 min
pH	-	8.2	8.1	6.5-8.5
Residual chlorine	mg/l	Nil	0.21	0.2 min
Total hardness	mg/l	299	155	200
Total hardness	mg/l	442	74	200
pH	-	6.9	7.7	6.5-8.5
pH	-	8.2	7.0	6.5-8.5
Residual chlorine	mg/l	Nil	0.40	0.2 min
Turbidity	NTU	223.7	0.37	1
Turbidity	NTU	105.3	0.49	1
Residual chlorine	mg/l	Nil	0.53	0.2 min
pH	-	6.7	7.0	6.5-8.5
Total hardness	mg/l	197	148	200
pH	-	8.2	7.7	6.5-8.5
Turbidity	NTU	74.2	0.22	1
Total coliform	MPN/100ml	420	Absent	Absent

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

7.x Engineering Design — Working Papers — Sheet 24

Design Computation

Discharge = $67.320 \times 9.412 = 633.595$ lps (permissible 703.387) — safe

Discharge = $70.787 \times 4.023 = 284.754$ lps (permissible 489.035) — safe

Residual pressure = $76.663 \times 6.378 = 488.948$ m (permissible 784.514) — safe

Residual pressure = $78.677 \times 2.337 = 183.896$ m (permissible 319.805) — safe

Head loss = $5.841 \times 1.509 = 8.814$ m (permissible 13.868) — safe

Head loss = $21.763 \times 10.599 = 230.660$ m (permissible 251.558) — safe

Discharge = $35.779 \times 10.194 = 364.748$ lps (permissible 562.236) — safe

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

7.x Engineering Design — Working Papers — Sheet 25

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Reference: drawing no. ST-621, revision A.

7.x Engineering Design — Working Papers — Sheet 26

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Residual chlorine	mg/l	Nil	0.79	0.2 min
Total coliform	MPN/100ml	871	Absent	Absent
pH	-	6.8	7.0	6.5-8.5
Turbidity	NTU	196.0	0.49	1
Total coliform	MPN/100ml	584	Absent	Absent
Iron	mg/l	4.09	0.07	1.0
Residual chlorine	mg/l	Nil	0.72	0.2 min
Total coliform	MPN/100ml	839	Absent	Absent
Total coliform	MPN/100ml	845	Absent	Absent
Residual chlorine	mg/l	Nil	0.24	0.2 min
Residual chlorine	mg/l	Nil	0.26	0.2 min
Residual chlorine	mg/l	Nil	0.36	0.2 min
Total hardness	mg/l	168	108	200
Residual chlorine	mg/l	Nil	0.49	0.2 min
Turbidity	NTU	82.4	0.84	1
Residual chlorine	mg/l	Nil	0.42	0.2 min
Total hardness	mg/l	114	160	200
Iron	mg/l	3.32	0.19	1.0
Residual chlorine	mg/l	Nil	0.60	0.2 min
Residual chlorine	mg/l	Nil	0.60	0.2 min
Residual chlorine	mg/l	Nil	0.46	0.2 min
pH	-	6.5	7.5	6.5-8.5
pH	-	6.6	7.7	6.5-8.5
Residual chlorine	mg/l	Nil	0.35	0.2 min
Turbidity	NTU	104.2	0.22	1

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

7.x Engineering Design — Working Papers — Sheet 27

Design Computation

Detention time = $41.058 \times 6.303 = 258.779$ hr (permissible 463.302) — safe

Velocity = $71.609 \times 8.081 = 578.647$ m/s (permissible 831.160) — safe

Residual pressure = $49.799 \times 1.862 = 92.727$ m (permissible 129.088) — safe

Residual pressure = $51.161 \times 4.414 = 225.838$ m (permissible 410.206) — safe

Velocity = $17.468 \times 1.907 = 33.306$ m/s (permissible 41.773) — safe

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

7.x Engineering Design — Working Papers — Sheet 28

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. PL-547, revision C.

7.x Engineering Design — Working Papers — Sheet 29

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-03	Sludge	96.7	30.4	40.04
P-03	Booster	96.2	25.2	33.05
P-08	Booster	261.5	79.9	284.50
P-04	Backwash	242.7	42.4	140.08
P-11	Raw water	220.5	17.2	51.60
P-11	Booster	144.7	19.8	39.07
P-03	Sludge	76.2	62.3	64.65
P-11	Booster	20.7	35.3	9.96
P-08	Raw water	63.4	19.6	16.92
P-07	Sludge	259.1	60.9	214.79
P-04	Backwash	160.6	69.1	151.12
P-10	Raw water	172.6	23.2	54.50
P-06	Backwash	206.1	25.2	70.86
P-02	Backwash	261.7	84.3	300.66
P-03	Raw water	75.3	21.8	22.36
P-09	Sludge	172.6	85.5	201.05
P-05	Booster	185.8	85.1	215.35
P-01	Sludge	211.1	74.1	213.03
P-02	Raw water	34.5	72.3	34.00
P-02	Clear water	212.9	66.3	192.45
P-03	Booster	235.3	83.8	268.74

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

7.x Engineering Design — Working Papers — Sheet 30

Design Computation

Detention time = $71.730 \times 11.365 = 815.207$ hr (permissible 1428.268) — safe

Head loss = $28.419 \times 3.933 = 111.775$ m (permissible 157.364) — safe

Velocity = $81.939 \times 9.038 = 740.607$ m/s (permissible 1008.439) — safe

Residual pressure = $22.725 \times 7.588 = 172.439$ m (permissible 272.654) — safe

Detention time = $84.305 \times 8.060 = 679.498$ hr (permissible 953.978) — safe

Head loss = $78.872 \times 1.676 = 132.211$ m (permissible 180.580) — safe

Residual pressure = $16.077 \times 4.627 = 74.395$ m (permissible 140.479) — safe

Discharge = $16.115 \times 10.110 = 162.930$ lps (permissible 225.524) — revise section

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

7.x Engineering Design — Working Papers — Sheet 31

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Reference: drawing no. WS-780, revision C.

7.x Engineering Design — Working Papers — Sheet 32

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	225.8	0.77	1
Turbidity	NTU	27.5	0.85	1
Residual chlorine	mg/l	Nil	0.53	0.2 min
Iron	mg/l	3.89	0.14	1.0
Total hardness	mg/l	220	149	200
Iron	mg/l	1.62	0.05	1.0
pH	-	7.4	7.0	6.5-8.5
pH	-	8.2	7.6	6.5-8.5
pH	-	8.3	8.0	6.5-8.5
Total coliform	MPN/100ml	131	Absent	Absent
Turbidity	NTU	25.0	0.65	1
Iron	mg/l	0.94	0.14	1.0
Turbidity	NTU	114.2	0.28	1
Iron	mg/l	3.16	0.05	1.0
Iron	mg/l	1.10	0.25	1.0
Total hardness	mg/l	216	129	200
Residual chlorine	mg/l	Nil	0.69	0.2 min
Total hardness	mg/l	221	63	200
Total hardness	mg/l	354	92	200

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

7.x Engineering Design — Working Papers — Sheet 33

Design Computation

Residual pressure = $31.665 \times 9.466 = 299.724$ m (permissible 514.378) — safe

Discharge = $70.103 \times 9.002 = 631.077$ lps (permissible 1196.351) — safe

Discharge = $20.960 \times 11.123 = 233.150$ lps (permissible 287.215) — safe

Residual pressure = $28.241 \times 7.549 = 213.178$ m (permissible 333.141) — safe

Head loss = $20.247 \times 6.128 = 124.071$ m (permissible 222.638) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

7.x Engineering Design — Working Papers — Sheet 34

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. ST-231, revision C.

7.x Engineering Design — Working Papers — Sheet 35

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-193	90.97	23.20	102.14	11.17
N-177	50.04	15.84	66.34	16.29
N-009	48.43	10.72	70.20	21.77
N-092	32.22	7.64	43.15	10.93
N-031	84.63	2.61	92.98	8.35
N-158	89.02	13.17	103.98	14.96
N-092	66.62	15.69	82.56	15.94
N-046	42.30	12.00	50.99	8.69
N-093	47.79	20.77	56.56	8.77
N-161	79.37	1.36	87.26	7.89
N-142	61.86	11.83	86.15	24.29
N-168	40.19	20.33	57.57	17.38
N-212	84.30	19.83	107.15	22.85
N-204	80.74	16.23	92.57	11.83
N-125	91.85	10.90	114.04	22.18
N-061	63.95	10.80	75.86	11.90
N-132	90.56	1.45	109.39	18.84
N-019	86.85	12.94	101.69	14.84
N-191	84.40	20.97	105.83	21.43
N-106	31.30	20.34	49.17	17.88
N-108	64.99	4.69	78.06	13.07

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

7.x Engineering Design — Working Papers — Sheet 36

Design Computation

Head loss = $40.862 \times 4.099 = 167.509$ m (permissible 283.698) — safe

Residual pressure = $44.439 \times 6.466 = 287.362$ m (permissible 409.168) — safe

Velocity = $13.093 \times 7.294 = 95.506$ m/s (permissible 152.174) — safe

Residual pressure = $49.797 \times 10.301 = 512.977$ m (permissible 960.365) — safe

Residual pressure = $31.407 \times 11.586 = 363.869$ m (permissible 486.198) — safe

Discharge = $23.619 \times 3.968 = 93.726$ lps (permissible 135.055) — safe

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

7.x Engineering Design — Working Papers — Sheet 37

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. WS-612, revision C.

7.x Engineering Design — Working Papers — Sheet 38

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-04	Backwash	48.9	27.0	17.96
P-11	Booster	36.0	21.1	10.35
P-06	Backwash	129.2	43.6	76.78
P-04	Backwash	82.8	18.7	21.16
P-09	Raw water	227.4	36.0	111.66
P-09	Sludge	146.8	43.9	87.85
P-05	Clear water	290.5	58.3	230.60
P-07	Clear water	261.6	42.3	150.77
P-03	Booster	48.9	38.9	25.95
P-05	Booster	39.4	27.3	14.68
P-01	Booster	279.9	24.3	92.55
P-10	Backwash	300.9	87.4	358.42
P-08	Booster	313.2	48.7	207.72
P-07	Booster	318.7	70.0	303.89
P-04	Backwash	36.6	28.0	13.95
P-09	Clear water	149.5	84.0	171.09
P-04	Booster	184.2	80.8	202.67
P-03	Clear water	200.1	20.8	56.58
P-08	Raw water	202.8	42.6	117.67
P-07	Raw water	65.7	37.6	33.67

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

7.x Engineering Design — Working Papers — Sheet 39

Design Computation

Discharge = $46.786 \times 8.608 = 402.742$ lps (permissible 553.482) — safe

Velocity = $11.539 \times 3.223 = 37.193$ m/s (permissible 50.591) — safe

Residual pressure = $66.934 \times 6.941 = 464.609$ m (permissible 741.400) — safe

Residual pressure = $64.245 \times 1.194 = 76.727$ m (permissible 117.199) — safe

Head loss = $54.182 \times 4.057 = 219.831$ m (permissible 292.827) — revise section

Head loss = $19.095 \times 1.786 = 34.110$ m (permissible 63.752) — safe

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

7.x Engineering Design — Working Papers — Sheet 40

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Reference: drawing no. PL-888, revision B.

7.x Engineering Design — Working Papers — Sheet 41

Bore Log Summary

Depth (m)	Strata	N value	Sample
7.2	Weathered rock	7	SPT
7.1	Clayey silt	33	DS
17.9	Medium sand	22	UDS
7.2	Medium sand	42	SPT
29.3	Silty clay	50	DS
25.6	Clayey silt	24	DS
5.2	Weathered rock	47	SPT
23.5	Silty clay	30	DS
22.4	Coarse sand	22	DS
24.9	Medium sand	60	SPT
29.4	Coarse sand	26	UDS
20.1	Gravelly sand	58	DS
20.8	Clayey silt	55	UDS
16.6	Medium sand	23	SPT
3.6	Clayey silt	31	SPT
25.2	Clayey silt	32	UDS
26.6	Gravelly sand	55	UDS
8.2	Clayey silt	53	SPT
11.3	Coarse sand	17	SPT
10.3	Weathered rock	60	SPT
13.8	Silty clay	38	DS
18.5	Coarse sand	51	UDS
8.5	Medium sand	32	SPT

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

8 FINANCIAL ESTIMATES AND COST PROJECTIONS

Rates are derived from the Schedule of Rates in force with lead and lift adjustments. Items outside the Schedule have been rate-analysed and are placed at the detailed estimate.

Sl.	Description of item	Amount (Rs. crore)
1	Intake well, jack well and raw water pumping	38.60
2	Raw water rising main, 1100 mm, 8.4 km	44.20
3	Water treatment plant, 46 MLD	71.40
4	Clear water transmission main	32.80
5	Service reservoirs, 4 nos., 12.4 ML	26.90
6	Distribution network, 214 km	38.10
7	Contingencies at 3 per cent	7.90
8	Quality control and supervision	4.90
	TOTAL PROJECT COST	264.80

The total project cost of the scheme is Rs. 264.80 crore.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 1

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. ST-595, revision A.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 2

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-147	87.99	1.94	103.04	15.05
N-133	32.34	7.86	57.45	25.11
N-133	31.88	2.70	49.43	17.55
N-139	55.46	8.86	65.69	10.23
N-119	61.41	15.01	69.92	8.50
N-182	66.37	19.58	83.70	17.33
N-205	87.34	15.22	95.10	7.76
N-180	54.95	5.75	76.39	21.44
N-027	78.48	0.69	102.99	24.50
N-092	55.96	11.66	68.41	12.44
N-209	40.20	7.36	62.28	22.09
N-099	30.64	2.73	51.74	21.09
N-082	88.20	23.88	96.23	8.03
N-010	45.00	3.61	61.35	16.34
N-158	45.57	14.49	61.12	15.55
N-141	86.18	20.45	102.20	16.02
N-114	61.52	16.06	82.24	20.72
N-158	62.87	1.14	76.27	13.40
N-162	87.72	7.37	96.71	8.99
N-198	64.21	10.70	73.37	9.16
N-200	89.79	5.38	107.93	18.14
N-023	84.93	2.19	99.02	14.09

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 3

Design Computation

Detention time = $13.164 \times 0.906 = 11.931$ hr (permissible 21.264) — safe

Discharge = $51.709 \times 3.451 = 178.470$ lps (permissible 285.007) — safe

Velocity = $48.619 \times 0.486 = 23.629$ m/s (permissible 34.872) — safe

Residual pressure = $45.075 \times 11.440 = 515.632$ m (permissible 563.698) — safe

Residual pressure = $9.422 \times 1.840 = 17.333$ m (permissible 30.706) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 4

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. WS-509, revision B.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 5

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
5,975	150	HDPE	K-9	365.3
16,663	100	HDPE	K-9	688.2
9,476	150	MS	PN 10	564.8
15,637	300	HDPE	K-7	167.9
14,956	500	DI	K-7	1575.6
12,058	150	MS	PN 6	1299.1
11,306	100	PVC-O	PN 10	365.4
7,004	300	DI	K-7	1131.3
598	600	PVC-O	PN 10	164.9
2,573	200	HDPE	PN 6	1572.9
9,712	300	HDPE	K-7	922.2
1,757	600	MS	K-9	1719.0
6,626	600	DI	PN 6	373.1
6,919	100	HDPE	K-7	1658.1
18,813	500	MS	PN 10	1049.2
9,406	500	DI	K-7	1772.6
21,100	200	PVC-O	K-7	94.0
3,074	600	DI	K-7	1174.0
5,686	300	HDPE	K-9	636.2
14,612	150	DI	K-7	846.8
11,757	400	MS	K-9	1761.8
12,311	250	MS	K-9	520.8

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 6

Design Computation

Residual pressure = $57.275 \times 10.644 = 609.630$ m (permissible 675.296) — safe

Detention time = $9.858 \times 5.919 = 58.351$ hr (permissible 94.323) — safe

Detention time = $49.282 \times 7.202 = 354.952$ hr (permissible 459.896) — safe

Discharge = $64.285 \times 3.345 = 215.059$ lps (permissible 363.141) — safe

Head loss = $75.110 \times 7.934 = 595.927$ m (permissible 971.437) — safe

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 7

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Reference: drawing no. WS-372, revision A.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 8

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-107	75.86	2.27	84.02	8.17
N-036	77.37	12.55	85.56	8.20
N-172	36.56	18.78	57.80	21.24
N-026	53.21	0.46	72.30	19.10
N-019	65.53	6.21	78.28	12.74
N-106	28.18	12.02	38.74	10.56
N-111	83.30	7.18	97.31	14.01
N-207	31.32	15.07	55.48	24.16
N-209	41.76	9.94	58.47	16.71
N-064	34.68	10.20	44.40	9.72
N-114	29.29	8.55	38.31	9.02
N-070	46.56	15.30	66.35	19.79
N-009	72.39	6.17	96.22	23.83
N-149	80.89	6.44	106.39	25.50
N-116	60.38	22.64	82.88	22.50
N-144	63.77	21.67	72.04	8.27
N-037	73.25	11.39	90.37	17.12
N-140	34.83	9.55	48.94	14.11

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 9

Design Computation

Head loss = $38.156 \times 10.587 = 403.947$ m (permissible 601.439) — safe

Discharge = $18.932 \times 6.737 = 127.549$ lps (permissible 178.342) — safe

Detention time = $77.902 \times 10.139 = 789.827$ hr (permissible 1246.964) — safe

Velocity = $57.547 \times 7.713 = 443.837$ m/s (permissible 835.057) — safe

Discharge = $17.152 \times 10.978 = 188.296$ lps (permissible 294.745) — revise section

Residual pressure = $13.017 \times 9.712 = 126.423$ m (permissible 184.257) — safe

Velocity = $84.244 \times 5.158 = 434.540$ m/s (permissible 728.713) — revise section

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 10

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. HY-549, revision C.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 11

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
21,284	100	MS	K-9	516.6
22,641	250	PVC-O	PN 16	519.1
8,150	150	PVC-O	PN 10	1572.1
7,467	600	MS	PN 10	884.7
5,228	300	HDPE	K-9	541.6
11,373	100	DI	PN 6	1165.1
3,258	500	PVC-O	PN 10	153.4
23,679	300	MS	PN 6	170.4
3,184	500	MS	K-9	784.8
4,555	150	MS	PN 10	1764.7
16,261	100	HDPE	PN 6	1524.1
784	300	HDPE	K-9	758.3
17,191	500	PVC-O	K-9	1650.4
1,152	600	PVC-O	K-9	341.0
10,208	200	MS	K-7	900.0
18,117	100	DI	PN 16	189.3
18,771	500	MS	PN 16	656.2
17,167	300	DI	K-9	728.9
1,761	600	DI	K-7	200.3
21,109	400	MS	PN 6	1767.2
7,235	300	PVC-O	PN 6	1127.3
22,125	200	HDPE	PN 16	135.2
3,282	300	MS	PN 16	1639.9

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 12

Design Computation

Detention time = $81.236 \times 9.829 = 798.451$ hr (permissible 1431.906) — safe

Velocity = $2.208 \times 5.437 = 12.005$ m/s (permissible 16.036) — safe

Discharge = $80.280 \times 8.168 = 655.759$ lps (permissible 1018.614) — safe

Velocity = $42.804 \times 9.391 = 401.991$ m/s (permissible 732.139) — safe

Residual pressure = $38.167 \times 8.177 = 312.105$ m (permissible 502.379) — safe

Detention time = $20.194 \times 7.471 = 150.864$ hr (permissible 198.003) — safe

Residual pressure = $84.055 \times 2.941 = 247.188$ m (permissible 372.565) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 13

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Reference: drawing no. HY-110, revision B.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 14

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Total coliform	MPN/100ml	47	Absent	Absent
Residual chlorine	mg/l	Nil	0.50	0.2 min
Total coliform	MPN/100ml	661	Absent	Absent
Total coliform	MPN/100ml	616	Absent	Absent
Total hardness	mg/l	289	60	200
Total hardness	mg/l	370	63	200
pH	-	7.8	7.4	6.5-8.5
Residual chlorine	mg/l	Nil	0.39	0.2 min
Iron	mg/l	2.06	0.20	1.0
Iron	mg/l	0.93	0.22	1.0
Total coliform	MPN/100ml	256	Absent	Absent
Iron	mg/l	1.25	0.08	1.0
Residual chlorine	mg/l	Nil	0.71	0.2 min
Residual chlorine	mg/l	Nil	0.37	0.2 min
Turbidity	NTU	45.5	0.37	1
Turbidity	NTU	34.3	0.66	1
Residual chlorine	mg/l	Nil	0.53	0.2 min
Iron	mg/l	2.15	0.11	1.0
Iron	mg/l	1.70	0.18	1.0

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 15

Design Computation

Residual pressure = $89.997 \times 11.053 = 994.704$ m (permissible 1649.078) — safe

Head loss = $60.219 \times 6.697 = 403.305$ m (permissible 567.044) — safe

Detention time = $9.746 \times 8.097 = 78.915$ hr (permissible 122.494) — safe

Residual pressure = $41.898 \times 10.109 = 423.543$ m (permissible 458.496) — safe

Velocity = $57.702 \times 0.668 = 38.527$ m/s (permissible 42.866) — safe

Discharge = $58.853 \times 5.732 = 337.367$ lps (permissible 383.100) — revise section

Discharge = $86.438 \times 4.110 = 355.222$ lps (permissible 488.117) — safe

Head loss = $30.655 \times 6.048 = 185.391$ m (permissible 264.152) — safe

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

8.x Financial Estimates And Cost Projections — Working Papers — Sheet 16

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. ST-736, revision B.

9 REVENUE STREAMS

Water tariff is levied by the Municipal Board at Rs. 6.20 per kilolitre for domestic connections with a telescopic slab above 20 kl per month, and Rs. 22.00 per kilolitre for commercial connections. At full coverage and 85 per cent collection efficiency the annual revenue is Rs. 11.60 crore against an O&M; requirement of Rs. 9.80 crore.

9.x Revenue Streams — Working Papers — Sheet 1

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. ST-963, revision A.

9.x Revenue Streams — Working Papers — Sheet 2

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Iron	mg/l	0.89	0.12	1.0
Iron	mg/l	3.88	0.06	1.0
Total hardness	mg/l	333	63	200
Turbidity	NTU	79.5	0.59	1
Total coliform	MPN/100ml	837	Absent	Absent
Residual chlorine	mg/l	Nil	0.70	0.2 min
Iron	mg/l	1.23	0.20	1.0
Total hardness	mg/l	291	112	200
Total coliform	MPN/100ml	691	Absent	Absent
Residual chlorine	mg/l	Nil	0.32	0.2 min
Total coliform	MPN/100ml	402	Absent	Absent
Residual chlorine	mg/l	Nil	0.28	0.2 min
Turbidity	NTU	77.1	0.63	1
Total coliform	MPN/100ml	311	Absent	Absent
Turbidity	NTU	152.3	0.87	1
Total hardness	mg/l	168	141	200
pH	-	6.5	7.8	6.5-8.5
pH	-	6.8	8.1	6.5-8.5
pH	-	7.6	7.7	6.5-8.5
Iron	mg/l	1.28	0.18	1.0
Total coliform	MPN/100ml	307	Absent	Absent
Iron	mg/l	1.86	0.04	1.0

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

9.x Revenue Streams — Working Papers — Sheet 3

Design Computation

Residual pressure = $36.630 \times 2.161 = 79.161$ m (permissible 83.270) — safe

Residual pressure = $22.085 \times 1.316 = 29.056$ m (permissible 36.157) — safe

Head loss = $6.821 \times 0.704 = 4.799$ m (permissible 5.672) — safe

Discharge = $46.395 \times 2.330 = 108.107$ lps (permissible 181.416) — safe

Velocity = $40.914 \times 8.809 = 360.397$ m/s (permissible 575.147) — safe

Detention time = $57.990 \times 4.259 = 247.004$ hr (permissible 351.011) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

9.x Revenue Streams — Working Papers — Sheet 4

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Reference: drawing no. HY-882, revision B.

9.x Revenue Streams — Working Papers — Sheet 5

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-174	34.31	19.60	59.12	24.81
N-173	63.25	5.16	72.41	9.17
N-137	58.02	3.56	82.37	24.35
N-212	76.66	2.91	100.80	24.14
N-041	63.77	5.66	78.18	14.42
N-069	34.79	19.18	58.67	23.88
N-085	47.09	15.01	56.66	9.57
N-167	66.47	23.36	85.48	19.01
N-024	54.28	22.26	79.28	25.00
N-160	40.74	22.23	59.32	18.58
N-111	84.09	5.25	102.76	18.67
N-074	33.53	14.91	52.93	19.40
N-212	94.14	12.89	117.55	23.41
N-007	71.39	17.33	82.71	11.32
N-129	81.38	7.69	104.18	22.80
N-218	30.67	4.42	40.19	9.52
N-116	56.02	20.74	71.11	15.09
N-109	87.73	10.16	109.83	22.11
N-015	93.96	10.43	118.14	24.18

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

9.x Revenue Streams — Working Papers — Sheet 6

Design Computation

Discharge = $6.212 \times 3.504 = 21.767$ lps (permissible 40.930) — safe

Detention time = $6.029 \times 11.053 = 66.640$ hr (permissible 77.267) — safe

Velocity = $26.821 \times 9.170 = 245.942$ m/s (permissible 357.831) — safe

Head loss = $68.121 \times 8.661 = 589.994$ m (permissible 1102.115) — safe

Head loss = $1.940 \times 1.304 = 2.529$ m (permissible 4.693) — safe

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

9.x Revenue Streams — Working Papers — Sheet 7

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. WS-347, revision C.

9.x Revenue Streams — Working Papers — Sheet 8

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-02	Backwash	196.0	69.4	185.41
P-10	Sludge	178.9	36.5	88.89
P-12	Raw water	24.7	60.8	20.49
P-06	Backwash	38.4	67.8	35.49
P-07	Backwash	70.7	79.4	76.56
P-06	Clear water	267.7	81.2	296.10
P-01	Sludge	164.3	34.9	78.01
P-07	Clear water	209.0	27.4	77.98
P-02	Raw water	131.6	25.8	46.34
P-03	Clear water	155.2	37.3	78.79
P-02	Raw water	309.4	71.8	302.66
P-09	Raw water	207.9	57.1	161.70
P-01	Sludge	81.3	83.5	92.45
P-05	Raw water	33.2	17.6	7.94
P-11	Booster	127.8	82.6	143.81
P-10	Clear water	197.6	56.4	151.77
P-04	Backwash	128.1	29.6	51.59
P-04	Raw water	138.6	66.2	125.04
P-07	Raw water	134.0	55.2	100.79
P-12	Sludge	90.2	40.9	50.33
P-05	Booster	126.2	20.4	35.07
P-04	Clear water	170.3	87.7	203.38
P-04	Sludge	101.2	15.6	21.50
P-03	Raw water	39.9	66.0	35.87
P-12	Raw water	268.6	59.4	217.55
P-09	Raw water	114.8	16.8	26.33

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

10 COST BENEFIT ANALYSIS AND INVESTMENT CRITERIA

The appraisal has been carried out over a 30 year evaluation period at a discount rate of 12 per cent.

Investment criterion	Value
Internal Rate of Return	14.6 per cent
Net Present Value at 12 per cent	Rs. 74.20 crore
Benefit Cost Ratio	1.44
Payback period	12.1 years

The Internal Rate of Return of the project works out to 14.6 per cent, against the threshold of 12 per cent applied to projects of this class. The project is therefore viable and is recommended for sanction.

The year-wise cash flow statement on which this computation rests is enclosed at the annexure.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 1

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. HY-886, revision A.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 2

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-131	88.23	23.24	106.78	18.55
N-038	51.26	19.55	73.46	22.20
N-002	74.60	21.94	91.37	16.77
N-097	66.38	6.94	88.73	22.36
N-016	81.76	13.44	100.60	18.84
N-085	83.47	18.40	97.30	13.83
N-049	58.94	6.27	67.58	8.64
N-198	84.87	11.47	104.18	19.31
N-037	59.40	21.97	67.65	8.26
N-056	52.55	4.14	74.67	22.12
N-035	85.51	2.13	97.73	12.22
N-198	87.02	22.11	103.81	16.79
N-046	42.35	14.46	66.35	24.00
N-162	91.94	18.14	116.14	24.20
N-148	87.21	7.69	97.24	10.03
N-198	59.26	10.05	83.65	24.39
N-137	78.73	20.50	87.60	8.88
N-200	37.39	1.61	47.25	9.86
N-042	35.57	1.04	59.05	23.49
N-117	78.26	15.08	89.08	10.82
N-086	48.31	18.84	62.01	13.70

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 3

Design Computation

Head loss = $63.200 \times 1.005 = 63.490$ m (permissible 98.447) — safe

Discharge = $4.127 \times 6.030 = 24.888$ lps (permissible 26.324) — safe

Velocity = $58.019 \times 0.780 = 45.258$ m/s (permissible 69.771) — safe

Discharge = $75.140 \times 4.028 = 302.697$ lps (permissible 444.648) — safe

Residual pressure = $50.861 \times 11.709 = 595.509$ m (permissible 935.039) — safe

Detention time = $35.382 \times 1.208 = 42.733$ hr (permissible 58.109) — safe

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 4

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Reference: drawing no. WS-988, revision B.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 5

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-130	85.81	17.89	97.51	11.70
N-164	47.73	0.52	63.52	15.79
N-117	71.27	3.22	96.04	24.77
N-207	73.19	14.72	82.98	9.79
N-133	69.87	16.07	86.97	17.11
N-202	58.76	23.42	74.29	15.54
N-161	47.22	1.50	54.22	7.00
N-168	77.12	21.26	102.22	25.10
N-190	90.02	19.43	100.32	10.30
N-043	37.70	8.60	57.26	19.56
N-051	69.44	20.69	81.41	11.96
N-212	80.38	0.45	97.37	16.99
N-059	71.03	18.88	79.37	8.34
N-186	36.66	18.89	51.19	14.52
N-218	47.48	11.97	70.53	23.05
N-178	54.07	0.53	70.97	16.89
N-074	47.76	18.01	72.72	24.96
N-116	81.46	8.29	95.48	14.02

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 6

Design Computation

Residual pressure = $8.428 \times 3.685 = 31.055$ m (permissible 55.330) — safe

Velocity = $33.648 \times 11.969 = 402.737$ m/s (permissible 509.454) — safe

Discharge = $84.457 \times 4.382 = 370.051$ lps (permissible 692.509) — safe

Detention time = $58.095 \times 7.460 = 433.364$ hr (permissible 822.281) — safe

Head loss = $76.459 \times 10.215 = 781.032$ m (permissible 1463.969) — safe

Residual pressure = $22.381 \times 1.430 = 31.996$ m (permissible 34.222) — safe

Velocity = $85.871 \times 4.349 = 373.446$ m/s (permissible 532.590) — safe

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 7

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Reference: drawing no. PL-216, revision C.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 8

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-05	Backwash	134.4	39.1	71.60
P-12	Booster	55.2	20.1	15.10
P-03	Sludge	234.7	31.7	101.40
P-08	Clear water	115.8	76.2	120.20
P-11	Raw water	136.3	19.0	35.34
P-05	Backwash	284.1	69.6	269.32
P-02	Sludge	273.9	39.4	147.23
P-09	Backwash	178.4	56.8	137.94
P-11	Backwash	72.8	86.4	85.73
P-09	Backwash	277.9	43.4	164.48
P-07	Raw water	191.7	79.9	208.56
P-01	Backwash	97.0	12.4	16.34
P-07	Backwash	63.8	21.4	18.60
P-06	Sludge	46.2	53.2	33.50
P-12	Sludge	246.0	58.5	195.97
P-09	Clear water	227.0	80.8	249.87
P-09	Raw water	219.5	24.5	73.38
P-01	Sludge	252.8	44.9	154.80
P-01	Clear water	41.2	53.7	30.19
P-04	Booster	16.0	27.8	6.08
P-11	Sludge	78.3	54.4	58.00
P-01	Clear water	223.3	34.6	105.28
P-06	Backwash	103.6	21.0	29.67

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 9

Design Computation

Discharge = $56.027 \times 0.524 = 29.349$ lps (permissible 32.528) — safe

Head loss = $50.534 \times 7.906 = 399.500$ m (permissible 610.634) — safe

Head loss = $53.713 \times 10.187 = 547.182$ m (permissible 825.153) — safe

Residual pressure = $10.516 \times 3.531 = 37.135$ m (permissible 64.554) — safe

Velocity = $58.995 \times 11.032 = 650.848$ m/s (permissible 709.883) — safe

Discharge = $23.984 \times 3.377 = 81.005$ lps (permissible 132.606) — safe

Velocity = $16.216 \times 0.678 = 10.988$ m/s (permissible 18.073) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 10

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. PL-960, revision A.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 11

Population Projection

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2051	Ward 9	10,197	150	1.530
2051	Ward 16	18,074	150	2.711
2056	Ward 13	42,667	70	2.987
2026	Ward 11	28,466	70	1.993
2026	Ward 20	39,285	135	5.303
2031	Ward 8	8,506	100	0.851
2036	Ward 4	28,227	135	3.811
2031	Ward 7	7,621	135	1.029
2036	Ward 20	21,591	70	1.511
2041	Ward 17	2,479	150	0.372
2056	Ward 18	39,488	100	3.949
2056	Ward 1	40,619	135	5.484
2051	Ward 11	27,459	100	2.746
2041	Ward 15	15,281	135	2.063
2051	Ward 15	8,028	100	0.803
2056	Ward 19	28,353	135	3.828
2046	Ward 1	12,747	70	0.892
2041	Ward 22	31,639	135	4.271
2046	Ward 15	21,014	150	3.152
2031	Ward 16	15,574	150	2.336
2036	Ward 13	34,877	70	2.441

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

10.x Cost Benefit Analysis And Investment Criteria — Working Papers — Sheet 12

Design Computation

Velocity = $3.877 \times 4.662 = 18.075$ m/s (permissible 32.844) — safe

Residual pressure = $69.275 \times 11.236 = 778.376$ m (permissible 925.258) — safe

Detention time = $38.693 \times 7.723 = 298.815$ hr (permissible 399.302) — revise section

Detention time = $13.170 \times 5.315 = 70.001$ hr (permissible 117.518) — safe

Discharge = $27.410 \times 9.740 = 266.959$ lps (permissible 311.241) — revise section

Residual pressure = $89.549 \times 7.727 = 691.977$ m (permissible 1222.981) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

11A Supplementary Notes — Sheet 1

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Reference: drawing no. WS-538, revision C.

11A Supplementary Notes — Sheet 2

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
8,293	300	HDPE	PN 10	1066.2
13,285	300	PVC-O	K-7	1232.9
11,127	250	HDPE	PN 10	62.4
17,048	600	PVC-O	K-7	121.2
6,957	600	PVC-O	PN 10	982.6
19,756	200	DI	K-7	338.4
3,625	200	HDPE	PN 6	1364.5
11,309	500	MS	K-9	953.9
7,513	250	PVC-O	K-9	726.9
17,662	100	DI	PN 10	1575.7
910	100	MS	PN 16	1664.2
20,947	150	MS	PN 16	336.4
9,240	500	HDPE	PN 6	1539.4
5,042	500	DI	PN 16	1645.8
4,441	300	HDPE	K-7	1492.2
18,276	250	DI	K-7	760.8
5,361	200	HDPE	PN 16	973.0
2,161	200	HDPE	K-7	471.5

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

11A Supplementary Notes — Sheet 3

Design Computation

Detention time = $2.088 \times 8.819 = 18.410$ hr (permissible 33.225) — safe

Velocity = $65.478 \times 5.519 = 361.398$ m/s (permissible 597.002) — safe

Head loss = $9.913 \times 10.537 = 104.449$ m (permissible 111.486) — safe

Detention time = $4.173 \times 5.872 = 24.500$ hr (permissible 38.868) — safe

Head loss = $10.861 \times 11.284 = 122.558$ m (permissible 183.938) — safe

Detention time = $79.315 \times 7.572 = 600.539$ hr (permissible 1049.859) — safe

Residual pressure = $43.867 \times 9.618 = 421.919$ m (permissible 752.152) — safe

Detention time = $56.183 \times 10.630 = 597.245$ hr (permissible 655.533) — revise section

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

11A Supplementary Notes — Sheet 4

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Reference: drawing no. ST-660, revision B.

11A Supplementary Notes — Sheet 5

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-156	60.68	1.60	68.09	7.41
N-146	83.06	6.23	102.66	19.61
N-104	57.31	8.41	65.09	7.78
N-134	32.41	10.92	41.11	8.70
N-095	66.89	19.57	90.77	23.88
N-101	70.69	3.41	92.16	21.46
N-178	83.47	15.07	108.86	25.39
N-060	40.58	9.64	58.09	17.51
N-184	54.89	6.62	70.31	15.42
N-198	87.64	14.19	97.23	9.59
N-130	77.90	17.27	101.57	23.67
N-060	30.49	21.45	41.50	11.01
N-054	58.79	10.42	79.61	20.82
N-149	42.96	21.09	56.24	13.28
N-203	72.89	23.76	89.58	16.69
N-162	66.57	21.15	90.24	23.67

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

11A Supplementary Notes — Sheet 6

Design Computation

Head loss = $84.869 \times 1.466 = 124.457$ m (permissible 138.410) — safe

Head loss = $69.446 \times 2.858 = 198.448$ m (permissible 311.877) — safe

Detention time = $66.138 \times 1.341 = 88.710$ hr (permissible 102.241) — safe

Head loss = $7.096 \times 11.900 = 84.440$ m (permissible 126.643) — safe

Velocity = $43.641 \times 8.160 = 356.100$ m/s (permissible 461.911) — safe

Residual pressure = $64.525 \times 10.439 = 673.565$ m (permissible 758.861) — safe

Velocity = $35.599 \times 5.565 = 198.122$ m/s (permissible 307.347) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

11A Supplementary Notes — Sheet 7

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. WS-185, revision B.

11A Supplementary Notes — Sheet 8

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Residual chlorine	mg/l	Nil	0.35	0.2 min
Total hardness	mg/l	278	166	200
Residual chlorine	mg/l	Nil	0.78	0.2 min
Total coliform	MPN/100ml	452	Absent	Absent
Turbidity	NTU	93.9	0.43	1
pH	-	7.2	7.1	6.5-8.5
Iron	mg/l	1.91	0.03	1.0
Turbidity	NTU	9.8	0.64	1
Total coliform	MPN/100ml	818	Absent	Absent
Turbidity	NTU	137.8	0.26	1
pH	-	6.9	8.0	6.5-8.5
Total hardness	mg/l	281	69	200
Total coliform	MPN/100ml	352	Absent	Absent
Total hardness	mg/l	442	190	200
Total coliform	MPN/100ml	894	Absent	Absent
Total coliform	MPN/100ml	699	Absent	Absent
pH	-	6.7	7.6	6.5-8.5
Total hardness	mg/l	401	61	200
pH	-	7.7	7.5	6.5-8.5
Total hardness	mg/l	404	162	200

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

12 RISK ASSESSMENT AND MITIGATION MEASURES

Sl.	Risk	Likelihood	Impact	Mitigation
1	Delay in balance land acquisition	Medium	High	Award proceedings advanced ahead of the zero date
2	Escalation in cement and steel	Medium	Medium	Price adjustment clause in the contract
3	Monsoon disruption to earthwork	High	Medium	Programme earthwork in the dry season
4	Contractor default	Low	High	Pre-qualification and performance security
5	Utility shifting delay	Medium	Medium	Deposit work sanctioned in advance

Each risk has been assessed for likelihood and impact and a mitigation plan documented covering avoidance, transfer and elimination as appropriate.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 1

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Reference: drawing no. HY-911, revision A.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 2

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
23,687	400	MS	K-9	1696.8
23,141	400	MS	K-9	536.6
9,555	250	HDPE	PN 16	1639.7
3,788	500	HDPE	K-9	1273.8
3,110	100	PVC-O	PN 10	735.0
22,564	500	PVC-O	PN 16	1725.5
23,205	500	DI	K-7	92.4
11,729	300	HDPE	K-9	714.5
18,897	400	PVC-O	PN 6	653.2
13,953	300	MS	PN 10	527.1
16,622	400	DI	PN 6	1013.0
17,759	250	MS	PN 6	1773.1
9,690	200	DI	K-7	1173.6
7,754	150	HDPE	K-9	769.3
23,075	100	HDPE	PN 10	1756.9
15,294	100	DI	PN 16	1512.4
4,939	400	MS	K-9	889.7
6,786	200	MS	K-9	1636.4
12,954	100	PVC-O	K-9	871.9
10,611	600	MS	K-7	846.9

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 3

Design Computation

Head loss = $31.161 \times 0.618 = 19.271$ m (permissible 28.469) — safe

Velocity = $61.431 \times 5.800 = 356.306$ m/s (permissible 529.534) — safe

Head loss = $12.235 \times 11.522 = 140.968$ m (permissible 168.475) — revise section

Head loss = $48.601 \times 6.610 = 321.235$ m (permissible 543.908) — safe

Discharge = $20.839 \times 9.887 = 206.031$ lps (permissible 331.772) — safe

Discharge = $14.587 \times 7.853 = 114.548$ lps (permissible 199.587) — safe

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 4

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Reference: drawing no. HY-348, revision A.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 5

Population Projection

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2051	Ward 9	23,411	135	3.160
2031	Ward 10	13,116	135	1.771
2056	Ward 5	7,962	150	1.194
2056	Ward 18	19,290	70	1.350
2056	Ward 23	14,262	70	0.998
2026	Ward 9	10,996	100	1.100
2041	Ward 8	7,426	150	1.114
2041	Ward 18	34,337	150	5.151
2026	Ward 13	18,909	150	2.836
2051	Ward 24	3,265	70	0.229
2031	Ward 10	6,484	70	0.454
2031	Ward 14	44,263	70	3.098
2041	Ward 18	43,019	100	4.302
2026	Ward 5	38,510	70	2.696
2036	Ward 17	39,923	70	2.795
2051	Ward 13	27,690	70	1.938
2031	Ward 16	10,023	70	0.702
2056	Ward 15	11,029	100	1.103
2026	Ward 17	12,583	135	1.699
2026	Ward 11	5,116	100	0.512
2041	Ward 19	27,086	135	3.657
2036	Ward 17	25,212	100	2.521

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 6

Design Computation

Velocity = $18.523 \times 6.866 = 127.178$ m/s (permissible 144.096) — safe

Head loss = $84.300 \times 1.097 = 92.484$ m (permissible 110.138) — safe

Discharge = $12.810 \times 2.108 = 27.008$ lps (permissible 31.700) — safe

Detention time = $55.988 \times 1.942 = 108.708$ hr (permissible 130.534) — safe

Detention time = $64.893 \times 5.575 = 361.767$ hr (permissible 597.433) — safe

Head loss = $11.676 \times 2.999 = 35.012$ m (permissible 58.929) — safe

Velocity = $21.073 \times 11.957 = 251.965$ m/s (permissible 320.212) — safe

Velocity = $6.573 \times 7.788 = 51.190$ m/s (permissible 77.185) — safe

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 7

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. ST-391, revision B.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 8

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-06	Clear water	150.5	82.6	169.30
P-06	Backwash	296.0	79.0	318.78
P-10	Raw water	208.4	35.1	99.68
P-09	Booster	62.8	84.4	72.26
P-01	Booster	214.7	59.4	173.70
P-08	Backwash	265.2	57.5	207.88
P-03	Booster	129.1	39.0	68.63
P-06	Clear water	108.2	53.2	78.48
P-08	Booster	292.0	61.4	244.20
P-07	Raw water	275.1	18.7	70.01
P-08	Clear water	162.2	35.1	77.58
P-09	Booster	204.7	86.4	241.03
P-12	Raw water	48.3	25.8	16.98
P-03	Sludge	261.4	38.5	137.14
P-04	Raw water	289.0	53.9	212.35
P-08	Booster	241.3	82.2	270.39
P-10	Backwash	277.7	13.8	52.22

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 9

Design Computation

Head loss = $33.828 \times 11.564 = 391.190$ m (permissible 557.102) — safe

Discharge = $89.281 \times 1.274 = 113.720$ lps (permissible 199.005) — safe

Detention time = $48.965 \times 4.569 = 223.725$ hr (permissible 250.891) — safe

Velocity = $66.870 \times 6.764 = 452.289$ m/s (permissible 636.068) — safe

Residual pressure = $84.377 \times 4.798 = 404.852$ m (permissible 711.152) — safe

Head loss = $18.667 \times 4.432 = 82.741$ m (permissible 123.864) — safe

Detention time = $80.026 \times 2.878 = 230.309$ hr (permissible 387.172) — safe

Residual pressure = $10.144 \times 10.172 = 103.182$ m (permissible 195.321) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

12.x Risk Assessment And Mitigation Measures — Working Papers — Sheet 10

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. PL-104, revision C.

13 PROJECT MANAGEMENT ORGANISATION

The Assam Jal Board implements through a circle office at Sivasagar headed by a Superintending Engineer. A third party inspection agency is engaged for pipeline laying, which is dispersed across 214 km and cannot be supervised departmentally at the required frequency.

13.x Project Management Organisation — Working Papers — Sheet 1

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. PL-438, revision B.

13.x Project Management Organisation — Working Papers — Sheet 2

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
13,832	100	PVC-O	K-9	82.9
14,044	400	PVC-O	PN 16	443.6
11,374	200	PVC-O	K-7	603.9
7,730	500	PVC-O	K-9	1187.6
3,676	500	DI	K-9	1593.9
15,099	400	MS	PN 16	1636.6
290	200	DI	PN 6	792.7
1,550	200	HDPE	K-9	49.6
23,330	500	HDPE	PN 10	1675.1
19,965	150	DI	K-9	454.6
19,181	250	HDPE	K-7	1428.8
11,889	500	PVC-O	K-7	1556.5
23,415	600	PVC-O	PN 6	493.3
8,881	500	PVC-O	PN 10	915.3
52	400	PVC-O	PN 10	1055.6
6,210	600	HDPE	PN 10	722.3
3,895	150	DI	K-7	1226.2
18,523	600	MS	PN 10	583.7
5,888	300	HDPE	PN 6	420.5
22,344	300	HDPE	K-9	1587.3
21,260	150	DI	PN 6	765.7
23,128	500	MS	K-7	670.2
6,389	300	DI	K-7	78.8
8,064	400	HDPE	PN 16	1142.1
18,588	250	MS	K-7	51.6
21,674	500	PVC-O	PN 6	152.8

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

13.x Project Management Organisation — Working Papers — Sheet 3

Design Computation

Velocity = $67.533 \times 6.543 = 441.889$ m/s (permissible 753.845) — safe

Head loss = $34.468 \times 4.345 = 149.772$ m (permissible 199.621) — safe

Detention time = $74.984 \times 5.530 = 414.653$ hr (permissible 748.509) — safe

Head loss = $55.803 \times 5.485 = 306.105$ m (permissible 563.807) — safe

Discharge = $23.249 \times 10.452 = 242.999$ lps (permissible 370.380) — revise section

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

13.x Project Management Organisation — Working Papers — Sheet 4

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Reference: drawing no. PL-857, revision A.

13.x Project Management Organisation — Working Papers — Sheet 5

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-087	65.29	23.19	73.91	8.62
N-107	64.13	16.54	87.85	23.71
N-149	86.88	7.36	94.69	7.82
N-112	57.91	4.20	82.70	24.79
N-112	88.50	23.44	110.47	21.97
N-082	30.06	6.21	50.16	20.10
N-065	54.43	15.96	80.00	25.57
N-012	33.36	21.91	41.88	8.52
N-071	76.57	22.45	99.24	22.68
N-127	40.97	18.27	49.62	8.65
N-055	65.38	11.39	83.72	18.34
N-037	35.19	8.04	53.16	17.97
N-218	65.47	15.45	82.33	16.86
N-091	72.76	5.44	95.30	22.54
N-001	83.97	3.62	98.15	14.18
N-175	31.66	1.91	54.55	22.88
N-039	76.04	5.07	93.08	17.05
N-186	52.72	16.40	75.50	22.78

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

13.x Project Management Organisation — Working Papers — Sheet 6

Design Computation

Head loss = $65.006 \times 5.240 = 340.634$ m (permissible 563.066) — safe

Discharge = $47.375 \times 10.609 = 502.574$ lps (permissible 688.248) — safe

Residual pressure = $78.491 \times 2.013 = 158.034$ m (permissible 287.620) — safe

Head loss = $12.002 \times 8.148 = 97.795$ m (permissible 175.972) — safe

Velocity = $49.339 \times 11.802 = 582.290$ m/s (permissible 936.517) — revise section

Head loss = $8.437 \times 2.407 = 20.309$ m (permissible 37.248) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

13.x Project Management Organisation — Working Papers — Sheet 7

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. HY-132, revision C.

13.x Project Management Organisation — Working Papers — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
1.3	Clayey silt	59	UDS
2.9	Weathered rock	50	SPT
11.4	Weathered rock	9	DS
26.1	Silty clay	52	UDS
0.6	Clayey silt	37	DS
27.0	Coarse sand	48	UDS
28.1	Medium sand	51	SPT
21.9	Medium sand	39	UDS
3.2	Gravelly sand	21	SPT
15.6	Weathered rock	30	SPT
8.3	Medium sand	53	UDS
13.4	Gravelly sand	59	SPT
16.2	Medium sand	20	DS
0.5	Coarse sand	12	UDS
23.9	Medium sand	7	UDS
18.2	Weathered rock	28	UDS
8.1	Coarse sand	18	UDS
19.9	Medium sand	27	SPT
23.7	Weathered rock	48	UDS
28.9	Medium sand	40	SPT
0.5	Silty clay	28	UDS
28.8	Weathered rock	48	UDS
7.5	Gravelly sand	53	UDS
21.1	Clayey silt	50	SPT

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

14 CONTRACT MANAGEMENT STRATEGY

The intake, rising main and treatment plant are proposed as a single design-build-operate package with a five year operation period. Distribution network laying is proposed as three geographic item rate packages.

14.x Contract Management Strategy — Working Papers — Sheet 1

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. PL-569, revision C.

14.x Contract Management Strategy — Working Papers — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
5.2	Coarse sand	32	DS
29.1	Clayey silt	29	SPT
21.9	Medium sand	53	SPT
8.8	Weathered rock	8	SPT
24.3	Silty clay	37	DS
13.8	Silty clay	37	DS
27.4	Coarse sand	27	DS
1.7	Coarse sand	29	SPT
29.3	Gravelly sand	27	SPT
2.6	Silty clay	42	UDS
0.9	Weathered rock	30	SPT
11.3	Clayey silt	17	UDS
20.2	Clayey silt	27	SPT
3.5	Coarse sand	42	DS
3.5	Medium sand	7	SPT
19.1	Silty clay	53	SPT
28.6	Gravelly sand	58	UDS
22.3	Coarse sand	54	DS
27.1	Clayey silt	58	UDS
25.7	Weathered rock	55	DS
23.7	Medium sand	17	UDS
21.2	Gravelly sand	28	SPT
10.5	Gravelly sand	15	SPT
22.9	Coarse sand	7	SPT
29.9	Silty clay	36	DS

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

14.x Contract Management Strategy — Working Papers — Sheet 3

Design Computation

Velocity = $77.458 \times 6.039 = 467.743$ m/s (permissible 655.078) — safe

Discharge = $10.757 \times 6.261 = 67.346$ lps (permissible 76.476) — revise section

Residual pressure = $40.819 \times 9.746 = 397.821$ m (permissible 699.114) — safe

Detention time = $76.590 \times 1.224 = 93.751$ hr (permissible 105.471) — safe

Discharge = $58.077 \times 1.473 = 85.519$ lps (permissible 107.669) — safe

Detention time = $83.020 \times 1.077 = 89.390$ hr (permissible 167.933) — revise section

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

14.x Contract Management Strategy — Working Papers — Sheet 4

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. ST-671, revision C.

14.x Contract Management Strategy — Working Papers — Sheet 5

Bore Log Summary

Depth (m)	Strata	N value	Sample
2.1	Coarse sand	15	DS
29.4	Weathered rock	48	UDS
6.7	Gravelly sand	46	UDS
27.3	Silty clay	11	UDS
27.9	Medium sand	24	SPT
8.6	Coarse sand	4	SPT
3.6	Coarse sand	28	DS
11.2	Weathered rock	46	DS
27.2	Clayey silt	43	DS
25.0	Medium sand	24	SPT
4.4	Weathered rock	22	UDS
9.4	Silty clay	40	UDS
10.1	Coarse sand	18	UDS
10.9	Gravelly sand	5	UDS
21.1	Coarse sand	32	DS
10.8	Medium sand	10	UDS
7.6	Weathered rock	51	UDS
28.7	Gravelly sand	4	UDS
22.9	Weathered rock	31	DS
5.7	Gravelly sand	55	SPT

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

14.x Contract Management Strategy — Working Papers — Sheet 6

Design Computation

Discharge = $23.198 \times 7.029 = 163.047$ lps (permissible 287.640) — safe

Velocity = $71.374 \times 6.517 = 465.166$ m/s (permissible 733.566) — safe

Detention time = $64.532 \times 11.338 = 731.678$ hr (permissible 825.103) — safe

Head loss = $16.595 \times 6.244 = 103.628$ m (permissible 163.068) — safe

Velocity = $27.234 \times 11.756 = 320.152$ m/s (permissible 358.152) — revise section

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

14.x Contract Management Strategy — Working Papers — Sheet 7

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Reference: drawing no. WS-829, revision B.

14.x Contract Management Strategy — Working Papers — Sheet 8

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
10,697	400	PVC-O	PN 6	1162.4
14,396	600	DI	K-7	876.7
4,075	200	PVC-O	K-7	763.7
20,058	150	PVC-O	PN 16	1556.9
5,677	400	MS	K-7	279.8
9,996	400	PVC-O	PN 6	1146.2
14,459	100	HDPE	K-7	1322.0
9,138	300	MS	PN 10	1254.7
6,991	300	PVC-O	PN 16	419.7
4,379	200	HDPE	PN 6	765.4
9,932	200	MS	K-7	710.3
22,780	100	HDPE	K-7	642.0
3,007	100	HDPE	PN 16	1047.6
23,597	300	DI	K-7	1031.6
16,922	600	HDPE	PN 16	1147.1
19,871	150	MS	K-9	577.0
9,811	100	PVC-O	PN 10	1048.8
1,838	100	PVC-O	PN 6	95.5
1,816	300	PVC-O	PN 6	913.2
3,239	150	HDPE	K-7	1307.8
7,085	600	HDPE	PN 10	1671.8
1,287	200	MS	PN 10	1654.8
6,527	150	DI	PN 6	1447.9
10,107	100	DI	PN 16	1777.1
12,517	100	DI	PN 6	1796.2
10,574	150	PVC-O	K-7	1139.8

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

15 IMPLEMENTATION SCHEDULE AND WBS

The zero date is the date of issue of the work order. The implementation period is 24 months.

WBS	Milestone	Month from zero date
1.0	Award and mobilisation	M0 – M2
2.0	Site preparation complete	M4
3.0	Substructure / foundation complete	M8
4.0	Superstructure complete	M16
5.0	Services and finishes	M21
6.0	Testing and commissioning	M24

15.x Implementation Schedule And Wbs — Working Papers — Sheet 1

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Reference: drawing no. HY-210, revision A.

15.x Implementation Schedule And Wbs — Working Papers — Sheet 2

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-01	Clear water	186.5	37.9	96.31
P-12	Clear water	56.2	54.6	41.83
P-02	Booster	65.0	80.8	71.55
P-12	Sludge	130.2	12.5	22.17
P-11	Clear water	46.1	15.3	9.64
P-05	Sludge	53.5	80.9	58.98
P-01	Backwash	189.9	32.2	83.22
P-04	Booster	119.8	72.9	118.97
P-06	Raw water	209.8	54.9	157.02
P-09	Sludge	136.5	70.5	131.02
P-11	Booster	223.9	60.5	184.59
P-11	Sludge	286.5	48.9	191.05
P-11	Clear water	129.2	14.4	25.39
P-08	Backwash	236.4	18.3	58.87
P-07	Raw water	66.9	37.6	34.27
P-06	Raw water	235.5	41.5	133.07
P-11	Backwash	72.7	18.7	18.51
P-04	Sludge	18.9	38.7	9.98
P-01	Backwash	177.3	67.0	161.81
P-11	Clear water	317.5	85.3	368.86
P-03	Booster	181.0	86.5	213.36
P-01	Raw water	150.6	69.1	141.81
P-05	Backwash	56.1	26.7	20.39
P-11	Sludge	223.9	84.6	257.96

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

15.x Implementation Schedule And Wbs — Working Papers — Sheet 3

Design Computation

Head loss = $55.965 \times 6.320 = 353.683$ m (permissible 396.291) — safe

Detention time = $71.820 \times 4.779 = 343.195$ hr (permissible 562.866) — safe

Residual pressure = $88.696 \times 3.445 = 305.599$ m (permissible 329.459) — safe

Velocity = $37.852 \times 0.676 = 25.588$ m/s (permissible 46.091) — safe

Discharge = $64.877 \times 8.277 = 536.986$ lps (permissible 869.084) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

15.x Implementation Schedule And Wbs — Working Papers — Sheet 4

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Reference: drawing no. ST-477, revision A.

15.x Implementation Schedule And Wbs — Working Papers — Sheet 5

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Residual chlorine	mg/l	Nil	0.64	0.2 min
Turbidity	NTU	235.5	0.67	1
Total coliform	MPN/100ml	409	Absent	Absent
Total coliform	MPN/100ml	326	Absent	Absent
Total coliform	MPN/100ml	265	Absent	Absent
Iron	mg/l	0.42	0.23	1.0
Total hardness	mg/l	270	184	200
Iron	mg/l	1.83	0.09	1.0
Iron	mg/l	3.99	0.03	1.0
Total coliform	MPN/100ml	709	Absent	Absent
Total coliform	MPN/100ml	45	Absent	Absent
Residual chlorine	mg/l	Nil	0.77	0.2 min
Turbidity	NTU	142.6	0.89	1
Iron	mg/l	3.92	0.24	1.0
Iron	mg/l	2.38	0.26	1.0
Turbidity	NTU	193.1	0.51	1
Iron	mg/l	0.51	0.26	1.0
pH	-	7.0	7.2	6.5-8.5
Residual chlorine	mg/l	Nil	0.28	0.2 min
Turbidity	NTU	175.1	0.27	1
Iron	mg/l	2.94	0.27	1.0

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

15.x Implementation Schedule And Wbs — Working Papers — Sheet 6

Design Computation

Discharge = $15.468 \times 9.611 = 148.661$ lps (permissible 160.174) — safe

Residual pressure = $41.727 \times 11.269 = 470.210$ m (permissible 517.552) — safe

Head loss = $42.059 \times 1.447 = 60.873$ m (permissible 82.489) — safe

Velocity = $73.953 \times 9.876 = 730.372$ m/s (permissible 1254.092) — safe

Discharge = $13.728 \times 8.233 = 113.022$ lps (permissible 150.125) — safe

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

15.x Implementation Schedule And Wbs — Working Papers — Sheet 7

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Reference: drawing no. PL-394, revision A.

15.x Implementation Schedule And Wbs — Working Papers — Sheet 8

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
26	300	PVC-O	PN 16	687.2
20,515	200	PVC-O	K-7	1100.7
11,464	400	DI	PN 6	931.2
94	300	DI	PN 10	1306.5
19,988	400	PVC-O	K-7	1330.8
17,499	250	PVC-O	K-7	377.4
23,668	400	PVC-O	PN 10	538.6
18,114	200	PVC-O	K-7	498.2
1,019	200	HDPE	PN 6	363.8
4,468	150	MS	K-7	1611.3
2,411	400	DI	PN 6	675.6
8,259	500	DI	PN 16	497.6
18,144	150	DI	PN 6	544.9
20,905	300	PVC-O	K-9	49.4
3,710	150	PVC-O	PN 10	1479.0
11,756	200	HDPE	PN 10	40.6
6,870	250	DI	PN 10	268.2
2,747	200	MS	PN 10	335.9
12,783	300	DI	K-7	254.7
18,643	500	MS	PN 16	134.3
22,110	250	MS	PN 6	1091.5
21,686	400	DI	K-9	582.3
6,552	500	HDPE	PN 10	119.1
2,151	200	HDPE	K-7	1007.8

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

15.x Implementation Schedule And Wbs — Working Papers — Sheet 9

Design Computation

Detention time = $52.468 \times 2.602 = 136.516$ hr (permissible 226.698) — safe

Head loss = $64.292 \times 11.292 = 725.950$ m (permissible 1345.264) — safe

Detention time = $8.982 \times 0.896 = 8.046$ hr (permissible 14.779) — safe

Detention time = $19.533 \times 0.813 = 15.887$ hr (permissible 18.013) — safe

Residual pressure = $75.533 \times 9.647 = 728.683$ m (permissible 1146.199) — safe

Residual pressure = $17.237 \times 8.031 = 138.433$ m (permissible 222.790) — safe

Velocity = $4.872 \times 2.166 = 10.552$ m/s (permissible 18.095) — safe

Discharge = $27.875 \times 6.317 = 176.091$ lps (permissible 227.355) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

15.x Implementation Schedule And Wbs — Working Papers — Sheet 10

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. PL-741, revision C.

16 STATUTORY CLEARANCES

Sl.	Clearance	Authority	Status
1	Consent to establish	State Pollution Control Board	Obtained, 12 February 2026
2	Building permit / layout approval	Local body	Obtained, 3 March 2026
3	Fire safety clearance	Directorate of Fire Services	Obtained, 19 March 2026
4	Ground water abstraction	CGWA	Not applicable

16.x Statutory Clearances — Working Papers — Sheet 1

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Reference: drawing no. ST-386, revision B.

16.x Statutory Clearances — Working Papers — Sheet 2

Population Projection

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2036	Ward 14	12,644	150	1.897
2046	Ward 14	3,225	150	0.484
2031	Ward 4	2,968	100	0.297
2031	Ward 5	28,221	135	3.810
2056	Ward 21	27,212	100	2.721
2056	Ward 13	25,159	135	3.396
2041	Ward 23	12,952	70	0.907
2041	Ward 19	35,423	70	2.480
2056	Ward 4	36,858	150	5.529
2026	Ward 5	41,567	150	6.235
2056	Ward 8	36,743	150	5.511
2051	Ward 17	13,866	150	2.080
2051	Ward 13	4,303	150	0.645
2046	Ward 15	43,639	135	5.891
2026	Ward 17	7,037	100	0.704
2031	Ward 18	5,839	70	0.409
2051	Ward 20	36,182	70	2.533
2046	Ward 16	32,594	150	4.889
2051	Ward 21	7,111	150	1.067
2031	Ward 13	3,659	150	0.549
2026	Ward 11	12,199	100	1.220

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

16.x Statutory Clearances — Working Papers — Sheet 3

Design Computation

Detention time = $39.933 \times 9.841 = 392.989$ hr (permissible 689.516) — safe

Discharge = $55.126 \times 1.041 = 57.394$ lps (permissible 79.340) — safe

Velocity = $19.271 \times 9.852 = 189.846$ m/s (permissible 267.849) — safe

Head loss = $69.086 \times 1.352 = 93.432$ m (permissible 111.032) — safe

Velocity = $51.655 \times 11.016 = 569.033$ m/s (permissible 695.261) — revise section

Head loss = $37.524 \times 9.554 = 358.504$ m (permissible 574.757) — safe

Head loss = $42.324 \times 9.183 = 388.647$ m (permissible 584.075) — safe

Residual pressure = $33.109 \times 3.365 = 111.407$ m (permissible 193.387) — safe

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

16.x Statutory Clearances — Working Papers — Sheet 4

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. HY-552, revision C.

16.x Statutory Clearances — Working Papers — Sheet 5

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
20,760	400	MS	K-7	259.8
2,832	400	MS	PN 6	911.0
2,105	100	PVC-O	PN 6	1722.4
5,458	150	MS	K-7	1271.1
1,462	300	MS	PN 6	1309.2
14,990	200	DI	PN 6	149.3
14,497	150	HDPE	PN 10	1284.0
2,983	500	DI	PN 10	1064.0
13,944	600	HDPE	K-7	1564.6
2,932	250	MS	K-7	850.2
20,057	300	HDPE	K-7	1099.3
3,486	600	PVC-O	K-9	721.6
12,461	250	DI	PN 6	1203.7
12,215	200	MS	K-7	523.8
5,597	600	DI	PN 16	589.8
17,313	100	MS	K-9	649.3
2,393	150	PVC-O	PN 10	1475.6
14,894	500	MS	K-7	839.9
4,759	300	DI	K-9	959.7
15,790	250	HDPE	K-7	221.8
14,287	100	MS	PN 10	1667.9
2,074	150	DI	PN 6	358.2
3,542	500	MS	PN 16	1078.1
19,210	500	HDPE	PN 6	1373.4

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

16.x Statutory Clearances — Working Papers — Sheet 6

Design Computation

Head loss = $27.991 \times 6.206 = 173.701$ m (permissible 221.283) — safe

Detention time = $14.127 \times 8.232 = 116.294$ hr (permissible 171.987) — revise section

Detention time = $25.122 \times 11.713 = 294.245$ hr (permissible 445.841) — safe

Velocity = $88.627 \times 2.426 = 214.964$ m/s (permissible 384.394) — safe

Velocity = $65.720 \times 1.706 = 112.117$ m/s (permissible 208.917) — safe

Discharge = $3.169 \times 7.635 = 24.195$ lps (permissible 29.977) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

16.x Statutory Clearances — Working Papers — Sheet 7

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. WS-697, revision A.

16.x Statutory Clearances — Working Papers — Sheet 8

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-183	81.30	11.14	90.41	9.11
N-159	90.18	6.54	113.30	23.12
N-161	58.74	11.19	65.91	7.16
N-213	73.90	21.71	97.81	23.91
N-163	38.91	12.28	47.81	8.91
N-003	61.06	22.64	80.77	19.72
N-066	68.39	1.71	84.28	15.89
N-114	70.97	9.16	96.95	25.98
N-102	38.99	8.30	54.27	15.28
N-218	36.87	17.03	46.09	9.21
N-122	65.02	16.51	73.29	8.28
N-193	60.77	21.40	80.68	19.91
N-068	87.64	11.15	98.94	11.30
N-038	95.25	14.52	104.80	9.56
N-099	90.03	7.17	97.13	7.10
N-137	36.32	17.57	60.14	23.81
N-173	35.40	12.62	56.22	20.81
N-174	36.88	1.75	54.99	18.11
N-180	50.10	20.21	71.18	21.08
N-096	77.38	0.49	96.32	18.94
N-101	70.63	16.75	78.31	7.68
N-035	49.18	0.86	56.42	7.24

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

16.x Statutory Clearances — Working Papers — Sheet 9

Design Computation

Detention time = $85.268 \times 2.852 = 243.155$ hr (permissible 384.687) — safe

Detention time = $21.432 \times 8.254 = 176.889$ hr (permissible 262.988) — safe

Detention time = $55.942 \times 8.965 = 501.509$ hr (permissible 900.870) — safe

Discharge = $39.849 \times 10.084 = 401.821$ lps (permissible 507.309) — safe

Residual pressure = $15.704 \times 4.141 = 65.032$ m (permissible 78.857) — safe

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

16.x Statutory Clearances — Working Papers — Sheet 10

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. HY-301, revision B.

17 QUALITY MANAGEMENT PLAN

Quality control covers pipe testing at the manufacturer's works, hydrostatic testing of laid mains in 500 m sections, and daily water quality testing at the plant laboratory against IS:10500. A third party agency audits at monthly intervals.

17.x Quality Management Plan — Working Papers — Sheet 1

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. HY-254, revision C.

17.x Quality Management Plan — Working Papers — Sheet 2

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-164	70.97	3.88	79.40	8.43
N-163	62.30	2.58	85.02	22.72
N-048	45.51	20.84	55.51	10.00
N-013	91.50	6.81	115.66	24.16
N-110	69.21	0.86	86.37	17.16
N-179	68.15	4.44	88.66	20.51
N-201	38.96	22.05	47.02	8.06
N-163	81.44	12.62	95.59	14.15
N-186	60.78	9.06	79.63	18.85
N-214	71.69	0.39	93.22	21.53
N-086	82.78	15.80	94.39	11.61
N-061	80.17	9.85	97.95	17.78
N-211	73.97	11.08	86.14	12.18
N-077	66.93	12.74	88.09	21.16
N-067	32.89	4.61	40.78	7.90
N-089	69.81	15.36	80.21	10.40
N-029	39.32	12.31	59.50	20.18
N-151	68.41	9.35	88.40	19.99
N-049	36.32	17.78	53.91	17.59
N-147	93.13	4.79	108.99	15.86
N-214	52.83	15.90	64.86	12.03
N-106	49.90	22.56	68.35	18.45
N-148	78.65	5.11	89.72	11.07
N-069	59.55	23.65	76.64	17.10
N-130	44.39	11.61	61.38	16.99

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

17.x Quality Management Plan — Working Papers — Sheet 3

Design Computation

Discharge = $55.273 \times 6.314 = 348.989$ lps (permissible 433.723) — safe

Velocity = $71.412 \times 5.754 = 410.913$ m/s (permissible 443.203) — safe

Detention time = $35.517 \times 5.164 = 183.407$ hr (permissible 258.148) — safe

Detention time = $70.025 \times 5.173 = 362.258$ hr (permissible 381.629) — safe

Discharge = $53.331 \times 8.416 = 448.839$ lps (permissible 651.030) — revise section

Head loss = $50.171 \times 8.079 = 405.335$ m (permissible 729.916) — safe

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

17.x Quality Management Plan — Working Papers — Sheet 4

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. WS-884, revision B.

17.x Quality Management Plan — Working Papers — Sheet 5

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
10,098	600	HDPE	K-9	739.0
15,544	500	HDPE	K-7	908.3
15,232	100	PVC-O	PN 16	152.2
10,513	150	DI	PN 16	1035.9
20,505	500	DI	K-7	816.7
8,628	300	DI	PN 6	219.6
18,872	500	PVC-O	PN 16	414.2
5,937	250	DI	PN 6	435.6
18,303	250	PVC-O	PN 10	94.6
18,207	150	DI	K-9	1220.2
8,012	250	PVC-O	K-9	1327.0
4,209	250	HDPE	PN 16	603.1
21,787	100	PVC-O	K-9	1448.9
23,032	400	PVC-O	PN 10	653.2
15,179	100	DI	K-7	1176.8
9,457	150	MS	K-9	311.4
10,721	400	MS	PN 6	337.1

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

17.x Quality Management Plan — Working Papers — Sheet 6

Design Computation

Head loss = $78.054 \times 2.842 = 221.833$ m (permissible 374.994) — safe

Velocity = $59.505 \times 1.154 = 68.652$ m/s (permissible 94.821) — safe

Discharge = $21.329 \times 11.466 = 244.546$ lps (permissible 330.225) — safe

Residual pressure = $40.156 \times 7.779 = 312.370$ m (permissible 559.717) — safe

Detention time = $29.043 \times 6.630 = 192.556$ hr (permissible 205.812) — safe

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

17.x Quality Management Plan — Working Papers — Sheet 7

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Reference: drawing no. PL-205, revision C.

17.x Quality Management Plan — Working Papers — Sheet 8

Population Projection

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2046	Ward 22	31,521	135	4.255
2026	Ward 8	7,771	135	1.049
2026	Ward 3	36,305	135	4.901
2041	Ward 18	45,623	100	4.562
2031	Ward 24	45,164	135	6.097
2036	Ward 7	7,342	150	1.101
2026	Ward 3	13,428	70	0.940
2046	Ward 23	8,206	135	1.108
2056	Ward 10	29,169	150	4.375
2041	Ward 12	14,578	70	1.020
2026	Ward 9	23,165	70	1.622
2041	Ward 24	7,444	150	1.117
2041	Ward 4	16,540	100	1.654
2046	Ward 14	20,645	150	3.097
2041	Ward 15	44,466	135	6.003
2056	Ward 5	5,283	135	0.713

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

17.x Quality Management Plan — Working Papers — Sheet 9

Design Computation

Residual pressure = $6.611 \times 9.275 = 61.314$ m (permissible 113.058) — safe

Residual pressure = $82.547 \times 4.092 = 337.818$ m (permissible 557.373) — safe

Detention time = $81.647 \times 6.950 = 567.480$ hr (permissible 679.535) — safe

Discharge = $16.582 \times 7.365 = 122.122$ lps (permissible 216.999) — safe

Residual pressure = $42.930 \times 1.819 = 78.095$ m (permissible 117.889) — safe

Head loss = $82.558 \times 11.077 = 914.525$ m (permissible 1135.715) — safe

Detention time = $13.587 \times 2.881 = 39.152$ hr (permissible 57.066) — safe

Discharge = $70.210 \times 10.725 = 752.972$ lps (permissible 1278.512) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

17.x Quality Management Plan — Working Papers — Sheet 10

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. ST-768, revision C.

18 OPERATIONS AND MAINTENANCE PLAN

The completed asset will be handed over to Assam Jal Board for operation and maintenance.

The annual recurring requirement for operation and maintenance is estimated at Rs. 9.80 crore. The source of this recurring provision is yet to be identified and will be worked out in consultation with the Finance Department.

18.x Operations And Maintenance Plan — Working Papers — Sheet 1

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. ST-584, revision A.

18.x Operations And Maintenance Plan — Working Papers — Sheet 2

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
4,356	600	DI	PN 6	988.7
11,753	250	PVC-O	PN 10	98.6
13,334	300	PVC-O	PN 10	948.0
57	150	MS	K-9	467.7
7,072	500	PVC-O	K-9	1659.4
1,917	250	PVC-O	K-9	1344.2
715	200	HDPE	K-9	768.6
227	400	DI	K-9	913.4
18,752	400	PVC-O	PN 10	757.4
17,079	300	DI	PN 6	1446.2
760	600	HDPE	PN 10	1290.9
6,084	100	DI	PN 6	734.6
2,360	150	PVC-O	K-9	215.6
23,939	400	PVC-O	K-9	1241.4
17,592	500	PVC-O	PN 10	1468.3
18,774	250	MS	PN 10	60.7
10,978	400	MS	PN 10	1564.0
4,008	250	HDPE	PN 16	950.5
18,720	400	MS	PN 16	1126.3
12,656	150	MS	K-7	1071.8
17,635	100	DI	K-7	1310.1
5,450	300	PVC-O	K-7	733.6
7,262	400	MS	PN 6	63.3
23,069	100	HDPE	K-9	947.4
2,051	500	PVC-O	PN 10	1110.3
6,507	500	PVC-O	K-7	244.9

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

18.x Operations And Maintenance Plan — Working Papers — Sheet 3

Design Computation

Velocity = $63.009 \times 0.562 = 35.427$ m/s (permissible 47.725) — safe

Residual pressure = $77.715 \times 2.567 = 199.460$ m (permissible 212.555) — safe

Velocity = $47.698 \times 0.469 = 22.368$ m/s (permissible 26.232) — safe

Head loss = $21.376 \times 7.189 = 153.676$ m (permissible 251.961) — safe

Velocity = $30.648 \times 2.800 = 85.807$ m/s (permissible 133.757) — revise section

Head loss = $31.310 \times 5.399 = 169.035$ m (permissible 282.600) — safe

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

18.x Operations And Maintenance Plan — Working Papers — Sheet 4

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. PL-139, revision B.

18.x Operations And Maintenance Plan — Working Papers — Sheet 5

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	215.4	0.29	1
Iron	mg/l	0.72	0.06	1.0
Total coliform	MPN/100ml	428	Absent	Absent
pH	-	6.8	7.3	6.5-8.5
Residual chlorine	mg/l	Nil	0.35	0.2 min
Residual chlorine	mg/l	Nil	0.37	0.2 min
Iron	mg/l	2.64	0.22	1.0
Turbidity	NTU	236.7	0.22	1
Total hardness	mg/l	155	71	200
Residual chlorine	mg/l	Nil	0.56	0.2 min
Turbidity	NTU	78.4	0.48	1
Total coliform	MPN/100ml	248	Absent	Absent
Residual chlorine	mg/l	Nil	0.40	0.2 min
Turbidity	NTU	68.8	0.47	1
Turbidity	NTU	211.7	0.44	1
Iron	mg/l	2.75	0.13	1.0
pH	-	6.6	7.1	6.5-8.5
Total coliform	MPN/100ml	548	Absent	Absent
Iron	mg/l	1.24	0.17	1.0
Turbidity	NTU	205.7	0.46	1
pH	-	8.2	7.0	6.5-8.5
Total coliform	MPN/100ml	868	Absent	Absent
Total hardness	mg/l	126	166	200
Total hardness	mg/l	395	165	200

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

18.x Operations And Maintenance Plan — Working Papers — Sheet 6

Design Computation

Head loss = $16.132 \times 4.493 = 72.484$ m (permissible 122.901) — revise section

Head loss = $15.557 \times 9.334 = 145.200$ m (permissible 244.005) — safe

Head loss = $76.723 \times 2.238 = 171.686$ m (permissible 215.534) — safe

Head loss = $6.106 \times 3.526 = 21.530$ m (permissible 27.138) — safe

Velocity = $49.844 \times 2.296 = 114.423$ m/s (permissible 196.168) — safe

Velocity = $44.735 \times 5.779 = 258.533$ m/s (permissible 361.788) — safe

Detention time = $26.455 \times 10.350 = 273.792$ hr (permissible 288.924) — safe

Discharge = $49.977 \times 3.207 = 160.272$ lps (permissible 181.754) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

18.x Operations And Maintenance Plan — Working Papers — Sheet 7

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. ST-923, revision A.

18.x Operations And Maintenance Plan — Working Papers — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
5.7	Medium sand	28	SPT
1.7	Gravelly sand	30	UDS
29.1	Coarse sand	44	SPT
3.8	Clayey silt	25	DS
29.5	Clayey silt	42	SPT
11.8	Coarse sand	48	UDS
0.9	Weathered rock	38	UDS
19.3	Weathered rock	18	SPT
1.6	Weathered rock	29	SPT
24.0	Weathered rock	40	UDS
22.1	Gravelly sand	9	UDS
23.9	Coarse sand	40	SPT
25.2	Clayey silt	51	UDS
25.3	Weathered rock	49	SPT
15.4	Weathered rock	53	SPT
5.6	Weathered rock	38	UDS
8.5	Medium sand	10	SPT

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

18.x Operations And Maintenance Plan — Working Papers — Sheet 9

Design Computation

Discharge = $72.793 \times 5.028 = 365.990$ lps (permissible 395.090) — safe

Discharge = $81.309 \times 10.810 = 878.938$ lps (permissible 1606.874) — safe

Velocity = $22.379 \times 0.836 = 18.708$ m/s (permissible 22.876) — safe

Head loss = $58.790 \times 11.411 = 670.857$ m (permissible 974.935) — safe

Detention time = $2.468 \times 4.586 = 11.318$ hr (permissible 20.331) — safe

Detention time = $9.794 \times 3.089 = 30.256$ hr (permissible 45.989) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

18.x Operations And Maintenance Plan — Working Papers — Sheet 10

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Reference: drawing no. WS-799, revision A.

ANNEXURE I — KEY MAP AND SERVICE AREA PLAN

Service area, intake location and reservoir zones. Drawing WS-001.

ANNEXURE I — Sheet 1

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Reference: drawing no. HY-434, revision C.

ANNEXURE I — Sheet 2

Bore Log Summary

Depth (m)	Strata	N value	Sample
14.4	Coarse sand	32	SPT
26.0	Clayey silt	34	UDS
27.7	Weathered rock	7	DS
7.3	Gravelly sand	16	UDS
9.3	Medium sand	39	SPT
28.4	Gravelly sand	52	SPT
20.1	Silty clay	40	UDS
22.9	Gravelly sand	30	DS
16.4	Clayey silt	47	DS
9.2	Silty clay	52	DS
9.7	Coarse sand	39	SPT
6.0	Weathered rock	32	DS
1.7	Coarse sand	4	UDS
12.7	Gravelly sand	19	DS
0.9	Gravelly sand	41	DS
9.9	Gravelly sand	21	SPT
27.8	Coarse sand	41	UDS
23.3	Weathered rock	31	DS
9.1	Clayey silt	57	UDS
2.7	Clayey silt	59	DS
5.8	Silty clay	17	UDS
16.3	Coarse sand	35	SPT
0.6	Medium sand	51	SPT
23.3	Clayey silt	59	DS
11.0	Weathered rock	14	UDS
5.3	Silty clay	11	UDS

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE I — Sheet 3

Design Computation

Head loss = $79.822 \times 5.431 = 433.542$ m (permissible 742.414) — safe

Discharge = $57.021 \times 9.666 = 551.157$ lps (permissible 674.939) — safe

Velocity = $88.585 \times 5.318 = 471.051$ m/s (permissible 824.545) — safe

Velocity = $41.656 \times 5.337 = 222.307$ m/s (permissible 359.854) — safe

Detention time = $58.204 \times 3.175 = 184.810$ hr (permissible 200.353) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

ANNEXURE I — Sheet 4

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Reference: drawing no. HY-904, revision C.

ANNEXURE I — Sheet 5

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Total hardness	mg/l	166	97	200
pH	-	6.4	8.0	6.5-8.5
Turbidity	NTU	130.9	0.26	1
Residual chlorine	mg/l	Nil	0.38	0.2 min
Total coliform	MPN/100ml	629	Absent	Absent
pH	-	7.3	6.9	6.5-8.5
Iron	mg/l	3.86	0.26	1.0
Residual chlorine	mg/l	Nil	0.35	0.2 min
Total coliform	MPN/100ml	241	Absent	Absent
Residual chlorine	mg/l	Nil	0.42	0.2 min
Total coliform	MPN/100ml	860	Absent	Absent
Iron	mg/l	2.83	0.24	1.0
Total coliform	MPN/100ml	727	Absent	Absent
Turbidity	NTU	100.7	0.41	1
Total coliform	MPN/100ml	626	Absent	Absent
Residual chlorine	mg/l	Nil	0.47	0.2 min
pH	-	6.5	7.0	6.5-8.5
Turbidity	NTU	123.0	0.56	1
Total coliform	MPN/100ml	114	Absent	Absent
pH	-	8.3	7.8	6.5-8.5

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

ANNEXURE I — Sheet 6

Design Computation

Head loss = $21.083 \times 11.217 = 236.478$ m (permissible 284.236) — safe

Discharge = $17.065 \times 7.713 = 131.620$ lps (permissible 185.561) — safe

Velocity = $33.818 \times 1.514 = 51.203$ m/s (permissible 75.569) — safe

Residual pressure = $81.298 \times 5.213 = 423.801$ m (permissible 712.243) — safe

Velocity = $50.417 \times 9.504 = 479.162$ m/s (permissible 905.217) — safe

Head loss = $9.198 \times 7.689 = 70.724$ m (permissible 126.512) — safe

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

ANNEXURE I — Sheet 7

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. HY-234, revision B.

ANNEXURE I — Sheet 8

Bore Log Summary

Depth (m)	Strata	N value	Sample
4.7	Medium sand	50	SPT
8.5	Coarse sand	49	DS
15.1	Silty clay	52	SPT
24.9	Medium sand	41	SPT
29.8	Gravelly sand	15	UDS
6.1	Weathered rock	37	UDS
10.2	Weathered rock	35	SPT
16.4	Gravelly sand	29	SPT
28.5	Silty clay	59	DS
28.5	Weathered rock	9	UDS
3.7	Clayey silt	45	SPT
21.1	Coarse sand	55	DS
28.4	Gravelly sand	12	DS
18.8	Weathered rock	27	SPT
9.3	Gravelly sand	52	DS
13.5	Clayey silt	55	SPT
10.0	Coarse sand	5	SPT
16.1	Medium sand	33	DS
27.5	Gravelly sand	59	UDS
14.4	Clayey silt	20	UDS
15.4	Silty clay	38	SPT
23.5	Gravelly sand	16	UDS
12.7	Silty clay	26	SPT
3.3	Clayey silt	26	SPT

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

ANNEXURE II — HYDRAULIC NETWORK ANALYSIS

EPANET output for the design year under peak and fire flow conditions.

ANNEXURE II — Water Quality Test Results, page 1

Parameter	Unit	Raw water	Treated	IS 10500 limit
pH	-	6.8	7.7	6.5-8.5
Residual chlorine	mg/l	Nil	0.70	0.2 min
Turbidity	NTU	16.5	0.37	1
pH	-	6.8	8.0	6.5-8.5
Iron	mg/l	1.32	0.11	1.0
Turbidity	NTU	98.8	0.73	1
Residual chlorine	mg/l	Nil	0.61	0.2 min
Residual chlorine	mg/l	Nil	0.30	0.2 min
Turbidity	NTU	158.8	0.53	1
Iron	mg/l	2.18	0.03	1.0
Turbidity	NTU	123.9	0.75	1
Iron	mg/l	3.71	0.20	1.0
Total coliform	MPN/100ml	461	Absent	Absent
Total hardness	mg/l	207	124	200
pH	-	7.2	7.5	6.5-8.5
Iron	mg/l	1.14	0.07	1.0
Turbidity	NTU	12.8	0.23	1
Total hardness	mg/l	318	110	200
Residual chlorine	mg/l	Nil	0.41	0.2 min
pH	-	8.1	7.7	6.5-8.5
Iron	mg/l	2.77	0.06	1.0
pH	-	7.2	7.2	6.5-8.5
Turbidity	NTU	232.6	0.79	1
Residual chlorine	mg/l	Nil	0.64	0.2 min
Iron	mg/l	3.58	0.07	1.0
Total hardness	mg/l	425	175	200
Total coliform	MPN/100ml	63	Absent	Absent
Total hardness	mg/l	150	91	200

ANNEXURE II — Bore Log Summary, page 2

Depth (m)	Strata	N value	Sample
18.2	Coarse sand	30	DS
2.4	Coarse sand	15	UDS
7.3	Gravelly sand	49	UDS
2.4	Coarse sand	59	DS
1.2	Silty clay	8	DS
8.0	Weathered rock	49	SPT
26.8	Clayey silt	17	DS
2.5	Gravelly sand	23	UDS
1.0	Weathered rock	19	SPT
7.5	Clayey silt	13	UDS
15.6	Weathered rock	54	UDS
2.0	Weathered rock	55	DS
19.2	Medium sand	41	SPT
29.9	Medium sand	27	DS
1.2	Coarse sand	39	UDS
6.6	Coarse sand	45	DS
29.7	Clayey silt	29	UDS
3.5	Gravelly sand	12	UDS
13.6	Clayey silt	28	SPT
25.8	Weathered rock	39	SPT
22.3	Gravelly sand	7	SPT
20.5	Silty clay	6	DS

ANNEXURE II — Population Projection, page 3

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2036	Ward 16	42,740	100	4.274
2046	Ward 16	21,428	135	2.893
2046	Ward 16	30,059	70	2.104
2026	Ward 5	5,193	150	0.779
2036	Ward 13	19,589	100	1.959
2036	Ward 19	31,790	150	4.769
2056	Ward 24	24,725	100	2.473
2026	Ward 11	45,438	150	6.816
2041	Ward 24	43,250	100	4.325
2051	Ward 19	40,140	70	2.810
2046	Ward 16	7,895	135	1.066
2026	Ward 21	17,129	70	1.199
2056	Ward 13	15,894	135	2.146
2056	Ward 22	38,951	100	3.895
2031	Ward 6	28,347	70	1.984
2031	Ward 17	38,838	135	5.243
2051	Ward 20	40,833	100	4.083
2041	Ward 5	39,443	150	5.916
2041	Ward 13	36,426	150	5.464
2036	Ward 16	16,100	70	1.127
2056	Ward 1	45,451	150	6.818
2041	Ward 10	35,411	150	5.312
2056	Ward 23	7,847	150	1.177
2046	Ward 18	2,321	70	0.162
2046	Ward 19	43,746	150	6.562
2041	Ward 24	12,102	70	0.847
2031	Ward 15	35,589	150	5.338
2026	Ward 14	9,233	150	1.385

ANNEXURE II — Pipeline Schedule, page 4

Chainage (m)	Dia (mm)	Material	Class	Length (m)
8,025	600	MS	PN 16	1093.7
21,429	400	HDPE	PN 6	1100.9
8,597	250	HDPE	PN 6	335.6
14,183	500	HDPE	K-7	1789.7
5,207	500	DI	PN 10	679.9
7,551	500	DI	K-7	671.4
10,886	300	HDPE	PN 16	1401.5
6,757	200	MS	PN 16	1053.7
19,544	300	DI	PN 6	944.4
23,075	100	HDPE	PN 16	1445.8
536	300	PVC-O	K-7	1307.0
13,857	600	DI	K-7	1386.1
17,682	300	HDPE	PN 16	841.8
11,222	100	MS	PN 16	1218.7
9,868	250	PVC-O	PN 6	1382.2
23,675	100	DI	K-7	175.9
21,982	150	DI	PN 16	866.3
6,569	300	HDPE	PN 10	172.1
3,242	250	HDPE	K-9	1423.2
23,988	250	PVC-O	K-9	94.4
11,524	150	HDPE	K-9	1565.0
12,461	250	HDPE	PN 6	247.8

ANNEXURE II — Hydraulic Node Analysis, page 5

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-142	33.32	22.97	47.34	14.02
N-098	62.32	9.16	83.90	21.58
N-200	64.01	12.59	80.18	16.17
N-093	32.10	1.70	48.57	16.47
N-109	37.26	17.93	53.77	16.52
N-207	48.96	11.75	67.46	18.50
N-176	31.74	22.02	54.31	22.56
N-026	86.16	18.60	93.45	7.29
N-060	44.56	2.60	52.14	7.58
N-126	45.57	6.62	70.42	24.85
N-120	72.61	13.25	84.45	11.84
N-186	57.31	19.46	73.38	16.06
N-213	51.53	9.98	74.79	23.25
N-088	88.70	3.17	107.35	18.66
N-046	86.28	14.05	99.47	13.20
N-209	62.66	8.38	71.84	9.18
N-039	50.42	13.65	68.92	18.51
N-018	79.16	10.87	97.49	18.33
N-079	89.83	15.46	97.53	7.71
N-173	93.12	12.69	116.51	23.39
N-176	64.96	16.08	72.92	7.96
N-147	65.96	13.46	89.50	23.54

ANNEXURE II — Pump Schedule, page 6

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-10	Sludge	264.2	33.9	122.01
P-12	Raw water	312.9	45.6	194.23
P-05	Backwash	184.7	70.3	177.01
P-09	Raw water	238.1	39.0	126.46
P-08	Clear water	257.1	84.6	296.33
P-11	Backwash	157.0	22.3	47.60
P-12	Raw water	210.5	27.0	77.59
P-08	Sludge	202.0	54.8	150.80
P-12	Backwash	207.8	42.6	120.64
P-06	Clear water	102.3	55.2	76.96
P-11	Raw water	240.1	67.1	219.45
P-04	Booster	233.0	35.1	111.41
P-10	Raw water	50.2	35.0	23.92
P-10	Sludge	87.8	63.2	75.63
P-09	Clear water	144.0	30.1	59.04
P-09	Backwash	133.6	23.8	43.39
P-06	Sludge	285.8	51.3	199.63
P-02	Raw water	257.2	72.7	254.73
P-06	Sludge	200.9	78.0	213.63
P-10	Clear water	317.7	55.4	239.60
P-09	Booster	164.3	87.8	196.57
P-03	Booster	99.2	14.2	19.20
P-03	Clear water	100.4	43.6	59.66
P-01	Backwash	114.6	39.1	61.06

ANNEXURE II — Water Quality Test Results, page 7

Parameter	Unit	Raw water	Treated	IS 10500 limit
Residual chlorine	mg/l	Nil	0.71	0.2 min
Turbidity	NTU	147.1	0.37	1
Total hardness	mg/l	477	185	200
Total hardness	mg/l	250	177	200
pH	-	8.4	7.9	6.5-8.5
Turbidity	NTU	51.2	0.54	1
pH	-	7.3	7.2	6.5-8.5
Residual chlorine	mg/l	Nil	0.50	0.2 min
pH	-	6.8	7.4	6.5-8.5
pH	-	7.0	7.6	6.5-8.5
Total hardness	mg/l	194	99	200
Total hardness	mg/l	291	134	200
Iron	mg/l	3.15	0.14	1.0
Total coliform	MPN/100ml	877	Absent	Absent
Total hardness	mg/l	326	189	200
Iron	mg/l	0.80	0.02	1.0
Total hardness	mg/l	203	112	200
Turbidity	NTU	38.0	0.71	1
pH	-	6.5	8.1	6.5-8.5
Total coliform	MPN/100ml	691	Absent	Absent
Iron	mg/l	2.64	0.23	1.0
Total coliform	MPN/100ml	192	Absent	Absent
Turbidity	NTU	10.0	0.48	1

ANNEXURE II — Bore Log Summary, page 8

Depth (m)	Strata	N value	Sample
27.2	Clayey silt	49	UDS
24.0	Medium sand	59	SPT
13.5	Coarse sand	52	UDS
19.4	Medium sand	34	SPT
15.8	Clayey silt	59	SPT
23.5	Clayey silt	10	DS
1.5	Weathered rock	4	UDS
7.1	Medium sand	16	SPT
27.5	Silty clay	41	DS
7.6	Weathered rock	60	DS
23.8	Coarse sand	17	SPT
10.2	Coarse sand	49	DS
18.0	Silty clay	12	SPT
8.4	Weathered rock	32	DS
13.6	Gravelly sand	38	DS
26.0	Weathered rock	38	UDS
16.6	Silty clay	18	DS
28.5	Coarse sand	38	DS
7.2	Weathered rock	20	DS
22.1	Silty clay	10	DS
13.9	Silty clay	16	UDS
14.0	Clayey silt	8	SPT
0.5	Weathered rock	45	SPT
2.5	Clayey silt	6	SPT
26.5	Gravelly sand	9	SPT
5.3	Medium sand	49	UDS
7.0	Silty clay	37	SPT
13.2	Coarse sand	33	UDS
4.1	Gravelly sand	56	UDS

ANNEXURE II — Population Projection, page 9

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2041	Ward 7	12,532	150	1.880
2046	Ward 12	22,350	150	3.353
2046	Ward 10	45,876	70	3.211
2031	Ward 13	9,542	100	0.954
2046	Ward 18	42,764	150	6.415
2041	Ward 3	15,548	135	2.099
2046	Ward 11	45,536	100	4.554
2036	Ward 8	43,023	150	6.453
2036	Ward 24	15,551	100	1.555
2036	Ward 1	39,149	100	3.915
2036	Ward 23	13,897	70	0.973
2036	Ward 5	22,717	150	3.408
2036	Ward 23	15,063	150	2.259
2036	Ward 14	18,524	135	2.501
2036	Ward 3	22,398	100	2.240
2026	Ward 15	43,722	135	5.902
2026	Ward 10	36,144	70	2.530
2051	Ward 21	36,727	150	5.509
2056	Ward 18	42,399	135	5.724
2031	Ward 1	43,525	100	4.353
2026	Ward 1	23,783	100	2.378
2041	Ward 8	19,028	150	2.854
2056	Ward 23	42,953	150	6.443
2026	Ward 11	35,744	70	2.502
2026	Ward 24	11,085	100	1.109
2036	Ward 10	4,293	150	0.644
2036	Ward 10	14,081	150	2.112
2046	Ward 7	22,505	135	3.038
2026	Ward 11	24,608	100	2.461

ANNEXURE II — Pipeline Schedule, page 10

Chainage (m)	Dia (mm)	Material	Class	Length (m)
6,401	250	MS	PN 6	643.3
6,063	250	HDPE	K-7	1585.7
11,034	400	PVC-O	K-7	507.4
4,862	100	HDPE	K-9	1131.9
14,396	200	DI	K-7	296.7
8,193	100	HDPE	PN 10	1517.9
2,793	200	PVC-O	PN 16	937.8
12,085	600	DI	PN 10	307.7
10,514	250	MS	PN 6	1371.7
11,783	500	MS	PN 6	1554.1
13,288	400	PVC-O	PN 6	800.9
15,539	500	HDPE	PN 6	1561.4
8,519	200	HDPE	K-7	90.9
5,333	500	DI	PN 6	1024.7
23,782	500	DI	PN 10	218.6
17,767	600	MS	PN 16	1793.1
371	250	MS	PN 6	1005.9
22,146	400	PVC-O	K-9	1237.8
5,309	400	HDPE	PN 6	272.9
8,020	400	PVC-O	PN 16	688.4
984	600	PVC-O	K-9	199.6
23,997	200	HDPE	PN 10	1675.7
9,811	250	PVC-O	PN 6	826.6
11,483	100	PVC-O	PN 16	418.2
404	500	DI	PN 6	1156.4
14,434	300	DI	PN 10	498.2

ANNEXURE II — Hydraulic Node Analysis, page 11

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-167	51.66	2.16	71.27	19.61
N-040	80.53	17.33	92.73	12.20
N-211	33.99	11.06	47.36	13.36
N-001	64.26	13.12	86.88	22.61
N-207	60.01	2.56	79.85	19.84
N-142	64.56	21.52	82.14	17.58
N-168	53.43	21.00	61.13	7.70
N-008	43.12	12.97	54.39	11.27
N-102	87.18	6.39	105.75	18.57
N-024	78.15	2.02	94.03	15.88
N-174	90.64	14.31	111.61	20.97
N-216	79.87	10.67	102.01	22.13
N-216	57.46	17.61	64.64	7.18
N-171	34.61	6.80	57.28	22.67
N-034	89.92	20.10	113.16	23.24
N-197	75.13	4.78	94.68	19.56
N-047	28.81	4.16	48.38	19.57
N-218	87.51	7.79	100.57	13.06
N-050	41.13	3.08	53.31	12.18
N-211	64.66	20.37	75.32	10.66
N-052	31.13	22.84	46.07	14.93
N-081	48.82	4.03	64.99	16.18
N-119	53.99	6.83	64.48	10.49
N-154	41.18	22.20	60.89	19.71
N-163	66.35	7.97	75.70	9.35
N-210	53.54	16.75	78.34	24.80
N-173	91.92	21.72	116.39	24.48

ANNEXURE II — Pump Schedule, page 12

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-07	Clear water	298.5	87.6	356.31
P-11	Sludge	311.7	36.3	154.02
P-05	Clear water	97.5	74.4	98.94
P-06	Backwash	40.9	31.5	17.57
P-11	Sludge	169.6	56.0	129.38
P-07	Backwash	189.2	45.7	117.91
P-07	Clear water	188.1	29.6	75.89
P-02	Booster	301.0	47.6	195.04
P-10	Backwash	83.9	34.6	39.56
P-05	Raw water	278.0	85.1	322.19
P-06	Raw water	36.3	29.7	14.67
P-08	Sludge	215.8	24.3	71.48
P-08	Raw water	256.4	73.1	255.51
P-06	Booster	234.4	64.3	205.22
P-11	Backwash	179.5	58.0	141.80
P-12	Sludge	119.3	56.6	91.97
P-04	Booster	84.6	32.3	37.20
P-11	Sludge	269.6	22.4	82.23
P-04	Backwash	163.1	75.0	166.73
P-11	Clear water	269.4	87.9	322.44
P-10	Sludge	264.9	78.8	284.58
P-02	Booster	63.7	19.9	17.23
P-11	Clear water	258.1	38.2	134.28
P-04	Clear water	199.4	43.8	119.09
P-08	Sludge	309.2	70.5	296.85
P-03	Sludge	78.9	65.8	70.77
P-10	Backwash	238.5	34.4	111.67

ANNEXURE II — Water Quality Test Results, page 13

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	85.4	0.89	1
Iron	mg/l	0.96	0.07	1.0
Total hardness	mg/l	336	185	200
pH	-	8.2	7.9	6.5-8.5
Residual chlorine	mg/l	Nil	0.43	0.2 min
Total coliform	MPN/100ml	791	Absent	Absent
Total hardness	mg/l	438	61	200
pH	-	7.5	7.5	6.5-8.5
Turbidity	NTU	132.3	0.68	1
Residual chlorine	mg/l	Nil	0.55	0.2 min
Iron	mg/l	3.23	0.27	1.0
Total hardness	mg/l	355	147	200
Iron	mg/l	1.16	0.23	1.0
pH	-	7.7	7.8	6.5-8.5
Total hardness	mg/l	253	139	200
Turbidity	NTU	229.2	0.69	1
Turbidity	NTU	159.9	0.31	1
Iron	mg/l	4.18	0.21	1.0
pH	-	8.1	7.7	6.5-8.5
Turbidity	NTU	167.6	0.42	1
Total coliform	MPN/100ml	797	Absent	Absent
Turbidity	NTU	112.0	0.43	1
Total coliform	MPN/100ml	570	Absent	Absent

ANNEXURE II — Bore Log Summary, page 14

Depth (m)	Strata	N value	Sample
12.3	Weathered rock	47	UDS
6.7	Weathered rock	25	SPT
4.3	Gravelly sand	7	UDS
28.5	Gravelly sand	19	SPT
0.8	Coarse sand	27	DS
1.3	Silty clay	31	DS
13.6	Weathered rock	21	UDS
5.0	Clayey silt	54	DS
3.6	Silty clay	41	UDS
19.9	Medium sand	52	UDS
27.3	Weathered rock	24	SPT
15.2	Gravelly sand	23	SPT
26.0	Gravelly sand	27	DS
17.9	Weathered rock	22	UDS
6.0	Weathered rock	23	SPT
28.0	Silty clay	26	UDS
18.4	Silty clay	43	UDS
20.9	Weathered rock	49	DS
7.0	Clayey silt	5	DS
11.0	Gravelly sand	23	UDS
23.9	Coarse sand	30	SPT
24.3	Weathered rock	30	DS
23.9	Gravelly sand	4	SPT
12.2	Weathered rock	43	UDS
25.5	Coarse sand	22	UDS
8.2	Medium sand	16	UDS
21.3	Medium sand	27	UDS

ANNEXURE II — Population Projection, page 15

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2026	Ward 20	39,015	150	5.852
2046	Ward 9	44,949	135	6.068
2051	Ward 8	42,547	70	2.978
2056	Ward 5	23,885	70	1.672
2031	Ward 16	28,843	70	2.019
2051	Ward 9	29,477	150	4.422
2051	Ward 22	29,386	135	3.967
2031	Ward 17	17,522	70	1.227
2031	Ward 8	37,085	135	5.006
2051	Ward 8	30,552	150	4.583
2056	Ward 7	26,036	150	3.905
2046	Ward 9	14,722	100	1.472
2041	Ward 6	18,440	70	1.291
2051	Ward 4	35,446	150	5.317
2051	Ward 11	19,253	100	1.925
2056	Ward 6	21,109	150	3.166
2046	Ward 1	2,871	70	0.201
2056	Ward 20	21,615	100	2.162
2026	Ward 9	26,433	150	3.965
2056	Ward 20	19,700	100	1.970
2036	Ward 11	26,966	70	1.888
2036	Ward 23	35,110	135	4.740
2041	Ward 17	42,296	150	6.344

ANNEXURE II — Pipeline Schedule, page 16

Chainage (m)	Dia (mm)	Material	Class	Length (m)
19,915	600	MS	K-9	1248.7
15,649	150	DI	PN 10	1470.8
1,319	100	MS	PN 6	1799.6
17,956	100	PVC-O	K-7	1560.3
481	300	PVC-O	PN 6	84.4
225	500	HDPE	PN 10	1040.8
16,173	150	DI	PN 16	409.8
21,827	250	DI	K-7	402.1
13,172	500	DI	PN 6	1181.6
1,999	200	DI	PN 16	1173.5
14,984	100	PVC-O	PN 10	470.3
23,425	300	MS	PN 16	1082.8
11,452	300	HDPE	K-9	416.6
11,801	600	DI	PN 6	1131.8
22,470	150	PVC-O	K-7	1322.4
3,282	600	HDPE	PN 6	1469.6
23,551	300	HDPE	PN 10	511.2
12,657	200	HDPE	K-9	115.5
13,147	300	HDPE	PN 6	814.5
23,258	100	DI	PN 10	506.9
3,216	400	PVC-O	PN 10	392.0
16,745	400	DI	K-9	136.9
19,404	500	DI	PN 10	654.1

ANNEXURE II — Hydraulic Node Analysis, page 17

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-075	58.89	2.22	77.43	18.54
N-033	90.18	15.64	108.13	17.95
N-058	74.80	21.91	84.97	10.17
N-112	63.29	3.30	80.83	17.54
N-007	81.31	0.42	89.72	8.41
N-161	43.14	9.99	52.49	9.34
N-029	42.86	23.23	67.36	24.49
N-033	83.95	11.39	105.67	21.72
N-171	52.08	22.11	62.17	10.09
N-171	51.30	19.59	73.69	22.39
N-164	55.76	12.06	73.25	17.49
N-031	45.26	18.65	55.32	10.05
N-132	95.05	2.63	103.51	8.46
N-140	29.34	22.74	38.68	9.33
N-013	29.07	21.13	44.47	15.40
N-014	59.61	6.68	83.42	23.81
N-102	42.72	7.99	67.21	24.50
N-170	53.56	11.01	67.91	14.35
N-173	53.19	0.55	67.60	14.41
N-175	83.00	3.75	95.45	12.46
N-133	69.05	10.74	90.04	21.00
N-204	66.60	12.90	82.59	15.99
N-062	87.22	6.96	99.97	12.75
N-033	30.62	13.70	50.54	19.92
N-163	34.83	11.38	43.98	9.15
N-061	54.19	10.32	69.07	14.88
N-063	82.92	5.70	105.86	22.94
N-098	66.30	21.17	73.85	7.54

ANNEXURE II — Pump Schedule, page 18

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-05	Backwash	282.4	54.4	209.28
P-03	Clear water	87.2	35.3	41.90
P-10	Backwash	201.9	75.3	207.10
P-06	Backwash	209.6	66.8	190.82
P-08	Backwash	140.0	38.4	73.32
P-08	Raw water	281.5	43.0	165.06
P-05	Clear water	255.3	40.6	141.28
P-10	Raw water	263.7	25.7	92.23
P-11	Sludge	135.5	74.7	137.92
P-01	Sludge	233.5	55.6	176.92
P-09	Sludge	186.7	87.1	221.46
P-03	Booster	202.7	39.4	108.80
P-10	Booster	205.7	53.1	148.65
P-12	Sludge	262.6	64.1	229.22
P-04	Backwash	318.0	32.0	138.44
P-11	Clear water	164.2	23.7	52.93
P-03	Booster	213.3	83.0	241.14
P-04	Clear water	83.4	72.5	82.41
P-08	Sludge	238.3	41.2	133.79
P-07	Sludge	105.3	87.7	125.73
P-03	Backwash	291.3	30.5	121.17
P-06	Backwash	254.4	13.3	46.26

ANNEXURE II — Water Quality Test Results, page 19

Parameter	Unit	Raw water	Treated	IS 10500 limit
Iron	mg/l	2.97	0.20	1.0
Turbidity	NTU	62.6	0.73	1
pH	-	7.3	6.9	6.5-8.5
Total hardness	mg/l	212	170	200
Total coliform	MPN/100ml	331	Absent	Absent
Total coliform	MPN/100ml	404	Absent	Absent
pH	-	6.6	6.9	6.5-8.5
Total coliform	MPN/100ml	547	Absent	Absent
Iron	mg/l	3.10	0.10	1.0
Total coliform	MPN/100ml	155	Absent	Absent
Turbidity	NTU	141.1	0.23	1
Iron	mg/l	2.46	0.03	1.0
Turbidity	NTU	231.1	0.83	1
Iron	mg/l	3.39	0.23	1.0
Total hardness	mg/l	173	153	200
Residual chlorine	mg/l	Nil	0.32	0.2 min
Residual chlorine	mg/l	Nil	0.45	0.2 min
Residual chlorine	mg/l	Nil	0.43	0.2 min
Turbidity	NTU	111.5	0.62	1
Residual chlorine	mg/l	Nil	0.20	0.2 min
Iron	mg/l	2.61	0.04	1.0
Turbidity	NTU	115.2	0.23	1
pH	-	8.0	7.1	6.5-8.5
Total coliform	MPN/100ml	491	Absent	Absent
Total hardness	mg/l	361	66	200
Residual chlorine	mg/l	Nil	0.22	0.2 min
Turbidity	NTU	209.7	0.35	1

ANNEXURE II — Bore Log Summary, page 20

Depth (m)	Strata	N value	Sample
23.0	Weathered rock	29	UDS
5.7	Gravelly sand	60	DS
15.3	Clayey silt	54	UDS
17.7	Weathered rock	31	SPT
28.7	Gravelly sand	6	DS
5.3	Clayey silt	58	DS
27.4	Gravelly sand	24	DS
7.6	Medium sand	55	UDS
29.4	Gravelly sand	60	SPT
15.3	Silty clay	12	DS
9.4	Gravelly sand	58	UDS
5.2	Coarse sand	5	DS
16.9	Silty clay	59	SPT
6.1	Medium sand	31	SPT
15.3	Coarse sand	44	SPT
18.4	Gravelly sand	51	UDS
25.3	Clayey silt	16	UDS
22.3	Coarse sand	47	DS
17.2	Weathered rock	37	UDS
14.3	Clayey silt	9	SPT
2.9	Weathered rock	35	DS
11.5	Weathered rock	32	DS
25.6	Coarse sand	29	SPT
21.1	Gravelly sand	13	DS
15.3	Gravelly sand	10	SPT
26.1	Weathered rock	38	DS
25.5	Clayey silt	44	UDS
14.0	Weathered rock	41	SPT

ANNEXURE II — Population Projection, page 21

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2031	Ward 12	6,001	135	0.810
2031	Ward 22	7,296	135	0.985
2051	Ward 22	14,634	100	1.463
2036	Ward 24	2,358	70	0.165
2026	Ward 2	41,895	100	4.189
2026	Ward 24	21,091	100	2.109
2041	Ward 24	15,716	100	1.572
2036	Ward 2	16,149	135	2.180
2056	Ward 21	26,429	70	1.850
2031	Ward 24	9,702	135	1.310
2046	Ward 16	39,152	135	5.286
2041	Ward 9	8,408	150	1.261
2031	Ward 22	41,301	70	2.891
2046	Ward 6	35,664	150	5.350
2046	Ward 16	42,291	100	4.229
2041	Ward 7	38,540	100	3.854
2056	Ward 21	28,524	100	2.852
2041	Ward 8	44,969	135	6.071
2046	Ward 24	32,897	150	4.935
2056	Ward 10	44,314	150	6.647
2036	Ward 19	39,234	135	5.297
2026	Ward 16	41,099	150	6.165
2026	Ward 15	40,429	150	6.064
2046	Ward 6	22,862	70	1.600
2036	Ward 22	45,481	150	6.822
2056	Ward 23	2,700	100	0.270
2026	Ward 4	16,302	100	1.630

ANNEXURE II — Pipeline Schedule, page 22

Chainage (m)	Dia (mm)	Material	Class	Length (m)
10,293	100	PVC-O	PN 6	865.0
21,185	250	DI	K-9	81.0
3,797	250	HDPE	K-9	672.8
79	600	MS	PN 6	1448.2
23,429	300	PVC-O	PN 10	1467.7
486	250	DI	PN 10	785.2
2,491	300	HDPE	K-9	1224.5
14,964	150	HDPE	PN 10	318.3
20,912	200	HDPE	K-7	983.9
4,728	600	DI	PN 6	216.3
7,020	500	DI	K-7	79.5
1,283	200	DI	PN 10	1189.1
18,103	300	MS	K-7	1772.0
12,775	100	DI	K-9	771.1
3,385	100	MS	K-9	1722.1
23,762	100	DI	PN 10	1239.1
6,293	150	HDPE	K-7	195.0
7,923	150	HDPE	PN 10	1756.2
21,954	600	HDPE	K-9	469.5
16,882	500	MS	PN 6	573.1
14,005	600	HDPE	PN 6	742.4
14,018	500	MS	PN 6	1146.6
3,375	100	MS	K-7	590.7
2,567	600	DI	PN 10	1160.4
13,500	400	HDPE	PN 10	179.3
20,720	200	DI	PN 16	826.1
2,722	100	DI	PN 10	714.7
8,229	600	MS	K-7	129.7

ANNEXURE II — Hydraulic Node Analysis, page 23

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-190	41.45	12.74	57.16	15.71
N-156	72.86	21.80	87.66	14.80
N-213	44.70	4.30	68.71	24.01
N-083	69.87	19.55	83.00	13.13
N-065	58.91	17.35	75.47	16.56
N-205	80.02	10.71	98.39	18.37
N-080	84.29	22.45	95.27	10.98
N-216	48.16	11.78	59.48	11.32
N-144	74.27	1.12	99.39	25.12
N-034	29.09	18.97	38.75	9.66
N-125	52.23	3.87	67.42	15.20
N-164	73.64	9.95	84.67	11.03
N-150	62.98	1.21	79.53	16.55
N-095	87.16	11.89	98.16	11.00
N-124	49.28	4.40	60.48	11.20
N-001	60.75	7.80	82.89	22.13
N-137	36.53	20.52	47.01	10.48
N-077	42.87	19.58	56.19	13.32
N-001	32.64	8.50	46.09	13.45
N-142	92.16	3.71	114.17	22.02
N-186	38.06	19.78	53.76	15.70
N-123	46.79	17.91	63.53	16.74
N-197	43.60	20.57	68.37	24.77
N-067	59.32	15.02	74.43	15.11
N-174	41.93	6.64	62.34	20.41
N-086	28.95	22.75	38.19	9.24
N-019	88.15	8.28	104.85	16.70

ANNEXURE II — Pump Schedule, page 24

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-03	Raw water	279.3	58.8	223.64
P-03	Clear water	22.9	51.8	16.18
P-07	Raw water	291.1	29.5	117.17
P-09	Backwash	266.3	69.1	250.66
P-09	Clear water	48.4	50.0	32.92
P-07	Sludge	15.6	67.3	14.33
P-03	Clear water	232.0	70.5	222.86
P-01	Backwash	290.6	52.7	208.62
P-01	Raw water	299.3	31.3	127.46
P-08	Booster	94.5	67.8	87.28
P-03	Backwash	181.3	74.3	183.47
P-11	Backwash	91.4	66.8	83.17
P-08	Clear water	295.3	38.5	154.94
P-07	Sludge	119.3	14.6	23.67
P-03	Raw water	274.4	46.3	172.99
P-08	Booster	121.2	45.6	75.35
P-09	Clear water	278.8	42.4	161.15
P-05	Clear water	34.9	79.5	37.84
P-02	Booster	208.5	46.3	131.55
P-05	Raw water	56.3	80.5	61.82
P-10	Sludge	225.6	40.5	124.58
P-05	Raw water	126.0	34.7	59.64

ANNEXURE II — Water Quality Test Results, page 25

Parameter	Unit	Raw water	Treated	IS 10500 limit
Iron	mg/l	3.01	0.05	1.0
Total coliform	MPN/100ml	723	Absent	Absent
Total hardness	mg/l	395	184	200
pH	-	7.9	7.4	6.5-8.5
Turbidity	NTU	50.2	0.27	1
Total coliform	MPN/100ml	809	Absent	Absent
Residual chlorine	mg/l	Nil	0.32	0.2 min
Iron	mg/l	3.65	0.04	1.0
Total hardness	mg/l	399	63	200
Iron	mg/l	1.27	0.24	1.0
pH	-	7.7	7.7	6.5-8.5
Residual chlorine	mg/l	Nil	0.49	0.2 min
Residual chlorine	mg/l	Nil	0.55	0.2 min
pH	-	6.8	7.5	6.5-8.5
Iron	mg/l	0.46	0.11	1.0
Turbidity	NTU	85.1	0.56	1
pH	-	7.4	7.0	6.5-8.5
Residual chlorine	mg/l	Nil	0.70	0.2 min
pH	-	8.3	7.6	6.5-8.5
Total coliform	MPN/100ml	262	Absent	Absent
Total hardness	mg/l	225	86	200
pH	-	6.5	8.1	6.5-8.5
Residual chlorine	mg/l	Nil	0.58	0.2 min
Residual chlorine	mg/l	Nil	0.30	0.2 min
Iron	mg/l	1.76	0.03	1.0
Turbidity	NTU	199.5	0.23	1
Turbidity	NTU	61.2	0.46	1
Turbidity	NTU	80.3	0.41	1
Total coliform	MPN/100ml	311	Absent	Absent
pH	-	6.7	6.9	6.5-8.5

ANNEXURE II — Bore Log Summary, page 26

Depth (m)	Strata	N value	Sample
17.7	Clayey silt	14	DS
3.1	Clayey silt	26	SPT
21.5	Medium sand	33	SPT
12.1	Gravelly sand	49	UDS
5.2	Silty clay	14	SPT
11.1	Silty clay	30	UDS
18.1	Clayey silt	19	UDS
7.6	Clayey silt	15	DS
4.9	Weathered rock	25	DS
19.8	Weathered rock	25	DS
13.4	Coarse sand	7	DS
23.5	Silty clay	38	UDS
10.1	Weathered rock	12	SPT
25.5	Gravelly sand	48	UDS
18.8	Medium sand	17	DS
28.8	Medium sand	37	DS
20.0	Clayey silt	45	DS
18.5	Silty clay	18	UDS
4.9	Gravelly sand	31	UDS
3.1	Gravelly sand	42	UDS
29.6	Silty clay	13	DS
0.9	Medium sand	23	SPT
22.9	Weathered rock	21	SPT
6.9	Silty clay	23	SPT
18.5	Coarse sand	36	SPT
13.5	Medium sand	33	DS

ANNEXURE II — Population Projection, page 27

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2031	Ward 13	34,222	150	5.133
2026	Ward 2	3,711	135	0.501
2031	Ward 23	4,911	135	0.663
2036	Ward 5	44,942	100	4.494
2031	Ward 7	24,390	70	1.707
2026	Ward 8	9,352	100	0.935
2036	Ward 12	12,434	135	1.679
2056	Ward 7	8,144	100	0.814
2056	Ward 18	27,419	135	3.702
2051	Ward 19	32,808	150	4.921
2051	Ward 19	23,585	100	2.358
2031	Ward 14	4,069	150	0.610
2041	Ward 20	18,441	70	1.291
2046	Ward 7	18,865	135	2.547
2031	Ward 19	22,712	135	3.066
2051	Ward 22	42,906	135	5.792
2036	Ward 19	16,673	150	2.501
2036	Ward 24	27,294	70	1.911
2041	Ward 2	21,580	100	2.158
2026	Ward 10	21,112	135	2.850
2036	Ward 7	9,524	150	1.429
2056	Ward 10	38,788	135	5.236
2026	Ward 21	22,261	70	1.558
2036	Ward 9	15,847	135	2.139
2041	Ward 3	15,306	70	1.071
2031	Ward 1	13,855	150	2.078
2031	Ward 4	6,123	150	0.918
2031	Ward 5	25,717	100	2.572
2041	Ward 9	18,246	150	2.737

ANNEXURE II — Pipeline Schedule, page 28

Chainage (m)	Dia (mm)	Material	Class	Length (m)
23,558	400	HDPE	PN 6	1403.8
21,842	150	PVC-O	PN 16	635.8
12,423	250	DI	K-7	1514.9
4,968	200	HDPE	K-9	1680.8
16,397	200	MS	PN 10	1597.4
16,968	300	DI	PN 10	1436.5
4,899	100	DI	PN 10	319.6
2,304	600	MS	K-7	1440.1
14,716	150	PVC-O	PN 16	746.3
19,977	150	PVC-O	PN 16	1432.2
3,454	600	DI	PN 10	1105.2
6,080	300	DI	PN 16	1110.9
8,030	200	MS	K-7	1270.3
8,450	200	HDPE	K-9	47.1
2,827	150	HDPE	PN 10	1356.3
15,682	500	HDPE	PN 6	467.1
16,664	250	PVC-O	K-9	1312.6
82	500	HDPE	PN 6	350.6
15,528	150	HDPE	PN 16	472.0
23,139	300	HDPE	PN 10	1171.6
15,778	600	DI	PN 6	718.3
2,748	200	PVC-O	K-9	1000.2
22,830	100	PVC-O	K-7	1374.6
44	100	MS	PN 16	695.1
4,536	600	PVC-O	K-7	321.0
11,994	400	PVC-O	K-9	760.7
3,649	400	HDPE	PN 6	1240.4
9,763	500	PVC-O	PN 16	69.8
6,251	250	PVC-O	K-7	1172.1

ANNEXURE II — Hydraulic Node Analysis, page 29

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-138	35.17	9.21	44.19	9.02
N-057	45.71	13.46	58.00	12.29
N-129	60.51	0.93	75.36	14.85
N-213	70.13	2.13	85.04	14.91
N-122	37.98	18.02	62.22	24.24
N-205	43.70	20.31	59.60	15.91
N-021	77.40	2.77	90.24	12.84
N-156	37.33	8.82	52.68	15.35
N-051	36.80	19.09	58.75	21.95
N-061	51.51	11.13	59.91	8.40
N-038	47.36	8.96	58.51	11.15
N-022	62.54	3.49	75.81	13.26
N-191	83.48	4.23	102.72	19.24
N-062	69.90	9.65	84.12	14.22
N-102	38.43	14.22	57.65	19.22
N-047	31.46	5.03	45.64	14.18
N-001	58.67	8.39	78.88	20.21
N-036	34.27	9.11	43.18	8.91
N-149	91.91	6.26	101.47	9.56
N-011	67.70	8.87	84.47	16.77
N-090	51.33	16.90	74.25	22.92
N-075	53.80	19.31	67.92	14.12
N-185	61.17	17.65	77.10	15.94
N-049	85.82	7.74	95.00	9.18
N-003	59.91	11.67	74.89	14.98

ANNEXURE II — Pump Schedule, page 30

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-06	Backwash	93.7	32.2	41.11
P-10	Raw water	312.8	22.7	96.64
P-03	Backwash	106.5	78.7	114.20
P-12	Booster	75.3	71.0	72.87
P-09	Sludge	305.7	15.3	63.90
P-01	Clear water	224.6	57.7	176.43
P-01	Backwash	91.8	45.6	57.09
P-09	Clear water	280.8	84.9	324.61
P-05	Sludge	36.5	39.1	19.41
P-01	Booster	98.8	56.0	75.33
P-03	Raw water	193.5	32.4	85.43
P-07	Booster	172.5	39.7	93.23
P-07	Raw water	191.1	41.0	106.85
P-07	Booster	92.0	45.0	56.42
P-08	Sludge	274.5	22.1	82.67
P-08	Booster	131.3	44.8	80.05
P-05	Booster	242.3	63.1	208.24
P-10	Sludge	23.7	80.5	26.04
P-05	Clear water	296.8	27.7	112.03
P-12	Sludge	306.8	55.5	232.06
P-09	Raw water	79.0	62.6	67.40
P-03	Clear water	254.2	28.6	99.07
P-10	Backwash	45.7	62.5	38.91
P-07	Backwash	152.2	86.8	180.02

ANNEXURE II — Water Quality Test Results, page 31

Parameter	Unit	Raw water	Treated	IS 10500 limit
Residual chlorine	mg/l	Nil	0.51	0.2 min
Total coliform	MPN/100ml	11	Absent	Absent
Iron	mg/l	2.23	0.07	1.0
Total hardness	mg/l	295	72	200
Turbidity	NTU	178.5	0.76	1
Turbidity	NTU	48.8	0.80	1
Iron	mg/l	0.74	0.08	1.0
pH	-	7.8	7.7	6.5-8.5
Turbidity	NTU	19.8	0.37	1
pH	-	7.8	7.5	6.5-8.5
Residual chlorine	mg/l	Nil	0.54	0.2 min
Residual chlorine	mg/l	Nil	0.77	0.2 min
pH	-	7.8	8.1	6.5-8.5
Total hardness	mg/l	326	176	200
Residual chlorine	mg/l	Nil	0.80	0.2 min
Iron	mg/l	1.13	0.20	1.0
Total hardness	mg/l	164	143	200
Total coliform	MPN/100ml	859	Absent	Absent
Iron	mg/l	4.17	0.23	1.0
pH	-	7.9	7.4	6.5-8.5
Total coliform	MPN/100ml	111	Absent	Absent
Turbidity	NTU	78.3	0.54	1

ANNEXURE II — Bore Log Summary, page 32

Depth (m)	Strata	N value	Sample
25.9	Weathered rock	26	UDS
19.2	Medium sand	7	UDS
14.1	Weathered rock	31	UDS
0.9	Gravelly sand	30	DS
1.6	Medium sand	23	DS
26.8	Weathered rock	54	DS
26.3	Medium sand	19	DS
26.0	Silty clay	30	DS
28.1	Weathered rock	24	DS
18.6	Coarse sand	52	UDS
4.2	Clayey silt	26	UDS
12.9	Silty clay	20	DS
2.7	Gravelly sand	55	DS
26.6	Medium sand	24	SPT
13.2	Clayey silt	9	UDS
3.3	Clayey silt	52	SPT
10.0	Gravelly sand	35	UDS
16.9	Silty clay	25	UDS
26.1	Medium sand	38	UDS
11.1	Weathered rock	25	SPT
27.3	Gravelly sand	9	DS
2.0	Weathered rock	51	SPT
28.8	Medium sand	17	SPT
30.0	Medium sand	14	UDS
16.5	Medium sand	53	DS
9.6	Weathered rock	16	DS
15.5	Medium sand	33	DS
11.8	Weathered rock	31	UDS

ANNEXURE III — TREATMENT PLANT DRAWINGS

Unit process layout, hydraulic flow diagram and structural drawings. WTP-010 to WTP-052.

ANNEXURE III — Sheet 1

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. WS-898, revision C.

ANNEXURE III — Sheet 2

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-064	44.09	8.39	51.51	7.42
N-031	34.52	2.63	58.58	24.06
N-147	78.32	6.42	102.37	24.05
N-192	54.01	2.70	64.85	10.84
N-117	92.01	6.37	107.64	15.63
N-215	40.17	23.00	47.49	7.32
N-132	82.75	20.89	92.40	9.65
N-213	32.58	11.70	51.62	19.04
N-162	73.79	7.91	95.53	21.74
N-189	92.29	20.95	103.63	11.34
N-174	71.40	18.99	80.83	9.43
N-068	32.48	18.90	43.57	11.09
N-019	61.28	1.29	82.29	21.02
N-182	40.42	19.12	55.38	14.96
N-009	30.20	22.80	47.00	16.80
N-096	95.42	17.25	119.89	24.47
N-101	75.08	17.10	93.16	18.08
N-040	89.62	10.71	102.02	12.40
N-125	29.51	17.82	52.61	23.11
N-062	71.58	21.06	93.84	22.26
N-056	59.57	5.56	69.75	10.19
N-208	35.23	5.93	42.82	7.59
N-066	38.05	22.58	63.55	25.50
N-057	63.35	6.92	71.83	8.48
N-106	40.38	2.78	63.95	23.56

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

ANNEXURE III — Sheet 3

Design Computation

Detention time = $63.440 \times 1.834 = 116.377$ hr (permissible 200.137) — safe

Discharge = $65.932 \times 8.871 = 584.863$ lps (permissible 774.906) — safe

Velocity = $7.947 \times 9.828 = 78.100$ m/s (permissible 132.876) — safe

Detention time = $28.016 \times 1.426 = 39.957$ hr (permissible 63.256) — safe

Head loss = $23.132 \times 11.448 = 264.812$ m (permissible 445.318) — safe

Residual pressure = $46.531 \times 8.033 = 373.793$ m (permissible 580.247) — safe

Detention time = $5.341 \times 5.337 = 28.505$ hr (permissible 30.108) — safe

Discharge = $33.645 \times 8.407 = 282.856$ lps (permissible 530.582) — safe

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

ANNEXURE III — Sheet 4

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. HY-199, revision C.

ANNEXURE III — Sheet 5

Population Projection

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2036	Ward 15	19,135	100	1.913
2031	Ward 19	37,478	135	5.060
2056	Ward 7	31,351	135	4.232
2056	Ward 15	41,968	100	4.197
2031	Ward 12	2,079	150	0.312
2056	Ward 9	34,722	135	4.687
2041	Ward 16	33,103	70	2.317
2041	Ward 4	29,592	70	2.071
2036	Ward 17	19,438	135	2.624
2026	Ward 17	27,259	100	2.726
2051	Ward 9	27,386	150	4.108
2026	Ward 16	19,279	150	2.892
2046	Ward 9	16,608	70	1.163
2046	Ward 10	24,198	135	3.267
2031	Ward 5	6,290	135	0.849
2041	Ward 11	13,053	100	1.305
2026	Ward 18	16,315	135	2.203
2031	Ward 9	14,057	70	0.984
2051	Ward 7	34,271	100	3.427
2046	Ward 5	9,132	70	0.639
2036	Ward 21	4,810	150	0.722

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

ANNEXURE III — Sheet 6

Design Computation

Head loss = $83.929 \times 2.637 = 221.293$ m (permissible 234.634) — safe

Detention time = $81.910 \times 11.573 = 947.909$ hr (permissible 1429.890) — safe

Detention time = $38.302 \times 4.322 = 165.557$ hr (permissible 272.060) — safe

Velocity = $37.339 \times 10.584 = 395.191$ m/s (permissible 735.577) — safe

Velocity = $7.717 \times 8.840 = 68.222$ m/s (permissible 119.874) — revise section

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

ANNEXURE III — Sheet 7

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. ST-929, revision B.

ANNEXURE III — Sheet 8

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-08	Raw water	106.0	67.8	97.92
P-11	Raw water	311.0	18.3	77.65
P-10	Clear water	38.9	19.1	10.14
P-02	Backwash	142.5	49.3	95.81
P-03	Booster	250.2	39.8	135.83
P-12	Sludge	18.1	72.1	17.77
P-01	Raw water	41.0	55.7	31.13
P-01	Booster	155.1	75.4	159.34
P-12	Clear water	276.9	78.1	294.86
P-08	Raw water	85.6	13.4	15.61
P-11	Booster	317.2	60.6	261.75
P-12	Booster	252.8	65.4	225.30
P-06	Backwash	120.0	37.9	62.01
P-06	Sludge	111.3	51.0	77.28
P-12	Clear water	285.0	29.0	112.61
P-02	Backwash	88.0	44.7	53.60
P-03	Sludge	33.4	15.9	7.22
P-07	Sludge	295.0	18.5	74.46

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

ANNEXURE III — Sheet 9

Design Computation

Residual pressure = $44.030 \times 8.436 = 371.433$ m (permissible 579.655) — safe

Residual pressure = $78.172 \times 6.044 = 472.441$ m (permissible 827.366) — safe

Head loss = $47.262 \times 10.389 = 491.027$ m (permissible 811.179) — safe

Detention time = $37.571 \times 1.355 = 50.890$ hr (permissible 58.149) — safe

Head loss = $10.513 \times 9.448 = 99.329$ m (permissible 119.886) — safe

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

ANNEXURE III — Sheet 10

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. PL-661, revision C.

ANNEXURE III — Sheet 11

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-149	51.43	11.83	61.94	10.50
N-160	80.10	3.79	94.43	14.33
N-127	68.15	23.99	77.44	9.29
N-199	37.59	11.12	51.03	13.43
N-193	46.19	15.44	62.75	16.56
N-217	66.28	9.53	83.51	17.23
N-098	31.87	18.67	51.03	19.16
N-081	65.24	3.63	76.18	10.94
N-089	54.16	18.07	72.26	18.11
N-051	68.32	2.31	83.40	15.08
N-091	68.49	18.56	76.04	7.55
N-115	33.91	22.92	52.55	18.64
N-080	61.84	10.62	72.26	10.42
N-027	81.67	12.17	105.06	23.39
N-211	76.74	18.15	93.85	17.11
N-125	66.58	8.39	75.95	9.38
N-218	53.65	6.05	64.48	10.83
N-211	51.79	8.75	59.50	7.71
N-191	67.47	10.60	87.54	20.07

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

ANNEXURE III — Sheet 12

Design Computation

Discharge = $19.746 \times 6.116 = 120.759$ lps (permissible 223.175) — safe

Residual pressure = $73.855 \times 7.013 = 517.963$ m (permissible 982.118) — safe

Head loss = $71.651 \times 0.713 = 51.065$ m (permissible 71.892) — safe

Head loss = $26.801 \times 2.656 = 71.195$ m (permissible 95.011) — safe

Residual pressure = $84.175 \times 3.262 = 274.595$ m (permissible 339.939) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

ANNEXURE III — Sheet 13

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. HY-316, revision A.

ANNEXURE III — Sheet 14

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	78.2	0.80	1
Total coliform	MPN/100ml	5	Absent	Absent
Turbidity	NTU	26.8	0.83	1
Total coliform	MPN/100ml	92	Absent	Absent
Total coliform	MPN/100ml	681	Absent	Absent
Iron	mg/l	2.17	0.04	1.0
Turbidity	NTU	134.7	0.71	1
pH	-	7.7	7.8	6.5-8.5
Residual chlorine	mg/l	Nil	0.44	0.2 min
Residual chlorine	mg/l	Nil	0.69	0.2 min
Residual chlorine	mg/l	Nil	0.43	0.2 min
Turbidity	NTU	106.0	0.46	1
Iron	mg/l	3.68	0.03	1.0
Turbidity	NTU	137.3	0.56	1
pH	-	8.0	7.8	6.5-8.5
Total hardness	mg/l	215	158	200
pH	-	6.7	7.5	6.5-8.5
Residual chlorine	mg/l	Nil	0.39	0.2 min
pH	-	7.3	7.8	6.5-8.5
pH	-	7.9	7.4	6.5-8.5

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

ANNEXURE III — Sheet 15

Design Computation

Detention time = $30.631 \times 3.268 = 100.113$ hr (permissible 178.244) — safe

Discharge = $15.190 \times 7.678 = 116.625$ lps (permissible 149.030) — safe

Detention time = $89.909 \times 0.429 = 38.589$ hr (permissible 63.700) — safe

Velocity = $32.530 \times 9.332 = 303.559$ m/s (permissible 410.246) — safe

Residual pressure = $35.467 \times 2.809 = 99.638$ m (permissible 153.465) — safe

Detention time = $1.181 \times 8.802 = 10.393$ hr (permissible 19.606) — safe

Detention time = $11.980 \times 3.896 = 46.675$ hr (permissible 72.093) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE III — Sheet 16

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. WS-132, revision A.

ANNEXURE III — Sheet 17

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
23,265	400	MS	K-7	725.7
13,653	200	PVC-O	K-7	1770.4
2,367	150	PVC-O	PN 16	1358.5
2,295	400	MS	K-9	562.1
3,208	600	DI	PN 16	356.2
9,889	100	HDPE	K-9	1152.5
2,250	150	HDPE	PN 6	1009.6
21,597	300	PVC-O	PN 10	681.0
11,999	300	HDPE	K-9	177.4
16,156	100	PVC-O	PN 10	1448.4
12,276	500	DI	K-9	465.3
6,484	500	HDPE	PN 6	686.6
18,540	100	PVC-O	PN 16	1001.8
11,957	200	DI	PN 6	1629.6
2,280	150	PVC-O	PN 16	228.6
12,393	250	PVC-O	PN 16	1330.3
12,930	400	DI	PN 10	817.1
21,126	200	MS	K-9	1073.0
20,375	400	DI	K-7	1519.3
18,874	250	MS	K-9	932.4
23,674	300	DI	K-7	665.7
14,895	400	PVC-O	PN 16	1794.1
13,138	600	DI	PN 10	1672.8
10,030	150	HDPE	PN 16	1255.5
7,369	500	DI	PN 6	1377.1
8,925	100	HDPE	K-9	1472.8

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

ANNEXURE III — Sheet 18

Design Computation

Head loss = $13.302 \times 1.582 = 21.045$ m (permissible 26.655) — safe

Discharge = $56.548 \times 5.425 = 306.771$ lps (permissible 546.907) — safe

Residual pressure = $47.877 \times 10.378 = 496.852$ m (permissible 643.660) — safe

Head loss = $70.843 \times 7.541 = 534.259$ m (permissible 686.571) — safe

Residual pressure = $63.005 \times 8.187 = 515.821$ m (permissible 948.394) — safe

Residual pressure = $43.598 \times 9.652 = 420.819$ m (permissible 528.365) — safe

Detention time = $37.119 \times 8.296 = 307.921$ hr (permissible 424.751) — safe

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

ANNEXURE III — Sheet 19

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. HY-110, revision A.

ANNEXURE III — Sheet 20

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-06	Sludge	130.7	65.0	115.78
P-06	Booster	153.4	86.5	180.82
P-12	Booster	43.4	43.8	25.89
P-12	Raw water	69.8	55.7	52.98
P-03	Sludge	155.3	87.4	184.93
P-05	Booster	70.7	41.4	39.89
P-01	Sludge	227.6	75.8	234.94
P-11	Backwash	114.3	44.9	69.95
P-10	Booster	255.2	20.6	71.49
P-02	Backwash	65.6	25.3	22.66
P-06	Clear water	171.8	44.8	104.80
P-12	Raw water	245.5	46.0	153.90
P-03	Sludge	152.3	62.1	128.77
P-11	Raw water	98.8	51.6	69.41
P-01	Raw water	187.1	39.0	99.53
P-06	Backwash	309.3	17.0	71.49
P-06	Raw water	73.4	50.9	50.90
P-03	Booster	235.3	45.8	146.92
P-07	Sludge	78.5	44.2	47.32
P-05	Backwash	244.6	14.7	48.99
P-11	Booster	85.0	54.1	62.64
P-05	Backwash	122.0	48.4	80.38
P-10	Backwash	102.1	61.1	85.06
P-02	Backwash	67.5	75.2	69.15

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

ANNEXURE III — Sheet 21

Design Computation

Residual pressure = $36.571 \times 1.807 = 66.083 \text{ m}$ (permissible 71.439) — safe

Detention time = $29.888 \times 4.028 = 120.402 \text{ hr}$ (permissible 145.679) — revise section

Velocity = $22.441 \times 9.700 = 217.666 \text{ m/s}$ (permissible 285.116) — safe

Head loss = $31.338 \times 5.418 = 169.783 \text{ m}$ (permissible 182.164) — safe

Detention time = $54.591 \times 5.512 = 300.906 \text{ hr}$ (permissible 454.060) — safe

Velocity = $16.112 \times 8.374 = 134.924 \text{ m/s}$ (permissible 154.975) — safe

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

ANNEXURE IV — DETAILED ESTIMATE

Item-wise estimate with quantities and rates, supported by measurement sheets.

ANNEXURE IV — Population Projection, page 1

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2031	Ward 14	7,400	100	0.740
2026	Ward 23	33,392	135	4.508
2036	Ward 14	38,833	100	3.883
2026	Ward 15	6,547	150	0.982
2056	Ward 5	45,992	135	6.209
2036	Ward 8	14,786	100	1.479
2036	Ward 15	15,221	100	1.522
2036	Ward 24	26,530	100	2.653
2041	Ward 17	19,337	100	1.934
2056	Ward 5	41,364	70	2.895
2041	Ward 13	38,945	150	5.842
2041	Ward 15	26,152	70	1.831
2056	Ward 15	23,018	100	2.302
2056	Ward 15	20,664	70	1.446
2051	Ward 15	34,675	100	3.467
2041	Ward 12	34,931	150	5.240
2046	Ward 10	28,498	70	1.995
2056	Ward 5	14,434	70	1.010
2051	Ward 20	37,003	150	5.550
2046	Ward 23	4,288	150	0.643
2036	Ward 10	38,835	135	5.243
2041	Ward 12	29,726	70	2.081
2036	Ward 15	36,522	135	4.930
2031	Ward 18	20,333	100	2.033
2056	Ward 6	26,444	100	2.644
2041	Ward 12	27,014	100	2.701
2026	Ward 7	3,261	100	0.326
2036	Ward 7	33,025	150	4.954

ANNEXURE IV — Pipeline Schedule, page 2

Chainage (m)	Dia (mm)	Material	Class	Length (m)
15,005	600	HDPE	PN 6	1681.4
11,275	300	PVC-O	K-7	1385.1
16,026	300	HDPE	K-9	1203.0
15,038	100	DI	PN 16	1086.8
463	200	MS	K-7	1351.4
19,339	400	PVC-O	K-9	1388.7
17,494	400	MS	PN 6	982.3
21,914	100	MS	K-9	644.9
873	400	HDPE	PN 6	1301.7
14,963	400	PVC-O	PN 16	497.1
23,880	200	MS	PN 16	204.2
14,881	250	DI	PN 6	971.0
16,576	400	DI	K-7	135.8
408	400	PVC-O	K-9	1221.3
17,767	150	DI	K-9	1110.5
13,414	100	HDPE	PN 6	513.7
400	300	PVC-O	K-9	1692.3
15,137	200	HDPE	PN 6	459.9
1,725	250	HDPE	K-7	996.4
19,098	400	PVC-O	PN 10	410.0
20,729	400	MS	K-9	90.2
14,858	600	MS	K-9	769.0
20,404	100	DI	PN 6	1084.8
4,290	250	PVC-O	K-9	351.7
18,583	250	DI	K-9	166.9
1,333	500	PVC-O	K-7	597.3
7,979	150	DI	PN 10	1624.5
18,620	150	MS	PN 10	462.9

ANNEXURE IV — Hydraulic Node Analysis, page 3

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-138	62.51	6.11	79.55	17.04
N-106	57.50	5.54	77.99	20.48
N-067	89.40	7.25	109.74	20.34
N-063	76.92	3.72	85.54	8.61
N-052	46.58	9.27	63.84	17.26
N-027	94.00	14.60	109.01	15.01
N-184	94.84	18.81	115.73	20.89
N-068	78.82	16.96	88.06	9.25
N-104	82.90	8.60	102.51	19.62
N-088	86.53	21.39	98.34	11.81
N-045	86.19	17.56	103.05	16.86
N-050	63.64	20.37	83.73	20.09
N-161	63.82	13.22	86.55	22.74
N-006	94.06	22.07	106.20	12.13
N-179	66.67	17.77	82.19	15.51
N-018	94.86	11.03	103.73	8.87
N-142	56.91	10.53	77.79	20.88
N-088	75.20	3.16	91.33	16.13
N-046	81.17	23.28	102.60	21.43
N-166	44.52	11.60	64.53	20.00
N-179	63.73	5.10	84.69	20.96
N-118	41.86	9.16	64.07	22.21
N-062	62.35	10.96	76.16	13.81
N-089	71.13	18.23	94.24	23.11
N-019	79.31	20.88	101.80	22.49
N-007	49.24	3.46	70.63	21.39
N-133	57.72	9.22	81.95	24.23

ANNEXURE IV — Pump Schedule, page 4

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-05	Backwash	199.5	51.1	138.94
P-03	Booster	198.8	29.8	80.84
P-08	Backwash	296.7	51.8	209.50
P-08	Raw water	98.5	32.2	43.15
P-03	Sludge	54.3	40.4	29.92
P-05	Raw water	179.6	72.1	176.47
P-12	Backwash	213.1	26.3	76.49
P-10	Booster	292.0	60.4	240.35
P-12	Backwash	17.0	20.8	4.81
P-12	Raw water	150.4	80.9	165.66
P-08	Sludge	192.9	66.6	175.03
P-08	Sludge	143.6	48.3	94.55
P-03	Sludge	278.7	70.7	268.66
P-10	Sludge	115.3	39.5	62.08
P-05	Raw water	229.0	68.0	212.08
P-10	Clear water	305.4	32.1	133.43
P-08	Booster	171.2	57.0	133.04
P-03	Raw water	301.6	79.3	326.06
P-10	Sludge	96.8	59.6	78.65
P-02	Clear water	202.4	67.1	185.01
P-08	Booster	217.5	47.4	140.36
P-12	Booster	63.6	39.7	34.45
P-10	Raw water	220.3	52.4	157.32

ANNEXURE IV — Water Quality Test Results, page 5

Parameter	Unit	Raw water	Treated	IS 10500 limit
Total coliform	MPN/100ml	490	Absent	Absent
Total hardness	mg/l	251	182	200
Total hardness	mg/l	247	114	200
Iron	mg/l	3.54	0.06	1.0
Residual chlorine	mg/l	Nil	0.61	0.2 min
Iron	mg/l	3.43	0.12	1.0
Total coliform	MPN/100ml	247	Absent	Absent
Turbidity	NTU	47.1	0.32	1
Total coliform	MPN/100ml	792	Absent	Absent
Residual chlorine	mg/l	Nil	0.75	0.2 min
Residual chlorine	mg/l	Nil	0.56	0.2 min
Total hardness	mg/l	461	142	200
Turbidity	NTU	201.9	0.82	1
Total coliform	MPN/100ml	751	Absent	Absent
pH	-	7.6	7.3	6.5-8.5
Turbidity	NTU	8.6	0.89	1
Total hardness	mg/l	239	76	200
Iron	mg/l	1.55	0.13	1.0
Turbidity	NTU	151.0	0.24	1
Turbidity	NTU	17.7	0.44	1
Total hardness	mg/l	328	95	200
pH	-	8.2	7.3	6.5-8.5

ANNEXURE IV — Bore Log Summary, page 6

Depth (m)	Strata	N value	Sample
8.8	Medium sand	9	DS
12.0	Gravelly sand	33	DS
14.8	Silty clay	37	UDS
8.6	Weathered rock	48	UDS
28.1	Medium sand	30	UDS
19.5	Silty clay	51	SPT
24.1	Clayey silt	53	UDS
14.0	Clayey silt	5	UDS
14.9	Clayey silt	24	SPT
13.7	Weathered rock	42	UDS
14.7	Weathered rock	50	DS
22.7	Clayey silt	19	UDS
5.4	Silty clay	35	UDS
20.2	Gravelly sand	59	UDS
27.7	Gravelly sand	19	SPT
2.3	Weathered rock	25	UDS
18.3	Medium sand	60	DS
4.9	Weathered rock	39	UDS
22.7	Medium sand	43	DS
19.5	Gravelly sand	32	SPT
13.5	Gravelly sand	28	UDS
20.6	Medium sand	50	UDS
27.3	Clayey silt	19	SPT
12.9	Gravelly sand	31	DS
3.2	Coarse sand	49	SPT
16.5	Coarse sand	18	SPT
15.2	Medium sand	22	UDS

ANNEXURE IV — Population Projection, page 7

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2056	Ward 8	40,024	150	6.004
2051	Ward 4	30,356	100	3.036
2051	Ward 5	36,837	100	3.684
2046	Ward 21	15,278	70	1.069
2036	Ward 16	9,962	135	1.345
2026	Ward 7	44,958	70	3.147
2031	Ward 5	36,710	70	2.570
2056	Ward 12	41,214	70	2.885
2036	Ward 23	11,486	100	1.149
2036	Ward 22	6,628	150	0.994
2036	Ward 24	15,007	70	1.050
2036	Ward 4	28,280	135	3.818
2026	Ward 18	10,466	135	1.413
2056	Ward 16	5,772	70	0.404
2046	Ward 9	33,865	70	2.371
2026	Ward 18	45,910	100	4.591
2026	Ward 3	27,586	70	1.931
2051	Ward 4	37,662	100	3.766
2046	Ward 2	3,623	100	0.362
2056	Ward 17	5,163	150	0.774
2056	Ward 2	13,031	150	1.955
2046	Ward 8	35,279	100	3.528
2056	Ward 7	43,432	135	5.863

ANNEXURE IV — Pipeline Schedule, page 8

Chainage (m)	Dia (mm)	Material	Class	Length (m)
22,330	250	PVC-O	PN 16	916.0
21,274	250	PVC-O	PN 6	393.2
22,632	300	HDPE	K-9	873.7
7,138	100	PVC-O	K-7	139.0
2,329	250	HDPE	PN 6	799.0
2,119	300	DI	K-9	847.4
2,495	600	PVC-O	K-9	394.4
18,246	400	MS	K-7	439.0
9,986	150	HDPE	K-7	1015.9
22,706	150	HDPE	K-9	752.7
2,030	100	HDPE	PN 6	1446.0
4,176	500	DI	PN 6	706.3
21,268	600	PVC-O	PN 6	1004.0
21,830	250	HDPE	PN 10	548.6
9,667	200	MS	K-9	757.0
22,996	250	DI	PN 10	1263.4
19,315	200	HDPE	K-9	345.2
9,055	500	DI	PN 16	1238.0
12,006	300	PVC-O	K-9	1323.7
3,648	300	MS	PN 16	388.8
11,722	600	HDPE	K-7	1640.9
22,891	600	HDPE	PN 16	670.2
14,506	250	DI	PN 10	91.5
4,785	400	DI	PN 16	317.3
17,766	250	PVC-O	K-7	401.2
5,029	600	MS	PN 6	1030.7
23,452	500	MS	PN 10	373.9
23,389	400	MS	K-7	1529.3
719	300	HDPE	PN 16	253.9
22,602	100	HDPE	PN 6	444.3

ANNEXURE IV — Hydraulic Node Analysis, page 9

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-153	33.73	16.09	42.76	9.02
N-095	79.88	5.15	101.59	21.71
N-075	66.45	6.48	75.09	8.64
N-035	32.90	14.78	44.66	11.76
N-162	40.86	16.09	48.25	7.39
N-048	37.64	14.74	52.70	15.06
N-134	43.71	20.95	66.83	23.12
N-204	78.47	23.78	101.82	23.35
N-026	50.83	0.86	71.46	20.63
N-053	65.78	19.65	75.00	9.21
N-142	88.04	10.43	108.54	20.50
N-094	82.07	0.76	104.04	21.96
N-155	54.68	17.66	68.48	13.80
N-090	59.52	4.37	80.89	21.37
N-146	70.08	8.93	84.75	14.66
N-023	59.19	7.42	76.57	17.37
N-014	57.45	2.73	79.18	21.73
N-079	61.82	12.92	69.85	8.04
N-140	72.14	7.62	83.19	11.05
N-067	58.02	11.15	74.55	16.52
N-162	30.86	19.26	46.93	16.08
N-112	43.12	10.83	51.74	8.63
N-021	43.22	0.97	67.71	24.49
N-033	84.02	16.82	96.88	12.86
N-016	91.50	2.49	109.63	18.12
N-208	76.72	18.54	88.03	11.31
N-190	95.25	22.28	110.25	15.00
N-159	78.95	5.12	92.25	13.31
N-040	51.70	17.95	62.86	11.16

ANNEXURE IV — Pump Schedule, page 10

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-02	Booster	195.2	47.8	127.23
P-08	Clear water	226.6	27.4	84.65
P-08	Sludge	15.5	28.3	5.98
P-10	Booster	214.1	49.5	144.39
P-08	Raw water	151.9	75.4	155.95
P-03	Raw water	38.0	28.3	14.65
P-03	Booster	32.0	36.1	15.72
P-10	Clear water	284.9	68.3	265.08
P-07	Clear water	185.8	78.8	199.62
P-12	Booster	310.1	87.5	369.67
P-03	Raw water	45.4	65.2	40.32
P-07	Sludge	176.9	37.6	90.73
P-03	Booster	152.1	64.3	133.37
P-11	Backwash	181.9	46.1	114.26
P-10	Clear water	259.4	37.3	131.66
P-01	Booster	167.0	52.6	119.78
P-10	Booster	86.5	37.0	43.62
P-12	Sludge	297.5	36.7	148.58
P-11	Raw water	258.1	24.4	85.67
P-10	Backwash	263.7	32.6	117.22
P-12	Raw water	261.2	86.2	306.63
P-12	Sludge	253.5	49.6	171.32
P-05	Booster	304.2	29.7	122.91
P-09	Raw water	199.2	53.2	144.51
P-10	Raw water	137.5	82.5	154.56
P-04	Booster	100.5	24.3	33.33

ANNEXURE IV — Water Quality Test Results, page 11

Parameter	Unit	Raw water	Treated	IS 10500 limit
Iron	mg/l	0.72	0.20	1.0
Turbidity	NTU	71.5	0.88	1
Turbidity	NTU	39.3	0.54	1
Iron	mg/l	2.82	0.11	1.0
Total hardness	mg/l	355	182	200
Total coliform	MPN/100ml	165	Absent	Absent
Turbidity	NTU	174.3	0.81	1
Iron	mg/l	3.61	0.25	1.0
pH	-	7.9	7.7	6.5-8.5
Iron	mg/l	0.51	0.11	1.0
Residual chlorine	mg/l	Nil	0.79	0.2 min
Iron	mg/l	4.07	0.08	1.0
Residual chlorine	mg/l	Nil	0.26	0.2 min
pH	-	7.8	7.8	6.5-8.5
Total hardness	mg/l	296	144	200
pH	-	7.0	7.4	6.5-8.5
Iron	mg/l	2.73	0.03	1.0
pH	-	8.1	7.3	6.5-8.5
Iron	mg/l	1.58	0.03	1.0
Total coliform	MPN/100ml	79	Absent	Absent
Iron	mg/l	1.78	0.06	1.0
Total coliform	MPN/100ml	539	Absent	Absent
Residual chlorine	mg/l	Nil	0.21	0.2 min
Residual chlorine	mg/l	Nil	0.41	0.2 min

ANNEXURE IV — Bore Log Summary, page 12

Depth (m)	Strata	N value	Sample
21.3	Coarse sand	60	UDS
14.6	Weathered rock	26	SPT
27.5	Medium sand	25	SPT
10.7	Medium sand	26	SPT
25.8	Clayey silt	8	UDS
7.5	Gravelly sand	24	UDS
13.1	Clayey silt	16	DS
5.5	Coarse sand	7	SPT
15.1	Gravelly sand	10	UDS
24.5	Medium sand	54	UDS
14.6	Gravelly sand	46	UDS
11.1	Weathered rock	8	UDS
2.8	Weathered rock	42	DS
14.4	Coarse sand	10	DS
27.9	Medium sand	52	UDS
24.0	Weathered rock	44	SPT
10.5	Medium sand	8	DS
8.7	Weathered rock	49	UDS
10.2	Weathered rock	46	SPT
11.7	Silty clay	15	DS
17.4	Coarse sand	7	UDS
29.9	Weathered rock	30	SPT
6.7	Clayey silt	50	DS
27.7	Medium sand	5	UDS

ANNEXURE IV — Population Projection, page 13

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2046	Ward 20	34,910	100	3.491
2046	Ward 8	42,501	135	5.738
2036	Ward 4	23,756	150	3.563
2036	Ward 9	25,692	150	3.854
2051	Ward 7	18,072	135	2.440
2036	Ward 6	42,529	100	4.253
2051	Ward 16	4,105	150	0.616
2031	Ward 1	35,177	100	3.518
2046	Ward 11	23,119	135	3.121
2031	Ward 5	16,585	100	1.659
2046	Ward 19	17,525	100	1.752
2051	Ward 17	8,274	70	0.579
2056	Ward 9	4,296	70	0.301
2031	Ward 16	16,612	100	1.661
2051	Ward 1	12,882	150	1.932
2046	Ward 16	29,215	135	3.944
2036	Ward 16	40,952	100	4.095
2056	Ward 23	22,581	100	2.258
2031	Ward 13	45,429	150	6.814
2051	Ward 8	6,321	135	0.853
2026	Ward 23	17,054	150	2.558
2036	Ward 24	28,647	100	2.865
2031	Ward 24	2,071	150	0.311
2036	Ward 10	29,719	100	2.972
2031	Ward 10	4,330	70	0.303
2036	Ward 19	30,422	70	2.130
2051	Ward 11	3,187	70	0.223

ANNEXURE IV — Pipeline Schedule, page 14

Chainage (m)	Dia (mm)	Material	Class	Length (m)
23,039	100	DI	K-9	1675.5
19,796	200	MS	PN 10	827.1
549	300	HDPE	PN 16	487.5
754	500	DI	PN 10	1657.1
10,039	600	MS	PN 6	675.5
20,049	200	MS	K-9	1416.5
16,466	250	PVC-O	PN 10	526.4
10,695	250	MS	PN 6	1563.6
4,111	300	HDPE	PN 6	1607.5
5,779	300	MS	PN 6	1200.3
15,955	400	MS	K-9	790.2
6,671	500	HDPE	PN 10	929.9
15,189	500	PVC-O	PN 6	523.8
9,152	600	HDPE	K-9	1085.4
23,913	600	HDPE	PN 6	1373.6
15,983	500	DI	PN 10	45.8
4,405	100	HDPE	PN 16	867.7
23,428	100	HDPE	K-9	1438.5
2,682	400	MS	PN 6	182.6
16,996	200	HDPE	K-7	598.8
22,646	500	HDPE	K-9	48.0
13,663	150	DI	PN 10	1578.5
17,205	150	PVC-O	K-7	1756.7
9,153	300	HDPE	PN 6	828.7
19,189	200	DI	PN 6	1760.8
19,005	300	HDPE	PN 16	257.4
22,800	300	MS	K-9	1213.3
16,786	250	MS	PN 6	1565.6
17,497	100	PVC-O	PN 16	1053.7

ANNEXURE IV — Hydraulic Node Analysis, page 15

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-003	47.54	19.12	59.54	12.00
N-161	69.38	6.38	78.07	8.69
N-220	28.14	21.60	46.75	18.61
N-080	90.44	6.17	111.23	20.78
N-099	68.82	7.44	93.57	24.75
N-096	55.45	16.00	67.19	11.75
N-166	57.35	6.32	73.94	16.59
N-138	32.55	12.62	46.65	14.11
N-080	74.23	11.05	90.53	16.30
N-154	78.21	5.61	97.04	18.83
N-142	80.99	4.05	90.72	9.72
N-165	77.46	23.13	94.63	17.16
N-165	68.12	0.75	77.01	8.89
N-093	38.51	9.77	61.34	22.83
N-161	60.86	14.85	83.62	22.77
N-015	69.44	10.99	86.73	17.29
N-064	60.77	20.93	70.42	9.65
N-121	83.37	4.98	98.43	15.06
N-203	34.93	4.60	44.31	9.39
N-049	33.69	13.11	45.03	11.35
N-119	75.47	10.77	90.82	15.35
N-185	31.59	4.04	55.43	23.84
N-079	92.53	13.57	114.88	22.34

ANNEXURE IV — Pump Schedule, page 16

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-08	Clear water	201.5	16.6	45.66
P-09	Backwash	55.1	47.1	35.37
P-06	Raw water	114.1	61.2	95.14
P-02	Raw water	29.5	55.7	22.38
P-06	Booster	189.7	14.2	36.72
P-11	Clear water	204.4	77.1	214.60
P-03	Raw water	311.3	49.8	211.29
P-02	Backwash	178.5	58.6	142.43
P-10	Sludge	196.8	28.5	76.52
P-05	Sludge	50.1	45.4	31.05
P-07	Sludge	190.1	16.0	41.39
P-08	Sludge	41.0	67.5	37.68
P-01	Clear water	280.7	22.3	85.25
P-07	Clear water	225.5	26.7	81.97
P-06	Clear water	248.0	13.9	47.09
P-04	Raw water	168.8	34.9	80.35
P-04	Clear water	185.6	62.1	157.12
P-05	Clear water	182.3	64.9	161.11
P-02	Clear water	195.8	16.4	43.88
P-01	Sludge	47.9	59.7	39.01
P-07	Backwash	277.3	27.1	102.40
P-06	Booster	87.3	58.9	70.12
P-03	Sludge	66.9	54.6	49.82

ANNEXURE IV — Water Quality Test Results, page 17

Parameter	Unit	Raw water	Treated	IS 10500 limit
Residual chlorine	mg/l	Nil	0.67	0.2 min
Iron	mg/l	1.71	0.12	1.0
Iron	mg/l	2.97	0.22	1.0
Residual chlorine	mg/l	Nil	0.30	0.2 min
pH	-	7.7	7.8	6.5-8.5
Total hardness	mg/l	347	127	200
Iron	mg/l	0.86	0.16	1.0
pH	-	8.0	7.6	6.5-8.5
pH	-	7.0	7.9	6.5-8.5
Total hardness	mg/l	370	126	200
Turbidity	NTU	220.0	0.80	1
Iron	mg/l	2.53	0.13	1.0
Iron	mg/l	1.48	0.25	1.0
Iron	mg/l	0.58	0.20	1.0
pH	-	6.8	7.4	6.5-8.5
pH	-	8.4	7.6	6.5-8.5
Total coliform	MPN/100ml	606	Absent	Absent
Turbidity	NTU	228.2	0.82	1
Total coliform	MPN/100ml	414	Absent	Absent
Total hardness	mg/l	91	88	200
Turbidity	NTU	118.9	0.59	1
Iron	mg/l	1.60	0.24	1.0
Iron	mg/l	2.94	0.21	1.0
Iron	mg/l	3.91	0.19	1.0
Turbidity	NTU	173.6	0.40	1
Total hardness	mg/l	251	117	200
Iron	mg/l	3.00	0.08	1.0
Total coliform	MPN/100ml	155	Absent	Absent
Residual chlorine	mg/l	Nil	0.79	0.2 min
pH	-	7.9	7.2	6.5-8.5

ANNEXURE IV — Bore Log Summary, page 18

Depth (m)	Strata	N value	Sample
15.0	Clayey silt	43	SPT
2.5	Silty clay	34	UDS
11.5	Medium sand	58	UDS
10.0	Gravelly sand	30	UDS
3.5	Silty clay	21	SPT
15.2	Gravelly sand	12	SPT
22.2	Weathered rock	34	DS
20.1	Medium sand	15	DS
12.6	Clayey silt	31	UDS
23.1	Silty clay	14	DS
9.3	Medium sand	31	DS
9.2	Silty clay	49	DS
0.9	Weathered rock	13	UDS
26.8	Medium sand	20	DS
21.5	Gravelly sand	44	UDS
6.0	Coarse sand	53	UDS
23.1	Coarse sand	30	UDS
19.1	Weathered rock	17	UDS
9.6	Silty clay	7	SPT
21.5	Gravelly sand	53	DS
2.8	Silty clay	17	DS
26.5	Weathered rock	6	UDS
27.3	Clayey silt	42	UDS
21.4	Clayey silt	29	UDS
7.6	Clayey silt	46	UDS
20.6	Silty clay	22	UDS
20.6	Silty clay	14	SPT
25.1	Gravelly sand	27	SPT
17.1	Silty clay	40	UDS
23.2	Gravelly sand	7	DS

ANNEXURE IV — Population Projection, page 19

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2026	Ward 2	44,333	100	4.433
2041	Ward 5	25,714	135	3.471
2031	Ward 15	4,233	135	0.571
2026	Ward 11	27,705	70	1.939
2036	Ward 22	36,912	70	2.584
2056	Ward 4	26,388	100	2.639
2026	Ward 21	20,583	100	2.058
2046	Ward 18	15,616	135	2.108
2051	Ward 11	4,900	135	0.661
2026	Ward 9	4,724	70	0.331
2056	Ward 11	26,151	150	3.923
2056	Ward 2	21,881	70	1.532
2046	Ward 6	28,616	150	4.292
2036	Ward 13	34,319	150	5.148
2031	Ward 16	34,811	70	2.437
2026	Ward 11	30,703	150	4.605
2051	Ward 3	16,165	70	1.132
2031	Ward 2	40,493	70	2.835
2051	Ward 14	39,072	70	2.735
2041	Ward 17	5,388	70	0.377
2031	Ward 2	43,983	135	5.938
2026	Ward 24	15,543	135	2.098
2056	Ward 19	30,945	100	3.095
2026	Ward 20	13,486	100	1.349
2026	Ward 2	3,589	70	0.251
2046	Ward 22	2,965	70	0.208
2026	Ward 6	3,233	100	0.323
2056	Ward 22	38,150	150	5.723

ANNEXURE IV — Pipeline Schedule, page 20

Chainage (m)	Dia (mm)	Material	Class	Length (m)
18,342	500	HDPE	K-7	937.2
8,515	600	HDPE	PN 10	1133.5
11,561	600	PVC-O	PN 6	748.4
8,807	500	PVC-O	PN 16	601.9
22,932	500	PVC-O	PN 10	813.0
12,943	250	DI	K-9	1757.3
2,745	100	HDPE	PN 10	1095.0
12,993	600	MS	PN 16	1046.1
4,404	400	HDPE	PN 10	1193.0
19,852	200	DI	PN 6	1146.8
21,716	250	DI	PN 6	625.2
16,383	400	HDPE	PN 6	711.8
8,807	150	MS	PN 16	337.2
5,370	250	MS	PN 6	522.2
23,907	150	PVC-O	PN 16	773.0
18,420	500	MS	K-9	738.9
11,561	200	HDPE	PN 6	1220.6
8,147	600	HDPE	PN 6	1722.5
2,732	150	HDPE	K-7	186.1
1,605	100	DI	PN 10	580.8
17,137	150	DI	K-9	1073.5
4,175	150	HDPE	K-9	133.2
23,677	100	MS	PN 6	808.4
16,474	300	HDPE	K-7	1740.8
19,787	250	DI	PN 10	1325.0
22,093	300	HDPE	PN 10	622.9
6,221	400	HDPE	PN 10	1184.4
20,875	150	HDPE	PN 10	1671.5

ANNEXURE IV — Hydraulic Node Analysis, page 21

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-191	33.40	1.96	50.49	17.09
N-026	65.10	2.67	82.08	16.98
N-044	84.53	8.42	108.92	24.39
N-150	69.70	19.42	81.21	11.51
N-048	74.61	10.04	92.38	17.76
N-073	49.63	3.70	72.03	22.40
N-199	87.81	18.39	107.82	20.02
N-007	36.79	0.37	54.78	17.99
N-079	60.88	14.88	73.88	13.00
N-166	83.77	16.28	104.15	20.37
N-210	86.11	11.72	111.08	24.97
N-171	78.71	13.17	87.05	8.34
N-172	77.17	3.11	94.34	17.17
N-154	32.91	21.30	47.82	14.91
N-104	43.38	2.39	54.60	11.23
N-114	88.76	18.38	95.96	7.20
N-075	57.18	7.71	78.70	21.52
N-166	83.38	8.35	96.96	13.57
N-076	89.05	14.12	113.76	24.70
N-195	82.07	2.14	89.27	7.20
N-185	87.02	2.57	107.19	20.17
N-167	70.55	10.84	86.36	15.80
N-166	41.11	19.52	54.65	13.54
N-040	68.60	19.47	94.21	25.61
N-171	70.26	2.68	95.22	24.96

ANNEXURE IV — Pump Schedule, page 22

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-10	Booster	268.7	70.9	259.53
P-01	Raw water	134.5	55.0	100.71
P-05	Clear water	258.0	87.9	308.83
P-05	Backwash	88.6	41.4	50.01
P-09	Raw water	136.7	32.4	60.28
P-01	Clear water	86.5	37.2	43.84
P-10	Backwash	253.3	52.9	182.49
P-08	Booster	160.8	56.8	124.45
P-10	Raw water	199.7	43.7	118.77
P-01	Booster	289.8	41.1	162.45
P-10	Clear water	134.0	43.3	79.09
P-08	Raw water	191.8	20.8	54.42
P-10	Booster	89.5	27.9	34.01
P-08	Sludge	123.2	35.3	59.32
P-03	Raw water	39.0	81.8	43.44
P-09	Backwash	116.5	30.0	47.64
P-07	Clear water	253.3	50.5	174.20
P-05	Sludge	166.4	84.4	191.33
P-07	Booster	180.6	48.6	119.56
P-12	Backwash	114.6	13.9	21.63
P-06	Sludge	47.5	27.8	18.01
P-05	Backwash	88.2	41.3	49.68
P-06	Clear water	98.6	66.1	88.82

ANNEXURE IV — Water Quality Test Results, page 23

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	32.1	0.67	1
Residual chlorine	mg/l	Nil	0.75	0.2 min
Residual chlorine	mg/l	Nil	0.35	0.2 min
Total hardness	mg/l	396	101	200
pH	-	7.3	8.0	6.5-8.5
Total hardness	mg/l	143	166	200
Iron	mg/l	1.21	0.19	1.0
Residual chlorine	mg/l	Nil	0.62	0.2 min
Total coliform	MPN/100ml	177	Absent	Absent
Turbidity	NTU	92.7	0.23	1
Residual chlorine	mg/l	Nil	0.59	0.2 min
pH	-	6.7	7.4	6.5-8.5
Iron	mg/l	3.50	0.04	1.0
Total hardness	mg/l	239	121	200
Residual chlorine	mg/l	Nil	0.28	0.2 min
Turbidity	NTU	24.8	0.42	1
Turbidity	NTU	99.5	0.69	1
Iron	mg/l	3.23	0.25	1.0
Residual chlorine	mg/l	Nil	0.28	0.2 min
Total coliform	MPN/100ml	459	Absent	Absent
Total coliform	MPN/100ml	400	Absent	Absent
Total coliform	MPN/100ml	611	Absent	Absent
Total coliform	MPN/100ml	457	Absent	Absent
Iron	mg/l	3.13	0.13	1.0
pH	-	7.5	7.6	6.5-8.5
Total hardness	mg/l	252	121	200
pH	-	7.2	7.3	6.5-8.5
Turbidity	NTU	173.3	0.55	1

ANNEXURE IV — Bore Log Summary, page 24

Depth (m)	Strata	N value	Sample
12.8	Weathered rock	19	SPT
4.9	Gravelly sand	50	DS
21.7	Gravelly sand	37	SPT
27.5	Coarse sand	47	DS
9.8	Coarse sand	24	DS
1.8	Coarse sand	49	UDS
26.2	Gravelly sand	51	DS
22.6	Coarse sand	52	SPT
18.3	Gravelly sand	28	DS
17.1	Medium sand	42	SPT
9.8	Gravelly sand	28	DS
29.0	Clayey silt	41	DS
14.2	Coarse sand	48	SPT
23.1	Silty clay	58	UDS
26.1	Silty clay	8	DS
1.6	Silty clay	48	UDS
12.9	Medium sand	41	DS
21.7	Coarse sand	16	DS
23.2	Silty clay	6	SPT
27.0	Coarse sand	41	UDS
7.5	Weathered rock	10	SPT
10.7	Clayey silt	17	SPT
28.4	Silty clay	44	DS

ANNEXURE IV — Population Projection, page 25

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2051	Ward 19	30,031	135	4.054
2046	Ward 7	35,555	150	5.333
2046	Ward 9	10,195	135	1.376
2056	Ward 6	27,987	70	1.959
2051	Ward 14	18,313	135	2.472
2056	Ward 14	25,019	100	2.502
2051	Ward 3	26,695	100	2.670
2056	Ward 19	34,982	100	3.498
2026	Ward 7	44,131	135	5.958
2031	Ward 11	35,233	135	4.756
2046	Ward 14	24,776	70	1.734
2031	Ward 5	24,801	70	1.736
2031	Ward 18	25,842	100	2.584
2051	Ward 7	18,532	135	2.502
2046	Ward 7	37,241	150	5.586
2036	Ward 20	29,042	135	3.921
2046	Ward 14	5,881	135	0.794
2036	Ward 22	9,815	70	0.687
2031	Ward 4	41,779	100	4.178
2046	Ward 12	44,063	150	6.609
2041	Ward 22	40,753	135	5.502
2036	Ward 22	36,579	70	2.561
2036	Ward 10	4,305	100	0.430
2056	Ward 24	41,467	100	4.147
2051	Ward 15	40,577	135	5.478
2056	Ward 13	19,810	150	2.971
2036	Ward 21	43,779	70	3.065
2056	Ward 23	19,425	150	2.914
2036	Ward 22	34,059	135	4.598

ANNEXURE IV — Pipeline Schedule, page 26

Chainage (m)	Dia (mm)	Material	Class	Length (m)
18,275	200	DI	PN 16	374.7
20,789	250	PVC-O	PN 6	693.6
4,517	150	HDPE	K-7	1695.4
1,843	100	HDPE	PN 10	1684.3
16,575	600	DI	K-7	61.6
8,629	250	MS	PN 6	536.5
6,133	150	MS	K-7	592.9
16,427	150	DI	PN 6	283.0
22,770	600	DI	PN 6	1208.4
1,990	300	HDPE	K-9	561.6
9,758	300	MS	K-9	214.1
3,921	100	MS	PN 6	1172.8
13,094	400	MS	PN 6	1406.7
3,468	200	DI	K-7	1468.4
22,174	100	MS	K-9	1788.8
17,115	500	DI	K-9	1571.8
7,143	600	MS	K-7	961.4
18,963	300	HDPE	PN 16	1256.3
4,493	200	HDPE	PN 10	717.0
21,961	500	MS	K-7	459.5
22,404	100	HDPE	PN 10	712.8
20,607	400	PVC-O	PN 10	82.5
16,190	200	DI	PN 6	577.9
10,753	500	DI	K-9	1142.7
20,022	400	MS	K-7	1231.9
1,141	250	HDPE	PN 16	495.8
5	600	MS	K-7	1507.8
10,252	200	HDPE	K-7	164.2

ANNEXURE IV — Hydraulic Node Analysis, page 27

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-208	73.96	13.11	98.33	24.37
N-107	46.52	23.28	60.00	13.48
N-013	86.12	0.74	108.16	22.04
N-144	95.45	1.83	115.55	20.10
N-198	65.89	11.38	86.77	20.88
N-170	59.29	12.82	72.16	12.87
N-191	34.48	3.60	44.79	10.32
N-037	53.61	12.19	66.24	12.63
N-162	48.97	21.14	68.58	19.61
N-108	34.55	16.55	56.31	21.76
N-125	89.44	10.80	110.92	21.48
N-209	85.94	2.34	111.27	25.33
N-016	39.56	20.77	48.14	8.58
N-155	56.24	4.77	69.48	13.24
N-050	34.58	15.51	54.27	19.68
N-019	36.17	15.05	45.60	9.44
N-018	53.85	20.79	68.00	14.15
N-198	74.71	4.09	96.30	21.59
N-123	68.52	5.27	79.52	11.00
N-014	78.96	1.93	90.18	11.21
N-011	28.64	12.24	49.51	20.87
N-139	75.83	6.76	97.33	21.50
N-211	77.65	11.55	89.54	11.89
N-129	86.13	5.18	94.19	8.06

ANNEXURE IV — Pump Schedule, page 28

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-07	Clear water	19.0	44.3	11.50
P-08	Clear water	270.0	35.1	129.11
P-11	Sludge	259.9	64.2	227.33
P-06	Booster	96.1	67.9	88.92
P-03	Backwash	116.5	17.8	28.25
P-07	Raw water	245.5	70.4	235.45
P-09	Clear water	217.6	17.4	51.59
P-03	Backwash	276.3	48.6	183.00
P-04	Clear water	194.5	46.0	121.88
P-05	Raw water	217.1	34.1	101.02
P-01	Booster	94.3	73.7	94.68
P-08	Clear water	56.5	67.2	51.75
P-07	Sludge	148.1	55.5	111.98
P-11	Backwash	241.3	38.0	124.86
P-03	Backwash	255.3	62.8	218.63
P-05	Backwash	169.2	49.8	114.87
P-03	Backwash	290.7	45.6	180.53
P-04	Sludge	311.4	58.0	246.15
P-09	Raw water	192.0	14.5	37.94
P-08	Booster	21.7	61.7	18.21
P-10	Backwash	303.8	68.9	285.21
P-01	Booster	150.9	14.9	30.63
P-08	Backwash	127.6	69.8	121.42
P-12	Backwash	227.9	48.4	150.36
P-09	Raw water	133.6	28.4	51.75
P-02	Clear water	60.1	77.3	63.24
P-03	Sludge	233.7	50.8	161.79
P-08	Sludge	127.3	60.2	104.30

ANNEXURE IV — Water Quality Test Results, page 29

Parameter	Unit	Raw water	Treated	IS 10500 limit
Total hardness	mg/l	233	109	200
Iron	mg/l	1.80	0.11	1.0
Residual chlorine	mg/l	Nil	0.32	0.2 min
Residual chlorine	mg/l	Nil	0.48	0.2 min
Residual chlorine	mg/l	Nil	0.27	0.2 min
Total hardness	mg/l	153	160	200
Residual chlorine	mg/l	Nil	0.77	0.2 min
Total coliform	MPN/100ml	699	Absent	Absent
Total hardness	mg/l	299	105	200
Residual chlorine	mg/l	Nil	0.31	0.2 min
Total hardness	mg/l	128	185	200
Turbidity	NTU	185.4	0.42	1
Turbidity	NTU	227.8	0.81	1
Total coliform	MPN/100ml	245	Absent	Absent
Iron	mg/l	3.83	0.17	1.0
Turbidity	NTU	95.9	0.59	1
Total hardness	mg/l	428	143	200
Total hardness	mg/l	187	163	200
Turbidity	NTU	9.8	0.65	1
Turbidity	NTU	20.2	0.79	1
Residual chlorine	mg/l	Nil	0.29	0.2 min
Iron	mg/l	3.20	0.28	1.0
pH	-	7.9	7.7	6.5-8.5

ANNEXURE IV — Bore Log Summary, page 30

Depth (m)	Strata	N value	Sample
22.0	Weathered rock	45	DS
1.9	Silty clay	19	SPT
4.8	Gravelly sand	36	DS
28.9	Medium sand	22	DS
1.8	Weathered rock	57	DS
17.3	Gravelly sand	48	DS
16.7	Coarse sand	33	SPT
26.8	Silty clay	6	SPT
3.9	Weathered rock	39	SPT
7.9	Clayey silt	57	SPT
26.8	Medium sand	9	DS
11.5	Coarse sand	33	DS
15.7	Weathered rock	59	DS
5.9	Gravelly sand	54	UDS
2.8	Silty clay	34	SPT
24.9	Clayey silt	28	DS
9.1	Gravelly sand	22	SPT
19.3	Coarse sand	20	DS
20.8	Silty clay	29	DS
20.6	Silty clay	42	DS
24.5	Gravelly sand	26	SPT
23.3	Clayey silt	33	UDS
14.6	Coarse sand	19	UDS
7.7	Gravelly sand	26	DS
14.1	Clayey silt	26	UDS
11.8	Clayey silt	35	SPT
3.0	Weathered rock	5	DS

ANNEXURE IV — Population Projection, page 31

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2026	Ward 17	16,101	100	1.610
2046	Ward 12	45,189	100	4.519
2031	Ward 13	8,665	100	0.867
2051	Ward 24	41,705	150	6.256
2046	Ward 19	32,261	70	2.258
2051	Ward 20	44,009	100	4.401
2041	Ward 1	4,696	70	0.329
2036	Ward 15	34,893	135	4.711
2031	Ward 20	5,384	135	0.727
2046	Ward 22	4,344	100	0.434
2026	Ward 16	39,356	150	5.903
2051	Ward 17	14,619	70	1.023
2051	Ward 2	39,211	135	5.293
2026	Ward 11	15,194	150	2.279
2046	Ward 16	26,285	70	1.840
2031	Ward 16	14,046	70	0.983
2046	Ward 6	4,677	100	0.468
2041	Ward 23	8,337	100	0.834
2026	Ward 7	26,662	135	3.599
2046	Ward 18	3,151	100	0.315
2046	Ward 16	43,195	100	4.319
2036	Ward 23	16,589	150	2.488
2051	Ward 3	23,947	70	1.676
2056	Ward 3	38,549	135	5.204
2036	Ward 9	35,962	70	2.517
2036	Ward 14	31,056	100	3.106

ANNEXURE IV — Pipeline Schedule, page 32

Chainage (m)	Dia (mm)	Material	Class	Length (m)
11,036	500	DI	PN 16	499.4
19,046	100	DI	PN 6	1330.1
10,907	250	DI	PN 6	758.3
16,015	500	DI	K-9	924.7
666	400	MS	K-9	1244.2
19,276	600	DI	PN 16	1139.2
4,755	250	PVC-O	K-7	117.9
9,654	600	MS	PN 16	410.8
13,695	500	PVC-O	K-9	1352.5
11,993	100	PVC-O	PN 16	1046.3
9,807	400	HDPE	PN 16	1184.5
1,419	200	DI	PN 10	795.2
10,945	300	DI	PN 10	1576.2
13,707	250	MS	K-9	215.6
14,734	300	PVC-O	PN 16	1542.2
16,087	600	PVC-O	PN 6	1357.2
15,211	500	PVC-O	PN 10	1093.1
17,024	600	HDPE	K-7	1280.8
6,900	400	MS	PN 6	1064.3
14,815	100	PVC-O	K-7	1459.2
16,390	400	PVC-O	PN 10	720.8
16,005	300	PVC-O	K-9	1075.3
1,507	300	MS	PN 10	592.1
4,888	400	DI	K-9	1237.9
5,437	300	DI	K-9	556.9
751	100	PVC-O	PN 16	1440.6
2,944	300	MS	K-9	1275.9

ANNEXURE IV — Hydraulic Node Analysis, page 33

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-113	43.76	9.75	68.83	25.07
N-183	33.58	10.52	53.31	19.74
N-109	49.96	0.86	70.49	20.52
N-216	89.78	0.82	106.17	16.40
N-044	65.45	11.02	78.08	12.63
N-167	87.43	19.46	102.28	14.85
N-203	77.11	14.59	88.58	11.47
N-120	60.64	19.37	68.16	7.53
N-053	84.23	18.73	100.31	16.08
N-095	63.44	6.42	86.27	22.82
N-010	53.69	16.08	68.04	14.35
N-077	60.26	2.72	68.11	7.84
N-005	44.68	18.78	52.74	8.05
N-050	39.95	1.17	56.60	16.64
N-168	80.28	3.48	94.65	14.37
N-161	40.58	4.52	62.43	21.85
N-024	57.94	5.16	65.83	7.89
N-027	79.42	15.58	97.69	18.27
N-086	34.20	1.23	58.42	24.22
N-219	85.91	1.41	97.18	11.27
N-159	30.05	21.98	45.26	15.21
N-057	55.76	2.18	72.39	16.62
N-004	93.51	23.70	112.26	18.75
N-134	94.78	23.31	113.52	18.73
N-203	70.89	24.00	90.42	19.53
N-142	32.91	10.91	57.80	24.90
N-123	41.12	20.53	56.98	15.86
N-061	34.35	14.46	43.00	8.65

ANNEXURE IV — Pump Schedule, page 34

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-11	Backwash	262.2	38.9	138.80
P-09	Backwash	250.2	34.9	118.81
P-04	Raw water	159.6	77.8	169.17
P-01	Clear water	85.7	87.0	101.60
P-02	Raw water	319.4	30.2	131.38
P-02	Booster	271.8	56.5	209.12
P-01	Booster	249.2	30.9	104.99
P-12	Backwash	182.3	75.0	186.25
P-01	Booster	229.5	34.8	108.79
P-05	Booster	301.0	13.6	55.86
P-07	Backwash	24.4	79.3	26.39
P-02	Clear water	210.3	46.9	134.35
P-10	Raw water	313.1	54.4	231.92
P-09	Sludge	225.8	28.6	87.86
P-01	Backwash	152.0	16.4	33.99
P-01	Sludge	82.0	47.0	52.47
P-12	Clear water	146.3	17.0	33.90
P-01	Raw water	236.0	75.3	242.18
P-06	Raw water	176.8	86.4	207.97
P-02	Sludge	273.1	80.9	300.88
P-12	Booster	194.3	60.3	159.57
P-08	Booster	220.2	48.5	145.66
P-11	Clear water	79.9	70.3	76.49
P-09	Clear water	262.6	79.2	283.35
P-09	Raw water	274.5	15.3	57.36
P-07	Clear water	37.7	37.1	19.07
P-06	Raw water	261.8	60.2	214.82

ANNEXURE IV — Water Quality Test Results, page 35

Parameter	Unit	Raw water	Treated	IS 10500 limit
Total coliform	MPN/100ml	280	Absent	Absent
Turbidity	NTU	202.4	0.63	1
Residual chlorine	mg/l	Nil	0.58	0.2 min
Total coliform	MPN/100ml	701	Absent	Absent
Turbidity	NTU	160.0	0.24	1
Total coliform	MPN/100ml	809	Absent	Absent
Residual chlorine	mg/l	Nil	0.57	0.2 min
Iron	mg/l	3.39	0.02	1.0
pH	-	6.7	7.5	6.5-8.5
Total hardness	mg/l	272	169	200
Residual chlorine	mg/l	Nil	0.70	0.2 min
Iron	mg/l	1.63	0.03	1.0
pH	-	6.8	7.9	6.5-8.5
Total hardness	mg/l	152	107	200
Total coliform	MPN/100ml	899	Absent	Absent
Residual chlorine	mg/l	Nil	0.72	0.2 min
pH	-	6.9	7.4	6.5-8.5
Iron	mg/l	3.34	0.16	1.0
pH	-	7.3	7.6	6.5-8.5
Turbidity	NTU	123.1	0.27	1
Total coliform	MPN/100ml	368	Absent	Absent
Total hardness	mg/l	173	163	200
pH	-	7.7	7.0	6.5-8.5
Turbidity	NTU	81.5	0.35	1
pH	-	8.1	7.6	6.5-8.5
Iron	mg/l	2.51	0.07	1.0
Total hardness	mg/l	373	169	200
Total coliform	MPN/100ml	620	Absent	Absent
Total hardness	mg/l	133	146	200

ANNEXURE IV — Bore Log Summary, page 36

Depth (m)	Strata	N value	Sample
20.1	Gravelly sand	56	UDS
23.0	Silty clay	51	SPT
25.8	Silty clay	18	SPT
25.0	Silty clay	12	DS
5.7	Medium sand	27	UDS
26.8	Silty clay	21	UDS
6.4	Gravelly sand	40	UDS
3.2	Coarse sand	38	UDS
11.7	Medium sand	53	UDS
0.8	Weathered rock	6	SPT
16.6	Weathered rock	39	DS
17.9	Weathered rock	39	UDS
16.3	Gravelly sand	55	DS
17.4	Coarse sand	42	DS
24.7	Coarse sand	28	DS
26.1	Clayey silt	22	UDS
9.7	Medium sand	4	DS
26.8	Gravelly sand	27	SPT
29.3	Weathered rock	38	SPT
6.2	Clayey silt	55	UDS
11.9	Silty clay	48	DS
13.8	Coarse sand	54	SPT
9.4	Coarse sand	59	SPT
11.9	Weathered rock	22	UDS
26.8	Silty clay	45	SPT
2.8	Gravelly sand	28	SPT
12.8	Weathered rock	21	DS
13.3	Gravelly sand	45	UDS

ANNEXURE IV — Population Projection, page 37

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2046	Ward 5	13,319	70	0.932
2056	Ward 24	38,978	100	3.898
2051	Ward 3	6,619	100	0.662
2051	Ward 2	45,376	100	4.538
2026	Ward 5	37,528	70	2.627
2041	Ward 10	10,381	135	1.401
2046	Ward 1	25,144	70	1.760
2051	Ward 22	24,089	100	2.409
2036	Ward 4	31,862	135	4.301
2056	Ward 23	4,266	70	0.299
2051	Ward 13	34,238	135	4.622
2046	Ward 21	19,039	100	1.904
2031	Ward 6	41,067	150	6.160
2041	Ward 23	28,936	150	4.340
2051	Ward 18	33,614	135	4.538
2046	Ward 7	3,270	150	0.490
2031	Ward 15	20,157	100	2.016
2031	Ward 8	19,272	135	2.602
2026	Ward 10	21,329	70	1.493
2026	Ward 23	34,938	100	3.494
2041	Ward 16	39,066	70	2.735
2031	Ward 4	23,590	100	2.359

ANNEXURE IV — Pipeline Schedule, page 38

Chainage (m)	Dia (mm)	Material	Class	Length (m)
20,527	300	HDPE	PN 6	1025.1
8,431	250	MS	PN 16	955.1
15,534	500	HDPE	PN 16	1143.8
20,431	200	DI	PN 16	872.7
4,658	200	HDPE	PN 6	306.4
799	300	PVC-O	PN 10	1565.7
8,999	300	DI	K-9	345.6
922	600	DI	PN 6	1633.6
20,519	200	HDPE	PN 6	625.1
19,141	500	DI	K-7	1393.0
1,070	600	DI	PN 16	102.5
23,574	500	HDPE	PN 6	1766.3
10,348	250	PVC-O	PN 6	973.7
18,409	500	HDPE	PN 6	1722.2
7,636	100	PVC-O	K-9	524.7
8,322	200	MS	PN 6	1058.7
2,650	100	PVC-O	K-9	1344.4
22,320	150	PVC-O	PN 10	563.0
18,574	100	PVC-O	PN 6	97.6
8,037	200	HDPE	PN 10	1685.9
5,123	100	MS	PN 16	866.5
3,790	150	HDPE	PN 16	343.6

ANNEXURE IV — Hydraulic Node Analysis, page 39

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-061	81.99	20.45	91.42	9.43
N-104	76.09	8.86	96.06	19.97
N-023	55.11	8.59	81.02	25.91
N-091	69.53	2.60	86.53	17.00
N-141	38.78	19.85	53.54	14.75
N-211	76.15	12.80	97.66	21.51
N-089	74.77	1.35	87.29	12.52
N-054	59.17	21.69	72.42	13.25
N-028	69.26	8.65	85.74	16.48
N-220	86.30	2.92	111.73	25.43
N-023	37.43	21.73	61.89	24.46
N-053	65.72	23.32	79.52	13.80
N-209	46.79	16.02	55.49	8.70
N-040	36.42	10.98	59.83	23.41
N-128	41.47	10.58	64.23	22.76
N-165	45.45	4.93	67.11	21.66
N-022	66.97	12.56	87.87	20.90
N-156	53.91	20.67	65.53	11.61
N-100	51.84	4.06	71.31	19.47
N-189	66.21	20.97	85.87	19.66
N-162	66.66	21.49	91.54	24.88
N-191	35.90	14.01	57.28	21.38
N-194	28.31	5.25	39.88	11.58
N-166	55.87	17.79	70.07	14.21
N-207	42.01	19.30	62.09	20.08
N-118	74.60	3.89	95.83	21.23
N-188	65.93	23.18	75.22	9.29
N-008	36.11	14.85	45.50	9.39
N-132	69.40	2.79	83.53	14.13
N-054	57.70	20.14	68.10	10.41

ANNEXURE IV — Pump Schedule, page 40

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-10	Raw water	49.2	67.9	45.49
P-03	Raw water	15.3	41.6	8.66
P-06	Backwash	52.4	79.0	56.33
P-07	Booster	298.6	75.9	308.94
P-09	Sludge	248.8	67.9	230.20
P-06	Sludge	282.9	53.5	206.32
P-06	Clear water	215.6	66.0	193.94
P-10	Raw water	126.0	19.4	33.32
P-07	Sludge	208.8	85.3	242.58
P-03	Clear water	182.4	36.4	90.54
P-04	Booster	100.5	15.2	20.80
P-12	Clear water	292.4	78.6	313.27
P-04	Clear water	256.7	49.6	173.46
P-07	Sludge	272.1	14.6	54.07
P-01	Clear water	315.7	48.4	208.17
P-04	Sludge	254.2	34.9	120.89
P-10	Raw water	96.1	24.0	31.42
P-04	Clear water	143.6	62.8	122.84
P-05	Backwash	168.5	35.4	81.30
P-11	Booster	153.8	17.0	35.64
P-05	Sludge	183.1	14.6	36.46
P-09	Clear water	164.2	46.6	104.18
P-06	Booster	26.8	61.8	22.54

ANNEXURE IV — Water Quality Test Results, page 41

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	32.9	0.82	1
pH	-	8.1	7.9	6.5-8.5
Total coliform	MPN/100ml	366	Absent	Absent
pH	-	8.1	7.0	6.5-8.5
Total coliform	MPN/100ml	241	Absent	Absent
Iron	mg/l	3.08	0.24	1.0
Iron	mg/l	3.28	0.12	1.0
Total hardness	mg/l	420	137	200
Total coliform	MPN/100ml	689	Absent	Absent
Iron	mg/l	3.24	0.23	1.0
Total hardness	mg/l	126	78	200
Total hardness	mg/l	272	140	200
Total coliform	MPN/100ml	260	Absent	Absent
Iron	mg/l	0.50	0.19	1.0
Residual chlorine	mg/l	Nil	0.73	0.2 min
Total hardness	mg/l	450	151	200
Total hardness	mg/l	131	149	200
Turbidity	NTU	165.1	0.25	1
Iron	mg/l	1.61	0.09	1.0
Total hardness	mg/l	135	150	200
Iron	mg/l	1.12	0.25	1.0
pH	-	6.8	7.2	6.5-8.5
pH	-	6.5	7.6	6.5-8.5
Total coliform	MPN/100ml	774	Absent	Absent
Iron	mg/l	1.36	0.05	1.0
Residual chlorine	mg/l	Nil	0.36	0.2 min
Turbidity	NTU	29.1	0.33	1

ANNEXURE IV — Bore Log Summary, page 42

Depth (m)	Strata	N value	Sample
25.5	Silty clay	16	DS
9.1	Coarse sand	6	UDS
2.7	Silty clay	47	DS
25.8	Gravelly sand	8	DS
24.7	Silty clay	53	UDS
2.7	Weathered rock	5	UDS
3.5	Silty clay	21	DS
22.9	Gravelly sand	22	UDS
8.2	Medium sand	47	DS
14.2	Weathered rock	18	SPT
21.2	Weathered rock	10	UDS
9.6	Weathered rock	22	UDS
24.4	Medium sand	11	UDS
17.2	Gravelly sand	26	SPT
29.9	Clayey silt	11	DS
1.1	Weathered rock	33	SPT
5.7	Gravelly sand	55	DS
28.4	Clayey silt	58	SPT
0.8	Coarse sand	19	UDS
3.3	Weathered rock	57	UDS
6.2	Weathered rock	59	SPT
3.0	Clayey silt	36	SPT
7.3	Silty clay	32	SPT

ANNEXURE IV — Population Projection, page 43

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2036	Ward 2	13,997	135	1.890
2056	Ward 12	9,013	135	1.217
2026	Ward 20	34,970	135	4.721
2051	Ward 19	13,373	150	2.006
2036	Ward 22	6,963	150	1.044
2041	Ward 4	21,152	150	3.173
2036	Ward 6	41,483	100	4.148
2051	Ward 1	28,832	150	4.325
2056	Ward 24	9,021	70	0.631
2031	Ward 10	7,249	135	0.979
2036	Ward 11	5,393	100	0.539
2036	Ward 4	10,800	135	1.458
2051	Ward 13	35,357	150	5.304
2056	Ward 14	4,548	100	0.455
2041	Ward 22	26,112	150	3.917
2041	Ward 23	45,413	70	3.179
2026	Ward 9	25,817	135	3.485
2026	Ward 8	10,235	150	1.535
2046	Ward 9	41,984	150	6.298
2036	Ward 11	41,515	150	6.227
2036	Ward 16	28,670	100	2.867
2056	Ward 23	40,796	150	6.119
2026	Ward 18	13,650	70	0.956

ANNEXURE IV-A — YEAR-WISE CASH FLOW STATEMENT

All figures in Rs. crore. Year 0 is the year of commencement.

Year	Capital cost	O&M cost	Gross benefit	Net cash flow
0	92.00	-	-	-92.00
1	104.00	-	-	-104.00
2	68.80	-	-	-68.80
3	-	5.56	31.76	26.20
4	-	5.56	31.63	26.07
5	-	5.56	31.51	25.95
6	-	5.56	31.38	25.82
7	-	5.56	31.25	25.69
8	-	5.56	31.12	25.56
9	-	5.56	31.00	25.44
10	-	5.56	30.87	25.31
11	-	5.56	30.74	25.18
12	-	5.56	30.62	25.06
13	-	5.56	30.49	24.93
14	-	5.56	30.36	24.80
15	-	5.56	30.24	24.68
16	-	5.56	30.11	24.55
17	-	5.56	29.98	24.42
18	-	5.56	29.85	24.29
19	-	5.56	29.73	24.17
20	-	5.56	29.60	24.04
21	-	5.56	29.47	23.91
22	-	5.56	29.35	23.79
23	-	5.56	29.22	23.66
24	-	5.56	29.09	23.53
25	-	5.56	28.97	23.41
26	-	5.56	28.84	23.28
27	-	5.56	28.71	23.15
28	-	5.56	28.58	23.02
29	-	5.56	28.46	22.90
30	-	5.56	28.33	22.77

ANNEXURE V — WATER QUALITY TEST REPORTS

Raw water analysis over four seasons against IS:10500 parameters.

ANNEXURE V — Pipeline Schedule, page 1

Chainage (m)	Dia (mm)	Material	Class	Length (m)
15,786	150	PVC-O	PN 10	1498.5
7,373	200	DI	K-9	1338.8
7,483	200	HDPE	PN 16	1734.6
10,549	100	MS	PN 6	1142.7
9,077	400	DI	K-7	184.4
9,588	250	MS	K-7	1223.2
17,451	200	MS	PN 16	1091.3
3,777	600	MS	PN 6	1484.3
2,373	250	DI	PN 16	297.6
16,441	400	PVC-O	PN 6	1208.9
138	150	MS	K-9	1006.4
23,621	300	DI	K-9	919.6
11,693	200	PVC-O	PN 10	1715.8
15,253	150	MS	PN 10	570.1
12,273	500	MS	PN 6	702.0
18,600	200	HDPE	PN 6	440.1
5,607	150	HDPE	K-7	996.0
8,944	250	MS	PN 6	610.2
11,312	500	DI	PN 10	854.7
2,584	150	HDPE	K-9	155.1
16,313	250	MS	PN 6	1661.0
15,195	600	MS	K-9	938.9
19,484	500	DI	K-9	1721.7
16,744	250	HDPE	K-7	998.3
7,742	500	HDPE	K-7	1682.0
20,661	250	PVC-O	PN 6	539.0
9,430	150	HDPE	PN 10	1347.9
6,782	200	PVC-O	K-7	107.5

ANNEXURE V — Hydraulic Node Analysis, page 2

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-119	32.71	4.45	58.33	25.62
N-219	91.84	2.82	110.74	18.90
N-074	33.15	18.59	42.87	9.71
N-164	62.69	21.39	87.87	25.18
N-148	63.57	15.64	87.09	23.52
N-014	47.21	22.70	63.91	16.69
N-011	87.82	19.68	95.92	8.10
N-111	83.31	8.12	100.68	17.37
N-092	77.33	2.33	93.97	16.64
N-073	91.47	11.93	115.52	24.06
N-173	42.05	16.40	61.51	19.46
N-040	43.57	1.50	65.75	22.18
N-151	67.69	20.14	84.27	16.58
N-025	50.68	17.95	73.49	22.81
N-057	33.71	22.86	48.23	14.52
N-075	77.81	17.24	90.23	12.42
N-043	29.09	2.98	43.96	14.86
N-213	52.05	6.37	68.63	16.59
N-135	44.60	19.90	70.48	25.87
N-014	80.88	15.66	91.61	10.73
N-203	74.82	16.77	90.56	15.74
N-190	30.52	3.92	56.22	25.71
N-151	57.25	21.46	64.95	7.70
N-156	72.50	10.32	88.09	15.59
N-181	44.91	3.18	65.85	20.95
N-214	63.83	2.36	71.48	7.66
N-005	28.75	11.04	43.51	14.76

ANNEXURE V — Pump Schedule, page 3

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-01	Sludge	241.0	47.4	155.52
P-01	Backwash	164.2	52.4	117.18
P-12	Backwash	297.9	32.9	133.52
P-05	Backwash	300.8	34.2	140.11
P-05	Raw water	193.2	38.9	102.51
P-05	Sludge	25.5	57.7	20.04
P-09	Backwash	243.1	23.5	77.74
P-11	Raw water	298.7	21.0	85.35
P-05	Booster	35.1	78.0	37.34
P-11	Backwash	192.7	87.4	229.53
P-02	Clear water	281.4	68.8	263.60
P-04	Raw water	51.6	36.0	25.32
P-04	Raw water	33.3	16.3	7.41
P-01	Backwash	137.1	37.5	70.14
P-06	Backwash	122.7	51.1	85.41
P-10	Sludge	32.1	79.6	34.85
P-11	Sludge	53.5	16.0	11.65
P-07	Clear water	110.5	60.0	90.33
P-05	Sludge	143.8	49.9	97.67
P-12	Raw water	103.3	62.0	87.25
P-09	Backwash	40.8	17.4	9.65
P-01	Booster	230.0	67.5	211.65
P-05	Booster	71.8	50.8	49.73
P-04	Raw water	72.5	19.0	18.76
P-12	Backwash	80.7	43.6	47.96
P-02	Raw water	80.5	73.8	80.87
P-12	Backwash	132.1	85.0	153.01

ANNEXURE V — Water Quality Test Results, page 4

Parameter	Unit	Raw water	Treated	IS 10500 limit
Iron	mg/l	2.05	0.25	1.0
Total hardness	mg/l	269	174	200
Total hardness	mg/l	93	187	200
pH	-	8.3	7.1	6.5-8.5
pH	-	6.7	8.0	6.5-8.5
pH	-	7.9	6.9	6.5-8.5
Turbidity	NTU	206.9	0.33	1
Total coliform	MPN/100ml	811	Absent	Absent
pH	-	6.7	7.8	6.5-8.5
Iron	mg/l	1.75	0.12	1.0
Residual chlorine	mg/l	Nil	0.63	0.2 min
Total coliform	MPN/100ml	53	Absent	Absent
Total hardness	mg/l	173	116	200
Iron	mg/l	2.36	0.23	1.0
Turbidity	NTU	162.4	0.80	1
Total coliform	MPN/100ml	460	Absent	Absent
Total coliform	MPN/100ml	361	Absent	Absent
Turbidity	NTU	192.3	0.30	1
Turbidity	NTU	235.1	0.63	1
Total coliform	MPN/100ml	494	Absent	Absent
Turbidity	NTU	79.7	0.45	1
Total coliform	MPN/100ml	383	Absent	Absent
pH	-	7.9	7.5	6.5-8.5

ANNEXURE V — Bore Log Summary, page 5

Depth (m)	Strata	N value	Sample
21.4	Silty clay	39	DS
21.7	Silty clay	33	UDS
3.3	Coarse sand	53	DS
17.2	Coarse sand	40	UDS
28.0	Coarse sand	25	UDS
28.7	Clayey silt	52	DS
17.7	Silty clay	8	SPT
17.5	Medium sand	11	DS
11.7	Gravelly sand	19	UDS
15.0	Clayey silt	14	SPT
16.8	Weathered rock	10	SPT
12.9	Medium sand	42	DS
12.8	Silty clay	56	SPT
26.6	Medium sand	33	UDS
27.1	Silty clay	47	SPT
10.5	Coarse sand	34	SPT
3.7	Medium sand	44	DS
20.5	Coarse sand	8	SPT
16.3	Clayey silt	54	DS
2.4	Clayey silt	16	UDS
9.7	Clayey silt	35	UDS
25.5	Medium sand	36	DS
24.5	Silty clay	39	UDS
20.7	Gravelly sand	25	UDS

ANNEXURE V — Population Projection, page 6

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2046	Ward 9	39,004	135	5.266
2056	Ward 9	27,355	70	1.915
2036	Ward 18	28,743	135	3.880
2046	Ward 14	22,478	100	2.248
2036	Ward 4	28,673	135	3.871
2026	Ward 21	38,518	150	5.778
2041	Ward 6	35,772	150	5.366
2051	Ward 1	22,274	150	3.341
2046	Ward 19	9,658	100	0.966
2031	Ward 20	12,956	135	1.749
2026	Ward 8	41,001	70	2.870
2026	Ward 24	16,061	135	2.168
2026	Ward 5	19,552	135	2.640
2056	Ward 15	40,062	100	4.006
2031	Ward 15	27,387	135	3.697
2056	Ward 20	8,482	135	1.145
2051	Ward 19	3,516	100	0.352
2051	Ward 13	3,025	135	0.408
2056	Ward 9	6,776	100	0.678
2051	Ward 3	36,636	100	3.664
2026	Ward 20	23,987	135	3.238
2041	Ward 16	11,950	100	1.195
2036	Ward 7	3,821	135	0.516
2051	Ward 7	45,138	135	6.094
2051	Ward 12	25,566	70	1.790
2036	Ward 5	16,396	150	2.459
2041	Ward 3	19,450	100	1.945
2046	Ward 2	26,723	100	2.672
2041	Ward 8	31,880	150	4.782

ANNEXURE V — Pipeline Schedule, page 7

Chainage (m)	Dia (mm)	Material	Class	Length (m)
601	400	PVC-O	PN 16	1072.5
20,955	500	MS	K-7	1251.3
779	150	MS	K-9	1799.9
7,684	400	DI	K-9	1228.4
19,456	300	DI	PN 6	1596.2
21,072	200	HDPE	PN 16	1618.6
1,085	150	HDPE	PN 6	1396.3
238	250	PVC-O	K-9	906.4
7,741	250	HDPE	K-7	957.0
8,636	300	HDPE	PN 10	915.6
20,523	250	DI	PN 10	1379.5
13,846	250	HDPE	PN 16	1027.5
7,633	250	MS	K-9	1103.1
12,672	250	PVC-O	PN 6	1687.2
3,752	300	MS	PN 6	659.5
2,765	250	MS	PN 6	1647.1
6,793	200	HDPE	K-9	1078.7
18,700	500	DI	PN 10	1074.3
14,422	200	DI	PN 10	54.8
23,477	300	PVC-O	PN 6	103.4
12,638	600	HDPE	PN 6	276.4
21,261	100	MS	K-7	1277.3

ANNEXURE V — Hydraulic Node Analysis, page 8

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-177	63.16	19.30	88.29	25.13
N-081	60.07	3.89	77.83	17.76
N-020	41.83	17.68	61.94	20.11
N-061	54.75	13.29	61.95	7.20
N-088	46.27	21.43	60.24	13.97
N-099	83.55	21.92	103.80	20.24
N-125	67.61	1.98	91.12	23.51
N-201	86.14	12.57	105.44	19.30
N-008	53.77	23.29	72.78	19.01
N-078	82.09	16.54	107.26	25.17
N-143	82.38	8.53	99.64	17.27
N-091	95.65	16.28	113.77	18.12
N-110	65.43	13.39	80.21	14.78
N-106	95.70	11.41	120.07	24.37
N-154	31.60	18.66	55.76	24.16
N-160	53.21	15.55	65.14	11.93
N-208	92.24	18.17	102.80	10.55
N-001	72.77	15.48	84.63	11.86
N-126	50.71	0.41	69.45	18.74
N-110	59.28	15.51	67.75	8.47
N-045	59.45	17.97	81.56	22.11
N-039	29.67	0.23	42.11	12.44
N-117	29.76	4.15	44.64	14.88

ANNEXURE V — Pump Schedule, page 9

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-06	Clear water	269.9	52.0	191.14
P-05	Clear water	92.5	47.9	60.35
P-01	Clear water	257.3	21.8	76.59
P-02	Booster	154.0	12.4	26.03
P-02	Clear water	290.4	87.1	344.56
P-03	Clear water	253.9	31.6	109.40
P-03	Raw water	313.0	16.8	71.58
P-04	Raw water	256.0	30.5	106.50
P-09	Raw water	277.3	18.5	70.06
P-03	Sludge	48.2	37.3	24.49
P-11	Raw water	19.2	63.1	16.52
P-05	Booster	181.5	87.0	215.12
P-04	Raw water	308.1	30.3	127.32
P-08	Clear water	158.0	38.5	82.98
P-08	Backwash	208.6	27.7	78.70
P-06	Clear water	169.0	68.8	158.42
P-03	Raw water	221.7	40.4	122.04
P-09	Sludge	16.3	30.6	6.80
P-02	Clear water	318.3	29.2	126.67
P-07	Booster	213.7	54.2	157.74
P-02	Raw water	316.7	24.8	106.92
P-02	Raw water	173.7	75.5	178.72
P-07	Booster	301.2	84.0	344.84
P-02	Booster	127.2	73.5	127.39
P-09	Booster	221.7	36.6	110.60
P-07	Raw water	80.3	85.9	93.96
P-08	Clear water	51.5	16.8	11.80
P-10	Backwash	75.8	58.7	60.61
P-02	Sludge	58.4	34.5	27.45

ANNEXURE V — Water Quality Test Results, page 10

Parameter	Unit	Raw water	Treated	IS 10500 limit
Iron	mg/l	3.53	0.06	1.0
Iron	mg/l	0.61	0.21	1.0
Turbidity	NTU	167.5	0.76	1
Total coliform	MPN/100ml	762	Absent	Absent
Residual chlorine	mg/l	Nil	0.65	0.2 min
Total coliform	MPN/100ml	335	Absent	Absent
Residual chlorine	mg/l	Nil	0.37	0.2 min
Residual chlorine	mg/l	Nil	0.64	0.2 min
Total coliform	MPN/100ml	515	Absent	Absent
Total hardness	mg/l	168	108	200
Residual chlorine	mg/l	Nil	0.26	0.2 min
pH	-	7.4	7.3	6.5-8.5
Iron	mg/l	3.84	0.12	1.0
Iron	mg/l	4.15	0.15	1.0
Total hardness	mg/l	297	93	200
Residual chlorine	mg/l	Nil	0.46	0.2 min
Total coliform	MPN/100ml	376	Absent	Absent
Total coliform	MPN/100ml	760	Absent	Absent
Total hardness	mg/l	152	86	200
Turbidity	NTU	83.7	0.81	1
pH	-	8.2	7.6	6.5-8.5
Turbidity	NTU	162.8	0.74	1
pH	-	6.6	7.6	6.5-8.5
Residual chlorine	mg/l	Nil	0.66	0.2 min
Turbidity	NTU	159.2	0.40	1
Iron	mg/l	2.37	0.05	1.0
Total hardness	mg/l	111	115	200
Turbidity	NTU	97.9	0.67	1

ANNEXURE V — Bore Log Summary, page 11

Depth (m)	Strata	N value	Sample
3.7	Weathered rock	45	DS
19.0	Silty clay	31	DS
28.9	Weathered rock	13	UDS
25.3	Coarse sand	11	SPT
20.2	Silty clay	19	DS
8.9	Gravelly sand	38	DS
12.2	Weathered rock	15	UDS
14.7	Silty clay	13	UDS
9.9	Coarse sand	49	SPT
27.6	Silty clay	22	SPT
17.7	Silty clay	23	SPT
13.5	Weathered rock	31	SPT
19.9	Silty clay	32	DS
11.1	Gravelly sand	15	SPT
17.7	Gravelly sand	57	SPT
17.2	Weathered rock	30	DS
24.0	Weathered rock	39	DS
21.9	Coarse sand	14	DS
4.3	Clayey silt	60	DS
23.6	Coarse sand	24	UDS
9.6	Silty clay	55	UDS
7.2	Clayey silt	42	UDS
28.1	Gravelly sand	6	SPT
22.1	Clayey silt	19	DS
22.7	Weathered rock	12	UDS
22.7	Clayey silt	55	UDS
14.7	Clayey silt	39	DS
27.8	Weathered rock	40	DS
29.6	Weathered rock	30	SPT
24.7	Coarse sand	17	UDS

ANNEXURE V — Population Projection, page 12

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2056	Ward 10	6,303	150	0.945
2046	Ward 20	38,093	70	2.667
2046	Ward 10	19,112	135	2.580
2051	Ward 24	26,726	135	3.608
2046	Ward 8	4,292	150	0.644
2056	Ward 19	42,796	150	6.419
2026	Ward 17	43,079	70	3.016
2031	Ward 2	8,579	70	0.601
2031	Ward 4	21,100	100	2.110
2041	Ward 21	19,369	70	1.356
2036	Ward 8	39,993	70	2.800
2041	Ward 8	32,479	70	2.274
2046	Ward 17	38,461	135	5.192
2026	Ward 6	3,884	135	0.524
2031	Ward 8	31,998	100	3.200
2036	Ward 16	3,364	100	0.336
2041	Ward 20	29,962	70	2.097
2051	Ward 19	21,922	70	1.535
2046	Ward 21	27,686	150	4.153
2041	Ward 22	22,213	150	3.332
2051	Ward 23	41,881	100	4.188
2041	Ward 14	34,699	100	3.470
2041	Ward 18	40,876	135	5.518

ANNEXURE V — Pipeline Schedule, page 13

Chainage (m)	Dia (mm)	Material	Class	Length (m)
18,941	100	DI	K-7	81.2
7,600	150	PVC-O	PN 10	323.9
12,669	300	DI	PN 6	60.0
22,595	200	HDPE	K-9	983.2
6,538	500	DI	PN 6	916.6
2,580	200	PVC-O	PN 10	1142.4
3,496	500	MS	K-7	1617.7
1,830	600	PVC-O	K-7	1265.8
9,745	500	HDPE	PN 16	103.0
8,840	200	DI	PN 6	1556.0
20,169	500	MS	K-9	467.3
10,703	100	HDPE	PN 6	1451.0
14,347	600	MS	PN 6	1302.4
7,734	300	DI	PN 10	1010.7
21,922	300	PVC-O	K-7	464.4
6,526	500	PVC-O	PN 16	689.2
10,674	400	DI	PN 6	247.1
9,371	600	PVC-O	K-9	564.0
22,907	100	MS	K-7	120.2
19,909	100	HDPE	PN 10	1604.7
3,713	150	DI	PN 16	1410.6
22,061	250	PVC-O	PN 16	983.1
12,409	100	PVC-O	PN 10	678.6
13,831	100	HDPE	K-7	1766.4
23,827	600	MS	PN 10	938.2

ANNEXURE V — Hydraulic Node Analysis, page 14

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-056	86.31	21.02	98.39	12.08
N-115	31.43	3.30	53.57	22.14
N-037	84.44	21.30	92.43	7.99
N-174	67.21	12.17	86.63	19.42
N-025	76.25	2.07	89.66	13.41
N-136	76.42	3.44	97.58	21.17
N-115	42.17	19.88	57.22	15.06
N-203	88.86	6.70	112.90	24.04
N-119	69.61	15.04	80.14	10.53
N-005	89.53	15.02	113.35	23.82
N-065	85.92	2.73	98.70	12.78
N-072	73.49	23.82	99.14	25.64
N-121	86.39	23.46	105.79	19.40
N-108	78.32	3.55	88.30	9.98
N-215	74.85	23.89	84.35	9.50
N-058	39.00	4.76	55.81	16.80
N-199	42.83	13.60	59.71	16.88
N-076	77.05	4.03	100.96	23.91
N-078	54.34	5.20	62.03	7.68
N-096	67.33	4.50	91.81	24.48
N-113	53.54	2.01	62.40	8.86
N-126	47.93	18.70	62.72	14.79
N-106	76.78	18.91	98.56	21.78
N-075	38.57	3.92	63.47	24.90

ANNEXURE V — Pump Schedule, page 15

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-06	Sludge	47.4	86.4	55.77
P-04	Clear water	68.6	60.5	56.50
P-05	Clear water	236.4	58.5	188.49
P-05	Raw water	102.1	46.2	64.21
P-07	Booster	66.2	73.4	66.20
P-03	Raw water	95.5	75.9	98.76
P-09	Sludge	161.9	68.3	150.61
P-09	Backwash	31.6	35.2	15.15
P-12	Clear water	153.5	75.2	157.26
P-06	Backwash	67.4	35.5	32.56
P-11	Backwash	304.1	15.9	65.85
P-07	Raw water	190.4	67.1	174.02
P-08	Booster	318.0	69.0	298.92
P-06	Booster	108.8	82.8	122.81
P-04	Sludge	174.3	74.2	176.31
P-09	Booster	57.0	73.6	57.18
P-05	Backwash	197.2	41.1	110.36
P-11	Clear water	243.6	16.5	54.84
P-08	Booster	100.3	21.4	29.32
P-11	Backwash	216.3	15.6	46.10
P-07	Backwash	286.6	82.3	321.22
P-04	Sludge	269.1	50.4	184.85
P-03	Clear water	211.4	61.7	177.63
P-02	Clear water	70.8	41.1	39.69
P-02	Sludge	158.3	84.0	181.08
P-10	Clear water	89.0	23.0	27.85
P-08	Booster	53.9	80.1	58.81

ANNEXURE V — Water Quality Test Results, page 16

Parameter	Unit	Raw water	Treated	IS 10500 limit
Total coliform	MPN/100ml	733	Absent	Absent
pH	-	6.6	8.0	6.5-8.5
Total coliform	MPN/100ml	276	Absent	Absent
Residual chlorine	mg/l	Nil	0.64	0.2 min
Total hardness	mg/l	229	158	200
Turbidity	NTU	82.2	0.58	1
Turbidity	NTU	32.2	0.24	1
Total coliform	MPN/100ml	682	Absent	Absent
Total coliform	MPN/100ml	469	Absent	Absent
Iron	mg/l	3.51	0.06	1.0
pH	-	8.0	7.1	6.5-8.5
Total coliform	MPN/100ml	694	Absent	Absent
Residual chlorine	mg/l	Nil	0.48	0.2 min
Total hardness	mg/l	398	139	200
Turbidity	NTU	12.0	0.73	1
Total coliform	MPN/100ml	647	Absent	Absent
Residual chlorine	mg/l	Nil	0.74	0.2 min
Iron	mg/l	1.77	0.07	1.0
Residual chlorine	mg/l	Nil	0.65	0.2 min
Residual chlorine	mg/l	Nil	0.52	0.2 min
Turbidity	NTU	148.5	0.68	1
Residual chlorine	mg/l	Nil	0.67	0.2 min
Turbidity	NTU	90.1	0.48	1
Total coliform	MPN/100ml	773	Absent	Absent
Iron	mg/l	1.47	0.20	1.0
Total hardness	mg/l	449	86	200
Iron	mg/l	2.20	0.24	1.0
pH	-	7.1	7.8	6.5-8.5
Total hardness	mg/l	418	128	200

ANNEXURE V — Bore Log Summary, page 17

Depth (m)	Strata	N value	Sample
28.3	Silty clay	6	SPT
30.0	Gravelly sand	18	DS
1.4	Gravelly sand	40	UDS
1.3	Coarse sand	42	UDS
29.6	Silty clay	23	SPT
1.1	Gravelly sand	30	SPT
7.7	Coarse sand	15	SPT
14.6	Silty clay	21	UDS
26.2	Silty clay	7	SPT
29.6	Gravelly sand	34	DS
5.2	Silty clay	22	DS
5.6	Clayey silt	8	DS
5.7	Silty clay	7	SPT
22.2	Weathered rock	31	SPT
6.5	Silty clay	48	DS
26.1	Silty clay	18	DS
25.7	Coarse sand	54	DS
7.4	Clayey silt	26	DS
25.8	Weathered rock	51	DS
21.9	Weathered rock	13	SPT
21.5	Weathered rock	10	SPT
11.2	Silty clay	34	UDS
5.2	Gravelly sand	26	SPT
29.9	Coarse sand	53	SPT
24.2	Gravelly sand	58	SPT
29.5	Silty clay	21	UDS
16.5	Medium sand	55	UDS
3.4	Gravelly sand	19	UDS
9.1	Medium sand	15	SPT

ANNEXURE V — Population Projection, page 18

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2031	Ward 18	17,648	70	1.235
2046	Ward 5	19,628	135	2.650
2031	Ward 13	10,030	70	0.702
2031	Ward 9	31,049	150	4.657
2051	Ward 6	19,947	135	2.693
2031	Ward 16	23,517	70	1.646
2031	Ward 12	21,834	135	2.948
2046	Ward 14	42,215	135	5.699
2056	Ward 2	29,154	150	4.373
2036	Ward 1	1,904	135	0.257
2041	Ward 19	36,661	70	2.566
2046	Ward 17	31,352	135	4.233
2031	Ward 14	2,406	150	0.361
2041	Ward 16	21,250	135	2.869
2051	Ward 19	12,213	70	0.855
2046	Ward 5	23,505	100	2.350
2041	Ward 4	20,739	135	2.800
2036	Ward 21	6,241	100	0.624
2046	Ward 1	30,445	150	4.567
2056	Ward 18	41,490	135	5.601
2026	Ward 15	19,111	70	1.338
2041	Ward 14	19,851	150	2.978
2041	Ward 18	16,200	135	2.187

ANNEXURE V — Pipeline Schedule, page 19

Chainage (m)	Dia (mm)	Material	Class	Length (m)
401	100	PVC-O	PN 10	684.7
16,892	600	HDPE	K-9	857.8
3,199	600	PVC-O	K-9	1779.8
23,561	400	PVC-O	PN 6	1432.9
7,490	100	DI	PN 16	1099.7
22,070	200	PVC-O	PN 6	245.4
8,511	600	PVC-O	PN 6	608.7
2,109	500	MS	PN 16	1348.9
5,645	300	DI	PN 6	279.2
17,856	150	DI	K-7	835.0
3,896	600	MS	K-9	1137.5
7,665	300	PVC-O	K-7	936.4
3,949	100	MS	PN 10	209.7
3,425	600	MS	PN 6	1046.7
16,740	250	DI	PN 6	675.5
7,082	150	HDPE	PN 10	187.0
14,188	600	HDPE	K-7	422.2
18,865	400	PVC-O	PN 10	609.6
19,223	300	PVC-O	PN 10	844.9
5,101	600	HDPE	K-9	689.5
14,584	100	HDPE	PN 10	1586.1
1,208	100	DI	PN 16	1058.8

ANNEXURE V — Hydraulic Node Analysis, page 20

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-122	54.93	8.83	73.56	18.63
N-035	85.87	18.58	97.03	11.16
N-168	45.25	2.09	62.27	17.02
N-019	77.50	23.51	97.63	20.13
N-133	37.36	11.93	56.74	19.37
N-062	34.08	14.08	50.58	16.50
N-085	63.63	11.25	83.42	19.79
N-045	74.58	11.66	98.62	24.04
N-024	51.82	18.79	68.01	16.18
N-138	79.57	5.55	96.19	16.62
N-072	31.56	8.61	55.00	23.44
N-004	55.29	7.61	74.08	18.79
N-132	95.06	11.07	107.90	12.84
N-142	84.46	18.20	109.25	24.79
N-055	38.03	5.78	49.98	11.96
N-052	39.29	5.91	59.81	20.52
N-132	95.17	9.94	121.16	26.00
N-093	89.72	12.67	102.34	12.62
N-086	87.96	3.41	102.61	14.65
N-216	72.16	0.42	95.57	23.41
N-013	88.09	18.16	96.72	8.63
N-020	71.78	6.19	81.13	9.34
N-126	84.65	6.65	100.69	16.04
N-155	58.38	12.42	81.71	23.33
N-027	90.06	15.77	97.84	7.78
N-086	67.01	1.43	74.48	7.47
N-189	31.20	20.07	48.45	17.25
N-163	41.88	19.59	63.07	21.18

ANNEXURE V — Pump Schedule, page 21

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-02	Raw water	248.5	46.9	158.70
P-08	Backwash	239.1	81.2	264.56
P-06	Sludge	227.3	75.4	233.47
P-01	Backwash	278.0	54.7	207.21
P-10	Backwash	276.2	23.2	87.20
P-11	Booster	132.4	17.6	31.69
P-01	Backwash	304.3	64.7	268.04
P-02	Clear water	151.4	52.5	108.31
P-11	Booster	148.1	45.8	92.32
P-12	Raw water	275.6	87.2	327.54
P-11	Booster	227.1	69.2	214.15
P-09	Backwash	227.1	83.1	257.19
P-11	Booster	81.0	43.2	47.61
P-10	Backwash	188.5	57.7	148.16
P-01	Raw water	217.7	65.2	193.43
P-02	Raw water	116.2	15.4	24.43
P-09	Clear water	19.0	14.8	3.84
P-05	Booster	63.4	81.2	70.06
P-06	Raw water	27.3	25.9	9.63
P-09	Backwash	283.0	76.1	293.30
P-10	Clear water	51.3	68.3	47.73
P-05	Backwash	288.6	59.3	233.09
P-06	Backwash	159.9	72.4	157.74
P-07	Backwash	153.5	19.8	41.44
P-04	Raw water	19.0	16.2	4.20
P-01	Booster	204.9	13.2	36.98
P-08	Booster	208.0	66.5	188.44
P-03	Sludge	129.0	21.0	36.86

ANNEXURE V — Water Quality Test Results, page 22

Parameter	Unit	Raw water	Treated	IS 10500 limit
Turbidity	NTU	34.7	0.31	1
Total hardness	mg/l	468	61	200
pH	-	8.0	8.1	6.5-8.5
Total coliform	MPN/100ml	514	Absent	Absent
Iron	mg/l	2.33	0.22	1.0
pH	-	7.2	7.3	6.5-8.5
pH	-	7.2	7.9	6.5-8.5
Turbidity	NTU	168.5	0.88	1
Total hardness	mg/l	476	114	200
Total coliform	MPN/100ml	386	Absent	Absent
Turbidity	NTU	17.8	0.79	1
Iron	mg/l	3.20	0.27	1.0
Turbidity	NTU	66.6	0.23	1
Total coliform	MPN/100ml	832	Absent	Absent
Total hardness	mg/l	232	93	200
Residual chlorine	mg/l	Nil	0.23	0.2 min
Total coliform	MPN/100ml	201	Absent	Absent
Total hardness	mg/l	457	129	200
pH	-	8.0	7.1	6.5-8.5
pH	-	8.4	8.1	6.5-8.5
pH	-	7.0	7.4	6.5-8.5
Turbidity	NTU	157.2	0.66	1

ANNEXURE V — Bore Log Summary, page 23

Depth (m)	Strata	N value	Sample
24.7	Gravelly sand	41	SPT
20.5	Gravelly sand	4	SPT
21.0	Clayey silt	44	SPT
13.0	Weathered rock	44	UDS
8.0	Silty clay	13	UDS
9.8	Coarse sand	22	SPT
8.2	Clayey silt	24	UDS
3.3	Medium sand	46	UDS
1.6	Coarse sand	30	DS
4.3	Clayey silt	5	UDS
17.5	Gravelly sand	52	DS
13.2	Coarse sand	52	SPT
25.5	Weathered rock	37	DS
7.0	Weathered rock	50	UDS
7.5	Silty clay	42	UDS
19.6	Silty clay	12	UDS
15.5	Weathered rock	12	DS
9.5	Medium sand	19	SPT
22.6	Medium sand	10	DS
2.6	Weathered rock	51	DS
27.5	Gravelly sand	15	DS
27.2	Coarse sand	60	SPT
29.2	Coarse sand	9	UDS
20.3	Clayey silt	44	SPT
27.4	Coarse sand	40	UDS
25.5	Clayey silt	13	SPT

ANNEXURE V — Population Projection, page 24

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2026	Ward 2	29,294	135	3.955
2051	Ward 4	4,854	70	0.340
2036	Ward 3	17,467	150	2.620
2056	Ward 17	15,130	70	1.059
2041	Ward 20	25,659	70	1.796
2041	Ward 8	31,207	150	4.681
2036	Ward 15	8,807	100	0.881
2041	Ward 15	29,926	135	4.040
2026	Ward 4	45,461	100	4.546
2056	Ward 24	34,638	100	3.464
2046	Ward 3	28,033	70	1.962
2041	Ward 9	27,516	150	4.127
2046	Ward 10	2,259	150	0.339
2051	Ward 5	45,480	150	6.822
2051	Ward 16	7,093	135	0.958
2031	Ward 14	21,465	100	2.147
2036	Ward 20	26,344	70	1.844
2046	Ward 12	40,565	70	2.840
2041	Ward 12	19,087	100	1.909
2036	Ward 4	32,108	70	2.248
2041	Ward 8	44,464	135	6.003
2051	Ward 21	11,817	70	0.827
2056	Ward 1	22,382	135	3.022
2046	Ward 19	13,771	100	1.377
2026	Ward 6	13,063	150	1.959

ANNEXURE V — Pipeline Schedule, page 25

Chainage (m)	Dia (mm)	Material	Class	Length (m)
10,822	500	HDPE	PN 16	213.0
17,684	600	MS	K-7	1287.8
18,605	150	HDPE	PN 10	1680.0
15,743	300	HDPE	K-7	1358.8
18,580	300	PVC-O	PN 6	1023.0
23,670	100	MS	K-9	341.3
16,222	300	DI	PN 10	1274.3
3,694	300	DI	PN 6	464.7
501	100	HDPE	PN 16	957.9
14,567	150	PVC-O	PN 6	1225.2
18,132	300	DI	K-9	1039.3
15,516	250	PVC-O	PN 6	85.6
4,692	100	DI	PN 10	1712.5
8,921	400	DI	PN 16	837.3
10,553	400	MS	PN 10	716.5
356	200	PVC-O	K-9	385.4
1,704	100	MS	PN 6	1574.8
10,295	600	PVC-O	PN 6	546.6
5,266	500	PVC-O	K-7	1209.5
20,603	600	DI	PN 10	1507.9
4,650	250	DI	PN 16	672.9
2,732	200	DI	PN 6	791.3

ANNEXURE V — Hydraulic Node Analysis, page 26

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-139	35.71	11.65	44.99	9.28
N-144	49.32	22.54	73.43	24.11
N-100	33.05	14.99	50.25	17.21
N-175	86.48	12.07	107.62	21.14
N-122	77.74	16.54	102.57	24.83
N-039	50.31	6.86	64.10	13.79
N-114	32.17	8.47	43.41	11.23
N-186	52.54	19.05	73.93	21.39
N-013	90.66	20.37	109.33	18.67
N-120	83.16	4.87	107.47	24.32
N-001	94.43	14.87	111.87	17.44
N-176	59.22	3.82	74.19	14.97
N-156	69.63	13.98	89.99	20.36
N-049	81.31	1.29	90.12	8.81
N-209	29.11	4.36	36.11	7.00
N-091	49.91	17.05	63.82	13.91
N-032	62.47	11.21	73.92	11.45
N-018	90.83	3.41	115.30	24.47
N-194	80.20	5.45	96.10	15.90
N-024	35.98	16.44	51.37	15.39
N-052	70.00	11.03	92.28	22.28
N-141	65.07	13.03	76.30	11.23
N-176	63.52	16.21	89.13	25.62
N-062	50.32	11.16	66.04	15.72
N-032	74.86	19.55	99.64	24.78
N-004	88.77	6.19	96.55	7.77
N-147	85.87	23.85	98.64	12.77
N-192	95.70	20.85	118.09	22.39
N-074	80.50	0.58	100.69	20.19

ANNEXURE V — Pump Schedule, page 27

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-02	Backwash	315.8	21.7	93.43
P-02	Raw water	274.7	57.7	216.12
P-04	Clear water	170.2	63.3	146.79
P-10	Clear water	102.2	12.5	17.34
P-01	Booster	26.1	12.5	4.45
P-11	Backwash	91.3	49.8	61.92
P-02	Booster	252.1	57.8	198.67
P-12	Clear water	182.7	84.6	210.61
P-02	Booster	92.2	84.5	106.16
P-07	Backwash	151.2	82.6	170.25
P-07	Backwash	186.3	12.3	31.10
P-11	Backwash	262.1	86.5	308.91
P-08	Sludge	68.7	13.9	13.05
P-06	Raw water	240.0	68.5	223.83
P-11	Raw water	220.2	76.7	230.20
P-11	Booster	72.9	56.3	55.98
P-03	Sludge	197.6	42.1	113.25
P-08	Sludge	103.7	75.8	107.15
P-03	Raw water	182.7	18.2	45.20
P-09	Clear water	261.6	43.3	154.47
P-01	Raw water	242.4	22.4	74.14
P-11	Clear water	226.5	25.0	77.25
P-07	Booster	75.9	42.8	44.26
P-03	Backwash	129.5	55.4	97.66
P-05	Backwash	250.4	20.3	69.22
P-06	Clear water	203.5	48.1	133.30
P-03	Backwash	93.7	22.1	28.21
P-04	Clear water	45.9	50.5	31.62

ANNEXURE VI — PIPELINE ALIGNMENT SHEETS

Longitudinal sections and alignment for the rising and transmission mains.

ANNEXURE VI — Sheet 1

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. PL-324, revision A.

ANNEXURE VI — Sheet 2

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
8,514	200	HDPE	K-9	1345.6
8,991	500	MS	PN 10	787.3
22,430	300	DI	PN 6	683.1
22,130	150	HDPE	PN 16	853.6
3,106	100	PVC-O	PN 16	947.7
3,086	150	DI	K-7	310.9
12,046	500	HDPE	K-9	1792.9
3,872	200	DI	K-7	231.6
10,499	500	HDPE	K-9	1083.5
19,750	400	MS	K-7	1359.6
20,785	100	DI	PN 16	861.9
10,873	300	MS	PN 16	315.4
11,333	250	PVC-O	K-7	55.2
8,660	600	MS	PN 6	58.1
1,011	250	DI	K-9	1163.3
6,434	200	MS	PN 10	356.2
16,358	250	PVC-O	K-9	1582.7
19,515	400	HDPE	K-7	299.2
22,402	300	HDPE	PN 6	1546.9
5,240	400	PVC-O	K-9	50.9
16,548	200	DI	PN 6	225.3

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

ANNEXURE VI — Sheet 3

Design Computation

Velocity = $7.693 \times 11.441 = 88.016$ m/s (permissible 105.903) — revise section

Velocity = $74.238 \times 6.310 = 468.411$ m/s (permissible 849.887) — safe

Discharge = $57.157 \times 10.601 = 605.900$ lps (permissible 973.577) — safe

Detention time = $79.887 \times 7.395 = 590.749$ hr (permissible 1078.714) — safe

Velocity = $44.028 \times 6.893 = 303.489$ m/s (permissible 534.291) — safe

Residual pressure = $86.815 \times 9.329 = 809.856$ m (permissible 1442.351) — safe

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

ANNEXURE VI — Sheet 4

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Reference: drawing no. WS-281, revision A.

ANNEXURE VI — Sheet 5

Bore Log Summary

Depth (m)	Strata	N value	Sample
5.7	Silty clay	37	SPT
19.0	Silty clay	48	UDS
14.1	Medium sand	36	SPT
22.0	Clayey silt	37	UDS
5.2	Clayey silt	9	DS
11.2	Silty clay	14	DS
19.7	Silty clay	24	UDS
1.2	Weathered rock	16	DS
27.3	Clayey silt	47	SPT
22.0	Silty clay	30	UDS
20.1	Medium sand	18	DS
22.2	Gravelly sand	56	UDS
1.1	Coarse sand	25	SPT
20.2	Weathered rock	40	DS
9.1	Weathered rock	25	SPT
5.7	Weathered rock	50	SPT
24.2	Silty clay	33	DS
13.0	Clayey silt	16	DS
15.0	Silty clay	42	DS

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

ANNEXURE VI — Sheet 6

Design Computation

Detention time = $5.177 \times 11.292 = 58.461$ hr (permissible 63.489) — safe

Residual pressure = $88.531 \times 9.639 = 853.367$ m (permissible 1247.716) — safe

Detention time = $12.265 \times 5.654 = 69.346$ hr (permissible 110.326) — safe

Discharge = $71.389 \times 10.351 = 738.960$ lps (permissible 781.588) — safe

Discharge = $79.017 \times 1.090 = 86.119$ lps (permissible 92.529) — safe

Residual pressure = $26.373 \times 10.661 = 281.166$ m (permissible 463.315) — safe

Head loss = $59.733 \times 4.980 = 297.479$ m (permissible 350.852) — safe

Velocity = $73.174 \times 10.263 = 750.954$ m/s (permissible 827.876) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE VI — Sheet 7

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Reference: drawing no. PL-631, revision C.

ANNEXURE VI — Sheet 8

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-03	Sludge	73.2	18.7	18.68
P-05	Backwash	293.9	70.1	280.75
P-10	Backwash	53.7	20.0	14.62
P-04	Raw water	206.0	16.4	46.03
P-03	Backwash	249.8	16.4	55.78
P-02	Sludge	37.2	86.4	43.78
P-04	Raw water	253.8	44.5	153.74
P-08	Raw water	152.7	27.6	57.43
P-07	Clear water	138.6	37.9	71.61
P-04	Sludge	242.5	67.0	221.52
P-07	Backwash	25.1	25.6	8.76
P-07	Booster	124.7	36.1	61.30
P-03	Sludge	44.5	58.0	35.17
P-01	Booster	103.9	82.0	116.15
P-12	Raw water	111.3	25.9	39.31
P-01	Backwash	27.1	46.4	17.13
P-04	Sludge	204.2	82.6	229.86
P-07	Backwash	127.0	31.2	54.05
P-06	Booster	283.4	17.7	68.26
P-07	Sludge	293.2	28.4	113.34
P-01	Backwash	107.2	73.5	107.39

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

ANNEXURE VI — Sheet 9

Design Computation

Residual pressure = $81.733 \times 8.420 = 688.158$ m (permissible 1285.940) — safe

Discharge = $46.585 \times 6.367 = 296.596$ lps (permissible 410.434) — safe

Discharge = $68.507 \times 9.997 = 684.842$ lps (permissible 863.016) — safe

Head loss = $45.493 \times 2.747 = 124.955$ m (permissible 133.332) — safe

Head loss = $83.960 \times 9.517 = 799.090$ m (permissible 931.281) — safe

Head loss = $76.888 \times 4.030 = 309.845$ m (permissible 403.136) — safe

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

ANNEXURE VI — Sheet 10

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. HY-465, revision A.

ANNEXURE VI — Sheet 11

Pipeline Schedule

Chainage (m)	Dia (mm)	Material	Class	Length (m)
2,825	400	HDPE	K-9	1184.0
9,168	100	DI	PN 16	1051.5
3,359	250	PVC-O	K-9	1404.4
9,460	250	DI	K-7	407.9
14,334	150	DI	PN 16	860.7
15,302	200	DI	K-9	1634.0
7,250	100	HDPE	PN 6	1486.3
20,152	250	HDPE	PN 16	223.3
16,251	250	MS	PN 16	1216.2
11,793	300	HDPE	PN 16	328.6
4,500	150	DI	PN 6	281.0
3,373	200	PVC-O	K-9	1613.0
4,830	400	MS	PN 16	1341.7
20,572	300	DI	K-7	495.0
19,784	500	HDPE	K-9	1191.0
14,767	150	HDPE	PN 10	329.5
10,142	300	PVC-O	K-9	815.2
6,016	300	HDPE	PN 16	1287.2
7,953	600	DI	K-9	126.5
9,697	200	MS	PN 10	686.5
1,174	600	DI	PN 6	1071.1

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE VI — Sheet 12

Design Computation

Head loss = $44.966 \times 3.863 = 173.703 \text{ m}$ (permissible 287.515) — safe

Detention time = $78.326 \times 2.310 = 180.911 \text{ hr}$ (permissible 240.751) — safe

Detention time = $62.920 \times 10.087 = 634.701 \text{ hr}$ (permissible 682.897) — safe

Detention time = $53.023 \times 0.401 = 21.241 \text{ hr}$ (permissible 33.158) — safe

Residual pressure = $87.213 \times 2.598 = 226.548 \text{ m}$ (permissible 390.178) — safe

Velocity = $42.086 \times 8.580 = 361.089 \text{ m/s}$ (permissible 546.629) — safe

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

ANNEXURE VI — Sheet 13

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. HY-210, revision A.

ANNEXURE VI — Sheet 14

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-164	52.42	16.73	73.45	21.03
N-046	56.25	1.11	68.39	12.14
N-009	92.11	14.10	114.32	22.21
N-050	62.37	19.17	70.65	8.28
N-121	64.35	22.66	71.91	7.56
N-129	44.02	21.25	62.74	18.71
N-144	52.28	22.49	67.94	15.66
N-200	50.10	19.51	75.12	25.02
N-058	39.75	14.52	52.83	13.09
N-149	34.56	0.92	52.24	17.68
N-181	54.45	15.19	75.83	21.38
N-156	88.80	2.86	109.78	20.98
N-049	83.36	16.15	94.34	10.98
N-018	39.15	21.56	55.37	16.22
N-072	72.95	3.73	85.60	12.65
N-018	60.24	13.25	69.00	8.76
N-132	29.00	7.39	42.50	13.50
N-135	81.93	21.21	103.32	21.39
N-101	54.26	20.81	67.36	13.10
N-012	66.12	1.31	77.86	11.74
N-044	67.17	2.10	80.17	13.00
N-100	79.98	17.21	98.77	18.79
N-169	34.81	18.09	42.93	8.12

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE VI — Sheet 15

Design Computation

Head loss = $46.128 \times 1.316 = 60.688$ m (permissible 75.935) — revise section

Detention time = $28.584 \times 2.655 = 75.888$ hr (permissible 112.028) — safe

Velocity = $72.302 \times 0.682 = 49.300$ m/s (permissible 85.770) — safe

Head loss = $6.909 \times 9.543 = 65.937$ m (permissible 71.028) — safe

Discharge = $61.970 \times 5.763 = 357.133$ lps (permissible 656.225) — safe

Residual pressure = $27.469 \times 0.532 = 14.610$ m (permissible 25.102) — safe

Residual pressure = $54.543 \times 6.455 = 352.102$ m (permissible 515.518) — revise section

Discharge = $36.095 \times 8.543 = 308.358$ lps (permissible 471.423) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

ANNEXURE VI — Sheet 16

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. ST-340, revision C.

ANNEXURE VI — Sheet 17

Population Projection

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2036	Ward 21	22,585	135	3.049
2046	Ward 11	8,662	150	1.299
2046	Ward 15	30,600	100	3.060
2041	Ward 14	8,109	100	0.811
2031	Ward 18	12,201	135	1.647
2056	Ward 10	29,076	150	4.361
2036	Ward 19	13,155	70	0.921
2051	Ward 13	11,125	100	1.113
2046	Ward 1	5,609	100	0.561
2031	Ward 21	29,107	70	2.037
2046	Ward 23	12,936	135	1.746
2036	Ward 3	3,643	100	0.364
2041	Ward 10	24,830	135	3.352
2031	Ward 23	40,320	100	4.032
2056	Ward 18	33,769	150	5.065
2031	Ward 2	12,712	150	1.907
2051	Ward 15	23,741	135	3.205
2036	Ward 10	15,753	150	2.363
2026	Ward 23	2,843	135	0.384
2056	Ward 24	16,895	70	1.183
2036	Ward 18	17,895	70	1.253
2026	Ward 15	27,615	100	2.761
2041	Ward 9	14,371	150	2.156

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

ANNEXURE VI — Sheet 18

Design Computation

Head loss = $50.423 \times 10.517 = 530.316$ m (permissible 666.665) — safe

Discharge = $2.984 \times 6.923 = 20.659$ lps (permissible 38.167) — safe

Discharge = $61.035 \times 3.789 = 231.248$ lps (permissible 407.054) — revise section

Detention time = $51.230 \times 4.381 = 224.447$ hr (permissible 388.626) — safe

Velocity = $26.449 \times 10.226 = 270.454$ m/s (permissible 487.798) — safe

Residual pressure = $24.395 \times 9.960 = 242.973$ m (permissible 447.505) — safe

Velocity = $61.989 \times 0.467 = 28.973$ m/s (permissible 52.862) — safe

Discharge = $79.374 \times 11.546 = 916.425$ lps (permissible 1098.235) — safe

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

ANNEXURE VI — Sheet 19

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

Reference: drawing no. HY-192, revision A.

ANNEXURE VI — Sheet 20

Bore Log Summary

Depth (m)	Strata	N value	Sample
25.5	Weathered rock	59	UDS
29.5	Medium sand	60	UDS
7.2	Coarse sand	37	DS
2.9	Silty clay	16	UDS
15.0	Medium sand	8	UDS
12.1	Weathered rock	23	DS
11.2	Clayey silt	58	DS
11.4	Gravelly sand	49	SPT
14.8	Coarse sand	42	DS
19.5	Weathered rock	10	SPT
6.4	Clayey silt	13	SPT
14.9	Weathered rock	36	DS
28.1	Gravelly sand	46	UDS
8.4	Medium sand	31	DS
22.2	Silty clay	48	UDS
19.0	Silty clay	15	DS
12.8	Medium sand	38	UDS
16.4	Weathered rock	4	DS
1.7	Gravelly sand	42	SPT
24.6	Gravelly sand	44	DS

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE VI — Sheet 21

Design Computation

Detention time = $68.395 \times 0.922 = 63.090$ hr (permissible 91.746) — safe

Detention time = $44.500 \times 4.522 = 201.222$ hr (permissible 305.211) — safe

Velocity = $67.101 \times 8.664 = 581.381$ m/s (permissible 676.172) — safe

Velocity = $78.688 \times 9.974 = 784.826$ m/s (permissible 1018.791) — safe

Residual pressure = $14.499 \times 7.012 = 101.663$ m (permissible 176.235) — safe

Residual pressure = $60.042 \times 3.566 = 214.091$ m (permissible 351.985) — safe

Detention time = $72.761 \times 6.137 = 446.561$ hr (permissible 653.340) — safe

Discharge = $54.905 \times 7.955 = 436.788$ lps (permissible 669.027) — safe

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

ANNEXURE VI — Sheet 22

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. ST-441, revision A.

ANNEXURE VI — Sheet 23

Bore Log Summary

Depth (m)	Strata	N value	Sample
18.1	Medium sand	33	SPT
25.1	Weathered rock	47	SPT
20.5	Silty clay	13	UDS
21.8	Coarse sand	44	SPT
19.9	Clayey silt	44	DS
14.3	Coarse sand	31	DS
27.2	Clayey silt	25	DS
6.4	Weathered rock	35	DS
7.2	Medium sand	46	SPT
4.2	Gravelly sand	50	SPT
18.5	Gravelly sand	7	DS
18.1	Clayey silt	36	DS
24.9	Coarse sand	51	UDS
21.6	Silty clay	58	DS
18.3	Silty clay	59	UDS
27.1	Weathered rock	52	UDS
16.7	Silty clay	36	UDS
10.8	Coarse sand	17	DS
2.1	Silty clay	46	DS
7.4	Medium sand	60	SPT
6.6	Coarse sand	32	SPT
28.8	Weathered rock	54	DS

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

ANNEXURE VI — Sheet 24

Design Computation

Head loss = $33.719 \times 7.124 = 240.196$ m (permissible 380.712) — safe

Detention time = $61.453 \times 1.472 = 90.440$ hr (permissible 150.837) — safe

Discharge = $59.322 \times 5.677 = 336.783$ lps (permissible 421.525) — revise section

Velocity = $64.989 \times 3.186 = 207.046$ m/s (permissible 368.748) — safe

Residual pressure = $72.139 \times 10.022 = 723.005$ m (permissible 1328.596) — safe

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE VI — Sheet 25

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Reference: drawing no. HY-107, revision C.

ANNEXURE VI — Sheet 26

Pump Schedule

Pump	Duty	Discharge (lps)	Head (m)	Rating (kW)
P-02	Raw water	150.6	75.4	154.77
P-05	Sludge	307.9	65.5	274.62
P-05	Sludge	76.5	26.0	27.11
P-01	Backwash	118.5	20.1	32.44
P-11	Booster	185.6	45.5	115.03
P-09	Raw water	222.3	15.3	46.28
P-03	Booster	111.3	16.0	24.23
P-10	Clear water	311.2	56.3	238.72
P-09	Backwash	174.4	74.7	177.57
P-04	Booster	101.1	61.8	85.09
P-11	Sludge	18.1	83.5	20.61
P-09	Clear water	249.6	62.0	210.97
P-09	Sludge	155.1	63.3	133.68
P-05	Backwash	290.8	36.5	144.80
P-12	Sludge	197.3	68.1	182.92
P-11	Raw water	72.6	45.8	45.26
P-10	Sludge	112.2	67.3	102.88
P-05	Raw water	235.9	62.7	201.55
P-09	Booster	141.0	35.6	68.42
P-09	Clear water	210.4	52.4	150.13
P-10	Booster	192.4	43.8	114.70
P-10	Raw water	287.9	56.5	221.69
P-12	Raw water	288.5	54.0	212.28
P-05	Backwash	102.7	31.6	44.17
P-08	Sludge	169.4	61.1	141.12
P-05	Raw water	67.7	32.0	29.55

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE VI — Sheet 27

Design Computation

Residual pressure = $83.206 \times 2.251 = 187.320$ m (permissible 281.087) — safe

Discharge = $2.997 \times 2.103 = 6.302$ lps (permissible 10.183) — safe

Detention time = $58.566 \times 5.225 = 306.014$ hr (permissible 513.013) — safe

Head loss = $67.380 \times 7.865 = 529.972$ m (permissible 644.275) — safe

Residual pressure = $62.143 \times 10.887 = 676.533$ m (permissible 966.252) — safe

Detention time = $9.123 \times 3.199 = 29.182$ hr (permissible 35.933) — safe

Residual pressure = $16.271 \times 7.315 = 119.028$ m (permissible 190.829) — safe

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

ANNEXURE VI — Sheet 28

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. WS-324, revision A.

ANNEXURE VI — Sheet 29

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-043	28.50	3.23	44.55	16.05
N-026	36.74	19.46	59.01	22.27
N-164	43.25	13.05	51.98	8.73
N-217	53.62	15.27	71.60	17.98
N-028	81.85	3.56	88.91	7.06
N-144	70.57	19.93	89.82	19.25
N-073	86.69	7.71	109.51	22.82
N-053	46.31	21.80	71.80	25.49
N-138	87.31	3.67	100.07	12.76
N-106	34.37	20.47	44.33	9.96
N-124	34.90	12.19	46.96	12.05
N-073	52.14	8.00	62.90	10.76
N-210	92.45	0.88	99.77	7.32
N-015	91.52	16.74	106.81	15.29
N-059	81.11	8.30	95.69	14.58
N-091	33.41	3.31	58.14	24.72

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

ANNEXURE VII — CLEARANCES AND CONSULTATION RECORDS

Copies of approvals and minutes of consultations with the Municipal Board and Gram Panchayats.

ANNEXURE VII — Sheet 1

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

Reference: drawing no. PL-811, revision B.

ANNEXURE VII — Sheet 2

Population Projection

Year	Ward	Population	Rate (lpcd)	Demand (MLD)
2031	Ward 6	22,414	135	3.026
2046	Ward 15	6,105	135	0.824
2041	Ward 7	15,048	150	2.257
2056	Ward 4	37,580	100	3.758
2046	Ward 6	23,970	70	1.678
2056	Ward 22	8,441	135	1.140
2051	Ward 5	18,502	150	2.775
2026	Ward 12	27,150	135	3.665
2036	Ward 21	40,899	150	6.135
2051	Ward 12	25,503	135	3.443
2056	Ward 23	30,162	135	4.072
2031	Ward 17	15,441	100	1.544
2046	Ward 17	35,438	135	4.784
2036	Ward 17	40,157	135	5.421
2031	Ward 21	34,481	100	3.448
2046	Ward 9	12,423	70	0.870

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE VII — Sheet 3

Design Computation

Detention time = $72.205 \times 11.778 = 850.413$ hr (permissible 952.565) — safe

Detention time = $39.481 \times 11.507 = 454.308$ hr (permissible 650.687) — safe

Discharge = $88.156 \times 3.697 = 325.900$ lps (permissible 348.651) — safe

Residual pressure = $27.576 \times 4.856 = 133.916$ m (permissible 240.926) — safe

Head loss = $29.917 \times 1.556 = 46.560$ m (permissible 88.370) — revise section

The distribution network has been analysed as a looped system. Residual pressure at every node satisfies the minimum prescribed at peak demand, and pipe sizes have been optimised against the head available from the service reservoir.

ANNEXURE VII — Sheet 4

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

Reference: drawing no. HY-989, revision A.

ANNEXURE VII — Sheet 5

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Residual chlorine	mg/l	Nil	0.31	0.2 min
Turbidity	NTU	48.2	0.40	1
Residual chlorine	mg/l	Nil	0.66	0.2 min
Total coliform	MPN/100ml	180	Absent	Absent
Iron	mg/l	1.17	0.02	1.0
Total coliform	MPN/100ml	57	Absent	Absent
Total hardness	mg/l	346	129	200
Iron	mg/l	3.23	0.13	1.0
Total hardness	mg/l	268	144	200
Iron	mg/l	3.46	0.02	1.0
Residual chlorine	mg/l	Nil	0.38	0.2 min
Turbidity	NTU	93.5	0.81	1
Turbidity	NTU	228.6	0.47	1
Iron	mg/l	3.99	0.14	1.0
Total hardness	mg/l	304	108	200
Turbidity	NTU	145.5	0.84	1
Residual chlorine	mg/l	Nil	0.80	0.2 min
Residual chlorine	mg/l	Nil	0.20	0.2 min
Turbidity	NTU	15.4	0.86	1
Iron	mg/l	0.72	0.19	1.0
Turbidity	NTU	107.5	0.68	1
Total hardness	mg/l	367	137	200

Pipe material has been selected on the basis of working pressure, laying condition and life-cycle cost. Ductile iron conforming to IS 8329 has been adopted for the transmission main and HDPE conforming to IS 4984 for the distribution laterals.

ANNEXURE VII — Sheet 6

Design Computation

Velocity = $38.885 \times 1.285 = 49.981$ m/s (permissible 70.428) — safe

Discharge = $16.858 \times 0.695 = 11.718$ lps (permissible 20.648) — safe

Velocity = $52.560 \times 11.424 = 600.453$ m/s (permissible 688.482) — safe

Velocity = $2.874 \times 7.967 = 22.900$ m/s (permissible 36.375) — safe

Detention time = $30.689 \times 8.997 = 276.096$ hr (permissible 306.247) — safe

Head loss = $83.507 \times 5.678 = 474.168$ m (permissible 576.670) — safe

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

ANNEXURE VII — Sheet 7

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

Reference: drawing no. PL-297, revision C.

ANNEXURE VII — Sheet 8

Water Quality Test Results

Parameter	Unit	Raw water	Treated	IS 10500 limit
Total coliform	MPN/100ml	732	Absent	Absent
Total hardness	mg/l	117	74	200
Iron	mg/l	2.63	0.19	1.0
Iron	mg/l	1.78	0.09	1.0
Turbidity	NTU	163.9	0.87	1
Iron	mg/l	3.28	0.17	1.0
pH	-	6.9	7.6	6.5-8.5
Iron	mg/l	3.28	0.03	1.0
Total coliform	MPN/100ml	875	Absent	Absent
Residual chlorine	mg/l	Nil	0.76	0.2 min
Residual chlorine	mg/l	Nil	0.70	0.2 min
Total hardness	mg/l	258	77	200
Total hardness	mg/l	403	134	200
Turbidity	NTU	133.0	0.24	1
Iron	mg/l	0.70	0.04	1.0
Turbidity	NTU	61.6	0.64	1
pH	-	7.4	6.9	6.5-8.5
Residual chlorine	mg/l	Nil	0.46	0.2 min
Total coliform	MPN/100ml	46	Absent	Absent
pH	-	6.6	7.1	6.5-8.5
Total hardness	mg/l	173	156	200
Turbidity	NTU	151.8	0.62	1

The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

ANNEXURE VII — Sheet 9

Design Computation

Head loss = $59.243 \times 8.329 = 493.453$ m (permissible 700.583) — safe

Detention time = $52.325 \times 1.464 = 76.583$ hr (permissible 131.199) — safe

Detention time = $7.001 \times 4.764 = 33.355$ hr (permissible 45.007) — safe

Residual pressure = $72.667 \times 3.029 = 220.096$ m (permissible 342.875) — safe

Residual pressure = $25.012 \times 11.591 = 289.922$ m (permissible 549.248) — safe

Discharge = $11.171 \times 5.013 = 56.004$ lps (permissible 92.815) — revise section

Head loss = $13.958 \times 4.611 = 64.365$ m (permissible 90.802) — safe

The treatment train comprises aeration, flash mixing, clariflocculation, rapid gravity filtration and disinfection by chlorination. Unit sizing and the design loading rates adopted are given in the design computations.

ANNEXURE VII — Sheet 10

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

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The per-capita supply level adopted is that prescribed by the CPHEEO Manual for a town with piped supply and full sewerage, with the applicable allowance for unaccounted-for water added to arrive at the gross demand.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

Reference: drawing no. HY-464, revision B.

ANNEXURE VII — Sheet 11

Hydraulic Node Analysis

Node	Ground level (m)	Demand (lps)	Head (m)	Residual (m)
N-010	42.28	20.28	59.49	17.22
N-176	41.28	4.79	49.88	8.60
N-048	39.28	11.76	55.47	16.19
N-182	55.54	8.70	64.72	9.18
N-183	77.90	10.92	96.58	18.67
N-168	77.70	19.11	99.12	21.42
N-206	51.85	1.34	76.48	24.64
N-018	73.45	21.94	99.13	25.68
N-130	54.26	2.91	65.15	10.89
N-169	67.32	10.95	87.31	19.99
N-151	77.49	21.84	100.63	23.14
N-092	30.85	11.88	41.38	10.53
N-172	71.98	19.32	95.87	23.89
N-168	88.15	18.36	112.86	24.71
N-206	89.91	17.45	100.39	10.48
N-081	55.07	6.58	69.41	14.34
N-209	55.94	2.77	79.13	23.20
N-095	44.28	15.61	51.33	7.05
N-088	90.65	12.40	112.93	22.28
N-123	33.84	16.52	45.34	11.49
N-102	76.37	22.33	92.07	15.71

Raw and treated water samples were tested for the parameters specified in IS 10500. The test reports of the accredited laboratory are enclosed at the Annexure and the treated water meets the acceptable limits for every parameter tested.

ANNEXURE VII — Sheet 12

Design Computation

Head loss = $30.260 \times 4.152 = 125.636$ m (permissible 174.167) — safe

Discharge = $9.886 \times 10.727 = 106.050$ lps (permissible 117.121) — safe

Detention time = $87.304 \times 9.277 = 809.905$ hr (permissible 1429.299) — safe

Head loss = $4.470 \times 2.360 = 10.550$ m (permissible 15.775) — safe

Velocity = $19.702 \times 0.773 = 15.232$ m/s (permissible 19.687) — safe

Head loss = $74.399 \times 6.936 = 516.044$ m (permissible 718.442) — safe

Detention time = $27.650 \times 10.368 = 286.672$ hr (permissible 339.080) — safe

The yield of the source has been assessed from the discharge records of the nearest gauging station for the period of record available. The lean-season yield exceeds the design draw-off with the margin required for a source of this class.

ANNEXURE VII — Sheet 13

Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

The design period has been adopted as 30 years in accordance with the CPHEEO Manual on Water Supply and Treatment. The base year population is taken from the latest Census and projected by the incremental increase method.

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Recurring operation and maintenance covers energy for pumping, treatment chemicals, establishment and periodic replacement. The energy component has been worked out from the pump duty and the operating hours assumed for the design year.

Reference: drawing no. PL-577, revision C.